Optimal Nonlinear Income Taxation

Lecture Notes

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1 The Environment

1.1 Households

There is a continuum of households indexed by their productivity type $z \in \mathcal{Z} = [z_0, \bar{z}]$, distributed according to a cumulative distribution function $F(z)$ with density $f(z) > 0$ on $\mathcal{Z}$.

Remark 1.1 (Bounded vs. unbounded support). The notation $\mathcal{Z} = [z_0, \bar{z}]$ allows $\bar{z} = \infty$: the support may be unbounded above. Much of the analysis in Sections 1–3 is stated for a generic interval $[z_0, \bar{z}]$ and applies in either case. The distinction matters for the optimal tax schedule at the top of the distribution: with bounded support, the marginal tax rate is zero at $\bar{z}$ (Section 8); with unbounded support and a Pareto tail, the limiting marginal tax rate is generally positive (the Diamond–Saez formula, also in Section 8).

Each household of type z has preferences over consumption c and labor supply n, represented by the utility function $U(c, n; z)$. We suppress the explicit dependence on z when it enters only through the budget constraint (as below), and write $U(c, n)$.

A household of type z produces output

$$y(z) = z\, n(z),$$

so that z is the marginal product of labor. For given output y, the labor required is $n = y/z$.

Assumption 1.1 (Regularity of preferences)**.** The utility function $U(c, n)$ is twice continuously differentiable and strictly concave:

$$U_{cc} < 0, \qquad U_{cc}U_{nn} - U_{cn}^2 > 0.$$

Assumption 1.2 (Normality)**.** Both consumption and leisure are normal goods. That is, when a household solves

$$\max_n \; U(zn + I,\, n)$$

for exogenous nonlabor income I, we have $c'(I) > 0$ and $n'(I) < 0$.

We verify Assumption 1.2 explicitly. The first-order condition of the household problem $\max_n U(zn + I, n)$ is

$$z\, U_c(zn + I, n) + U_n(zn + I, n) = 0.$$

Totally differentiating with respect to I:

$$z\, U_{cc}(zn'(I) + 1) + z\, U_{cn}\, n'(I) + U_{nc}(zn'(I) + 1) + U_{nn}\, n'(I) = 0.$$

Collecting terms in $n'(I)$:

$$\big[z^2 U_{cc} + 2zU_{cn} + U_{nn}\big] n'(I) = -\big[zU_{cc} + U_{nc}\big].$$

Hence

$$n'(I) = -\frac{zU_{cc} + U_{nc}}{z^2 U_{cc} + 2zU_{cn} + U_{nn}}, \tag{1}$$

and

$$c'(I) = zn'(I) + 1 = \frac{zU_{cn} + U_{nn}}{z^2U_{cc} + 2zU_{cn} + U_{nn}}. \tag{2}$$

The denominator is the quadratic form

$$(z, 1)\begin{pmatrix} U_{cc} & U_{cn} \\ U_{cn} & U_{nn} \end{pmatrix}\begin{pmatrix} z \\ 1 \end{pmatrix} = z^2U_{cc} + 2zU_{cn} + U_{nn} < 0,$$

which is strictly negative by the strict concavity of U (Assumption 1.1 implies the Hessian is negative definite). Therefore, the normality conditions $n'(I) < 0$ and $c'(I) > 0$ require

$$zU_{cc} + U_{nc} < 0 \qquad \text{(leisure is normal)}, \qquad zU_{cn} + U_{nn} < 0 \qquad \text{(consumption is normal)}. \tag{3}$$

1.2 Government

The government has an exogenous revenue requirement $B \geq 0$. It must design a tax-and-transfer system subject to the economy-wide resource constraint

$$\int_{z_0}^{\bar{z}} \big[y(z) - c(z)\big] f(z)\, dz \geq B. \tag{4}$$

The government evaluates outcomes using a social welfare function

$$\int_{z_0}^{\bar{z}} \psi(z)\, U(c(z), n(z))\, f(z)\, dz,$$

where $\psi(z) \geq 0$ are Pareto weights with $\int \psi(z) f(z)\, dz = 1$. A preference for redistribution is captured by $\psi(z)$ declining in z: the planner values utility gains to low types more than to high types.

1.3 First-best benchmark

If the government could observe each household's type z directly, it would solve

$$\max_{c(\cdot), y(\cdot)} \int_{z_0}^{\bar{z}} \psi(z)\, U\left(c(z), \frac{y(z)}{z}\right) f(z)\, dz \quad \text{s.t.} \quad \int_{z_0}^{\bar{z}} \big[y(z) - c(z)\big] f(z)\, dz \geq B,$$

where $\psi(z)$ is the Pareto weight on type z. With Lagrange multiplier μ on the resource constraint, the first-order conditions are

$$[c(z)]: \quad \psi(z)\, U_c\left(c(z), \frac{y(z)}{z}\right) = \mu, \tag{5}$$

$$[y(z)]: \quad \psi(z)\, U_n\left(c(z), \frac{y(z)}{z}\right) \frac{1}{z} = -\mu. \tag{6}$$

Dividing (6) by (5), the Pareto weights cancel and we obtain

$$-\frac{U_n\left(c(z), \frac{y(z)}{z}\right)}{z\, U_c\left(c(z), \frac{y(z)}{z}\right)} = 1. \tag{7}$$

The left-hand side is the marginal rate of substitution between output and consumption, which equals the marginal rate of transformation (unity). *There are no distortions at the first best*: the implicit marginal tax rate is zero for all types. This holds for *any* Pareto weights $\psi(z)$, not just the utilitarian case—at the first best, the planner can redistribute via lump-sum transfers without distorting the labor-leisure margin.

The extent of redistribution, however, does depend on $\psi(z)$. The first-order condition (5) gives $U_c(c(z), y(z)/z) = \mu/\psi(z)$: marginal utilities of consumption are inversely proportional to Pareto weights. Under a utilitarian objective with $\psi(z) = 1$ for all z, this equalizes U_c across all types—the planner redistributes until marginal utilities of consumption are equalized.

Remark 1.2 (Utility gradient at the first best (utilitarian case)). Under utilitarian weights $\psi(z) = 1$, the FOCs (5)–(6) become $U_c = \mu$ and $-U_n/z = \mu$. Differentiating $U_c = \mu$ (constant) with respect to z:

$$U_{cc}\, c'(z) + U_{cn} \frac{y'(z)z - y(z)}{z^2} = 0.$$

Similarly, differentiating $-U_n/z = \mu$:

$$\frac{1}{z^2} U_n - \frac{1}{z} U_{nc}\, c'(z) - \frac{1}{z} U_{nn} \frac{y'(z)z - y(z)}{z^2} = 0.$$

Solving these two equations simultaneously:

$$c'(z) = -\frac{1}{z} \frac{U_n\, U_{cn}}{U_{nn}U_{cc} - U_{cn}^2},$$

$$\frac{y'(z)z - y(z)}{z^2} = \frac{1}{z} \frac{U_n\, U_{cc}}{U_{nn}U_{cc} - U_{cn}^2}.$$

The utility gradient is then

$$\begin{aligned} \frac{dU}{dz} &= U_c\, c'(z) + U_n \frac{y'(z)z - y(z)}{z^2} \\ &= \frac{1}{z}\big[U_n U_{cc} - U_c U_{cn}\big] \frac{U_n}{U_{nn}U_{cc} - U_{cn}^2} \\ &= -\frac{\mu}{z} \underbrace{\big[zU_{cc} + U_{cn}\big]}_{<0 \text{ (leisure normal)}} \underbrace{\frac{U_n}{U_{nn}U_{cc} - U_{cn}^2}}_{<0} < 0. \end{aligned} \tag{8}$$

Under utilitarian weights, higher-productivity households have *lower* utility at the first best: the planner equalizes marginal utilities ($U_c = \mu$ for all z) and makes more productive types work harder, so they do not enjoy the fruits of their higher productivity. With general Pareto weights $\psi(z)$, the gradient depends on $\psi'(z)$ as well, and the sign can go either way.

2 The Nonlinear Income Tax Problem

The first-best allocation from Section 1 requires the government to observe each household's type z directly. In practice, the government observes only income $y = zn$, not the decomposition into productivity z and labor n separately. The government's instrument is therefore a *nonlinear income tax schedule* $T : \mathbb{R}_+ \to \mathbb{R}$, mapping pre-tax income y to a tax payment $T(y)$ (with $T(y) < 0$ representing a transfer).

2.1 The household's problem given T

Facing a tax schedule $T(\cdot)$, a household of type z solves

$$\max_{y \geq 0} \; U\Big(y - T(y),\; \frac{y}{z}\Big). \tag{9}$$

Let $y^*(z;T)$ denote the solution and define the induced allocation

$$y(z) = y^*(z;T), \qquad c(z) = y(z) - T(y(z)).$$

If T is differentiable at $y(z)$ and $y(z) > 0$, the first-order condition is

$$\big[1 - T'(y(z))\big]\, U_c\Big(c(z), \frac{y(z)}{z}\Big) + \frac{1}{z}\, U_n\Big(c(z), \frac{y(z)}{z}\Big) = 0, \tag{10}$$

which says the marginal rate of substitution between output and consumption equals the net-of-tax return:

$$-\frac{U_n(c, y/z)}{z\, U_c(c, y/z)} = 1 - T'(y).$$

2.2 The government's problem over tax functions

The government chooses $T(\cdot)$ to maximize social welfare subject to the resource constraint, *taking into account* that households optimize in response:

$$\begin{aligned} \max_{T(\cdot)} \quad & \int_{z_0}^{\bar{z}} \psi(z)\, U\Big(y^*(z;T) - T(y^*(z;T)),\; \frac{y^*(z;T)}{z}\Big)\, f(z)\, dz \\ \text{s.t.} \quad & \int_{z_0}^{\bar{z}} T(y^*(z;T))\, f(z)\, dz \geq B. \end{aligned} \tag{11}$$

This is a difficult problem for two reasons:

(i) **Infinite-dimensional control.** The choice variable is an entire function $T : \mathbb{R}_+ \to \mathbb{R}$, not a finite-dimensional parameter.

(ii) **Implicit dependence on T.** The mapping from the tax schedule to household behavior, $z \mapsto y^*(z;T)$, is determined implicitly through each household's optimization problem (9). Small changes in T can produce discontinuous jumps in behavior (e.g., when bunching occurs), and the map $T \mapsto y^*(\cdot\,;T)$ is generally non-smooth.

2.3 From tax functions to allocations: the taxation principle

The *taxation principle* (a special case of the revelation principle) provides a way around these difficulties. It establishes an equivalence between two formulations of the problem.

Definition 2.1 (Incentive compatibility)**.** An allocation $\{c(z), y(z)\}_{z \in \mathcal{Z}}$ is *incentive compatible* (IC) if no type gains by mimicking another: for all $z, \tilde{z} \in \mathcal{Z}$,

$$U\Big(c(z), \frac{y(z)}{z}\Big) \geq U\Big(c(\tilde{z}), \frac{y(\tilde{z})}{z}\Big).$$

Proposition 2.1 (Taxation principle)**.** *An allocation* $\{c(z), y(z)\}_{z\in\mathcal{Z}}$ *can be implemented by some nonlinear tax schedule* $T(\cdot)$ *if and only if it is incentive compatible (Definition 2.1) and resource feasible* (4)*.*

The key insight is the direction from allocations back to taxes: given any incentive-compatible allocation $\{c(z), y(z)\}$, one can construct a tax schedule $T(y) = y - c(z)$ for y in the range of $y(\cdot)$, extended appropriately off the equilibrium path, that implements the allocation. This means instead of optimizing over the space of tax functions, the planner can equivalently optimize over the space of incentive-compatible allocations—a more tractable problem.

3 The Mechanism Design Formulation

By the taxation principle (Proposition 2.1), we can optimize over incentive-compatible allocations $\{c(z), y(z)\}$ rather than tax functions. In this section we characterize the IC constraint (Definition 2.1) and show that, under a single-crossing condition, it reduces to a local envelope condition plus monotonicity.

3.1 The announcement problem and local IC

A household of true type z facing the mechanism $\{c(\cdot), y(\cdot)\}$ chooses which type $\tilde{z}$ to announce:

$$\max_{\tilde{z}} \; U\left(c(\tilde{z}), \frac{y(\tilde{z})}{z}\right). \tag{12}$$

At an IC allocation, truth-telling ($\tilde{z} = z$) is optimal. The first-order condition at $\tilde{z} = z$ gives the *local* IC condition:

$$U_c\left(c(z), \frac{y(z)}{z}\right) c'(z) + U_n\left(c(z), \frac{y(z)}{z}\right) \frac{y'(z)}{z} = 0. \tag{13}$$

3.2 The envelope condition (local incentive compatibility)

Define the value function of the announcement problem:

$$V(z) = \max_{\tilde{z}} \; U\left(c(\tilde{z}), \frac{y(\tilde{z})}{z}\right).$$

At an incentive-compatible allocation, $V(z) = U(c(z), y(z)/z) \equiv \mathcal{U}(z)$. By the envelope theorem, we differentiate with respect to z *only through the place where* z *appears directly* (i.e., in $y(\tilde{z})/z$), not through the optimally chosen $\tilde{z}$:

$$\frac{dV}{dz} = -U_n\left(c(\tilde{z}), \frac{y(\tilde{z})}{z}\right) \frac{y(\tilde{z})}{z^2}.$$

Evaluating at truth-telling $\tilde{z} = z$:

$$\boxed{\frac{d\mathcal{U}}{dz} = -U_n\left(c(z), \frac{y(z)}{z}\right) \frac{y(z)}{z^2}.} \tag{14}$$

Since $U_n < 0$ (the marginal disutility of labor) and $y(z)/z^2 > 0$, we have $d\mathcal{U}/dz > 0$: more productive households enjoy higher utility. This is the *local* (first-order) incentive compatibility constraint.

Remark 3.1 (Derivation via direct differentiation). We can also derive (14) without the envelope theorem. Differentiating $\mathcal{U}(z) = U(c(z), y(z)/z)$ with respect to z:

$$\frac{d\mathcal{U}}{dz} = U_c\, c'(z) + U_n \left[\frac{y'(z)}{z} - \frac{y(z)}{z^2}\right] = \underbrace{U_c\, c'(z) + U_n \frac{y'(z)}{z}}_{=0 \text{ by } (13)} - U_n \frac{y(z)}{z^2},$$

confirming (14).

3.3 Single crossing and global incentive compatibility

The local condition (14) is a necessary condition for incentive compatibility, but it is not automatically sufficient: a household might prefer a *non-local* deviation. The single-crossing property guarantees that local incentive compatibility implies global incentive compatibility.

Definition 3.1 (Marginal rate of substitution in (c, y)-space)**.** For a household of type z at a point (c, y), the slope of the indifference curve in (c, y)-space is

$$S(c, y, z) \equiv \left.\frac{dc}{dy}\right|_{\bar{U}} = -\frac{U_n(c, y/z)/z}{U_c(c, y/z)} = \frac{1}{y}\,\phi(c, y/z), \tag{15}$$

where

$$\phi(c, n) \equiv -n\,\frac{U_n(c, n)}{U_c(c, n)} \tag{16}$$

is the expenditure on leisure measured in units of consumption.

Derivation of (15). Along an indifference curve $U(c, y/z) = \bar{U}$, totally differentiate with respect to y:

$$U_c\,\frac{dc}{dy} + U_n\,\frac{\partial(y/z)}{\partial y} = 0 \qquad \Longrightarrow \qquad U_c\,\frac{dc}{dy} + \frac{U_n}{z} = 0.$$

Solving for the slope gives the first equality in (15):

$$S(c, y, z) = \left.\frac{dc}{dy}\right|_{\bar{U}} = -\frac{U_n(c, y/z)}{z\,U_c(c, y/z)}.$$

For the second equality, substitute $n = y/z$ and multiply numerator and denominator by $n/y = 1/z$:

$$S = -\frac{U_n}{z\,U_c} = \frac{1}{y}\cdot\left(-\frac{y}{z}\right)\frac{U_n}{U_c} = \frac{1}{y}\left(-n\,\frac{U_n}{U_c}\right) = \frac{1}{y}\,\phi(c, n). \qquad \square$$

Assumption 3.1 (Single-crossing property)**.** $\phi(c, n)$ is strictly increasing in n for every fixed c.

Lemma 3.1 (Implication of single crossing)**.** *Under Assumption 3.1, for any fixed point (c, y) with $y > 0$, the indifference curve slope $S(c, y, z)$ is strictly decreasing in z: more productive types have flatter indifference curves through any given (c, y) bundle.*

Proof. Fix (c, y) and consider two types $z_1 < z_2$. To deliver output y, the required labor is $n_i = y/z_i$, so $n_1 > n_2$. By Assumption 3.1, $\phi(c, n_1) > \phi(c, n_2)$. Since $S(c, y, z_i) = \phi(c, n_i)/y$, we conclude $S(c, y, z_1) > S(c, y, z_2)$. $\square$

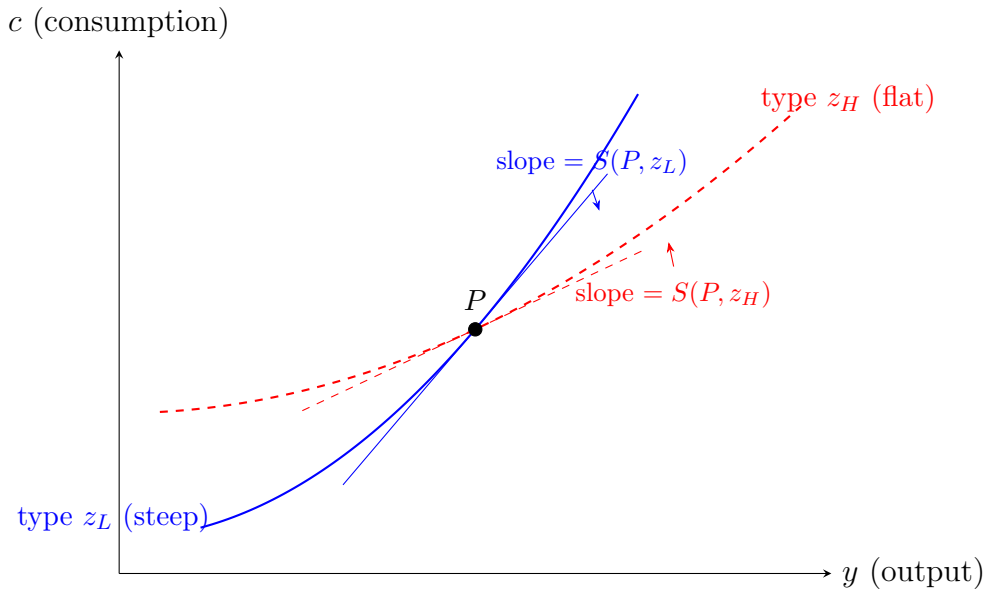

Figure 1: The single-crossing property (Lemma 3.1). At any point P, the indifference curve of the less productive type z_L (blue, solid) is steeper than that of the more productive type z_H (red, dashed). Since $n_L = y/z_L > y/z_H = n_H$, the low type works harder to produce the same output, requiring more compensation per unit of additional y.

Remark 3.2 (Verification for separable preferences). For $U(c, n) = u(c) - v(n)$ with $v' > 0$ and $v'' > 0$, we have $\phi(c, n) = n\, v'(n)/u'(c)$. Then $\partial\phi/\partial n = [v'(n) + n\, v''(n)]/u'(c) > 0$, confirming Assumption 3.1.

We now prove the main result: under single crossing, the local (first-order) approach to incentive compatibility is valid.

Proposition 3.1 (Sufficiency of the first-order approach)**.** *Under Assumption 3.1 (single crossing), a differentiable allocation $\{c(z), y(z)\}$ is globally incentive compatible if and only if:*

(i) ***Envelope condition:*** $\mathcal{U}'(z) = -U_n\Big(c(z), \frac{y(z)}{z}\Big)\,\frac{y(z)}{z^2}$ *for all z;*

(ii) ***Monotonicity:*** *$y(z)$ is nondecreasing in z.*

The proof, which uses three steps illustrated in Figures 6 and 7, is given in Appendix A.

4 The Planner's Problem as a Hamiltonian

4.1 Statement of the problem

Equipped with Proposition 3.1, the planner replaces the global IC constraint with the local envelope condition (14) (plus monotonicity, which we verify ex post). The planner solves:

$$\max_{\mathcal{U}(z_0),\, y(\cdot)} \int_{z_0}^{\bar{z}} \psi(z)\,\mathcal{U}(z)\, f(z)\, dz \tag{17}$$

subject to:

$$\frac{d\mathcal{U}}{dz} = -U_n\left(c(z), \frac{y(z)}{z}\right)\frac{y(z)}{z^2}, \tag{18}$$

$$\int_{z_0}^{\bar{z}} \big[y(z) - c(z)\big] f(z)\, dz \geq B, \tag{19}$$

where $c(z) = C(\mathcal{U}(z), y(z))$ is implicitly defined by $\mathcal{U}(z) = U(c(z), y(z)/z)$.

The control variables are the output schedule $y(\cdot)$ and the initial utility level $\mathcal{U}(z_0)$. Given these, the state $\mathcal{U}(z)$ for $z > z_0$ is determined by integrating the envelope ODE (18) forward, and consumption is recovered from the utility definition at each z. The initial value $\mathcal{U}(z_0)$ is free because the planner can choose the utility level of the bottom type (equivalently, the level of $c(z_0)$ and $y(z_0)$). The optimality condition for this free endpoint is precisely the transversality condition $\lambda(z_0) = 0$ derived below.

4.2 Substituting out consumption

The constraint $\mathcal{U}(z) = U(c(z), y(z)/z)$ implicitly defines consumption as a function of the state $\mathcal{U}$ and the control y:

$$c(z) = C\big(\mathcal{U}(z), y(z)\big). \tag{20}$$

The partial derivatives of this function follow from implicit differentiation of $\mathcal{U} = U(C(\mathcal{U}, y), y/z)$:

$$C_{\mathcal{U}}(\mathcal{U}, y) = \frac{\partial c}{\partial \mathcal{U}} = \frac{1}{U_c(c, y/z)}, \tag{21}$$

$$C_y(\mathcal{U}, y) = \frac{\partial c}{\partial y} = -\frac{U_n(c, y/z)}{z\, U_c(c, y/z)}. \tag{22}$$

Comparing (22) with the tax wedge (34):

$$C_y(\mathcal{U}, y) = 1 - T'(y(z)). \tag{23}$$

This is intuitive: the marginal consumption gain from an extra unit of pre-tax income equals one minus the marginal tax rate.

4.3 The Hamiltonian

Let μ denote the multiplier on the resource constraint (19) and $\lambda(z)$ the costate variable for the state $\mathcal{U}(z)$. The Hamiltonian is

$$\begin{aligned} H(\mathcal{U}, y, \lambda; z) = {} & \psi(z)\,\mathcal{U}(z)\, f(z) + \mu\big[y(z) - C(\mathcal{U}(z), y(z))\big] f(z) \\ & - \lambda(z)\, U_n\left(C(\mathcal{U}(z), y(z)), \frac{y(z)}{z}\right)\frac{y(z)}{z^2}. \end{aligned} \tag{24}$$

The first line is the "static" contribution: social welfare plus the resource surplus at z, weighted by density. The second line is the "dynamic" contribution reflecting how the allocation at z affects the evolution of the state $\mathcal{U}$.

4.4 Optimality conditions

Optimality in the control y. Setting $H_y = 0$:

$$\begin{aligned} 0 = \mu\big[1 - C_y\big] f(z) - \lambda(z)\frac{U_n}{z^2} \\ - \lambda(z)\left[U_{nc}\, C_y + U_{nn}\frac{1}{z}\right]\frac{y(z)}{z^2}, \end{aligned} \tag{25}$$

where all functions of $(c, y/z)$ are evaluated at the allocation $(C(\mathcal{U}(z), y(z)),\, y(z)/z)$.

Costate equation. The law of motion for λ is $\lambda'(z) = -H_{\mathcal{U}}$:

$$\frac{d\lambda(z)}{dz} = \mu\, C_{\mathcal{U}}\, f(z) - \psi(z)\, f(z) + \lambda(z)\, U_{nc}\frac{y(z)}{z^2}\, C_{\mathcal{U}}. \tag{26}$$

Boundary (transversality) conditions. Since the problem is unrestricted at both endpoints of the type distribution:

$$\lambda(z_0) = 0, \qquad \lambda(\bar{z}) = 0. \tag{27}$$

4.5 Worked example: quasilinear preferences

To see how the Hamiltonian machinery produces a closed-form tax formula, we work through the special case $U(c, n) = c - v(n)$ with $v' > 0$, $v'' > 0$. Because $U_c = 1$ and $U_{cn} = 0$, many terms drop out and the algebra is transparent.

Simplifications. With $U_c = 1$ we have $C_{\mathcal{U}} = 1/U_c = 1$ and $C_y = -U_n/(zU_c) = v'(n)/z$, so that $1 - C_y = 1 - v'(n)/z$. Since the household FOC gives $v'(n) = z(1 - T')$, we confirm $C_y = 1 - T'$. The separability $U_{cn} = 0$ eliminates the non-separable terms in both the H_y condition and the costate equation.

Step 1: Tax formula in terms of $\lambda(z)$. With $U_{cn} = 0$ and $U_n = -v'(n)$, the $H_y = 0$ condition (25) simplifies. The $U_{nc}\, C_y$ term in the second line vanishes, leaving only the U_{nn}/z term. Writing $n = y/z$:

$$\mu\, T'(y)\, f(z) = \lambda(z)\left[\frac{v'(n)}{z^2} + \frac{v''(n)}{z}\cdot\frac{n}{z}\right] = \frac{\lambda(z)}{z^2}\big[v'(n) + n\, v''(n)\big].$$

Using $v'(n) = z(1 - T')$ and dividing both sides by $\mu(1 - T')f(z)$:

$$\frac{T'(y(z))}{1 - T'(y(z))} = \left[1 + \frac{n\, v''(n)}{v'(n)}\right]\cdot\frac{\lambda(z)}{z\, f(z)\, \mu}. \tag{28}$$

This already has the structure of the general tax formula (46): the tax wedge is proportional to an elasticity term times the costate $\lambda(z)$ (normalized by $zf(z)$ and μ). The factor $1 + nv''/v'$ will later be identified as the elasticity ratio $(1 + \varepsilon^u)/\varepsilon^c$ (Section 5), but for now it is simply a curvature measure of v. The costate $\lambda(z)$ remains to be determined.

Step 2: Solving for $\lambda(z)$ from the costate equation. With $C_{\mathcal{U}} = 1$ and $U_{nc} = 0$, the costate equation (26) reduces to

$$\lambda'(z) = [\mu - \psi(z)]\, f(z).$$

This is a first-order ODE with constant coefficients (given ψ and f). Integrating from z to $\bar{z}$ and using $\lambda(\bar{z}) = 0$:

$$\lambda(z) = \int_z^{\bar{z}} \big[\psi(\zeta) - \mu\big]\, f(\zeta)\, d\zeta. \tag{29}$$

The remaining boundary condition $\lambda(z_0) = 0$ pins down μ:

$$0 = \int_{z_0}^{\bar{z}} [\psi(\zeta) - \mu]\, f(\zeta)\, d\zeta = \underbrace{\int \psi\, f\, d\zeta}_{=1} - \mu,$$

so $\mu = 1$. This is special to quasilinear preferences: the marginal utility of consumption is constant ($U_c = 1$), so one dollar of public funds has the same social value as one dollar of private consumption.

Step 3: The closed-form formula. Substituting $\mu = 1$ and (29) into (28):

$$\frac{T'(y(z))}{1 - T'(y(z))} = \left[1 + \frac{n\, v''(n)}{v'(n)}\right] \cdot \frac{1}{z\, f(z)} \int_z^{\bar{z}} \big[\psi(\zeta) - 1\big]\, f(\zeta)\, d\zeta. \tag{30}$$

Define the welfare cdf $\Psi(z) = \int_{z_0}^{z} \psi(\zeta)\, f(\zeta)\, d\zeta$. Then

$$\int_z^{\bar{z}} [\psi(\zeta) - 1]\, f(\zeta)\, d\zeta = \big[1 - \Psi(z)\big] - \big[1 - F(z)\big] = F(z) - \Psi(z).$$

For a redistributive planner with $\psi(z)$ declining, the welfare mass $\Psi(z)$ accumulates faster than the population mass $F(z)$ at low z, so $\Psi(z) > F(z)$ for interior types and the tax rate is positive. Rewriting:

$$\boxed{\frac{T'(y(z))}{1 - T'(y(z))} = \underbrace{\left[1 + \frac{n\, v''(n)}{v'(n)}\right]}_{A(z)} \cdot \underbrace{\frac{\Psi(z) - F(z)}{z\, f(z)}}_{B(z)\cdot C(z)}.} \tag{31}$$

This is the Diamond (1998) ABC formula, derived here directly from the Hamiltonian. Three forces shape the tax schedule:

- $A(z) = 1 + nv''/v'$: the curvature of v governs the efficiency cost of distorting labor supply. Higher curvature (more concave labor supply) means less elastic workers and higher optimal taxes.
- $\Psi(z) - F(z)$: the gap between the welfare and population CDFs measures the planner's desire to redistribute. When $\psi(z)$ is front-loaded, Ψ leads F and the planner taxes to transfer resources downward.
- $1/(z\, f(z))$: the inverse hazard rate of the type distribution. A thicker tail (more mass above z relative to density at z) raises the tax, because many agents contribute revenue while only those at z are directly distorted.

Remark 4.1 (What this example reveals). The quasilinear case exposes the essential three-step structure of the Hamiltonian approach: (i) the $H_y = 0$ condition delivers the tax formula in terms of the unknown costate $\lambda(z)$, trading off the mechanical revenue gain $\mu T' f$ against the incentive cost $\lambda \cdot$ [curvature of v]; (ii) the costate equation determines $\lambda(z)$ as the cumulative redistribution pressure $\int_z^{\bar{z}} [\psi - \mu]\, f\, d\zeta$; (iii) the boundary conditions $\lambda(z_0) = \lambda(\bar{z}) = 0$ pin down μ and, with bounded support, deliver zero marginal taxes at both endpoints (see Remark 1.1 for the unbounded case). In the general case (Section 6), the same three steps apply but with additional terms from income effects ($U_{cn} \neq 0$) and non-constant U_c.

5 Labor Supply Elasticities

We now derive the key elasticity concepts that appear in the optimal tax formula derived in the next section. These elasticities characterize how labor supply responds to changes in incentives, and they are the central "sufficient statistics" governing the efficiency cost of taxation.

5.1 Setup

The household solves

$$\max_{c,n} U(c,n) \quad \text{s.t.} \quad c + T(zn) = zn,$$

where $T(\cdot)$ is the tax schedule. With Lagrange multiplier λ (marginal utility of wealth) and after-tax wage $w = z(1 - T'(zn))$, the optimality conditions are

$$U_c(c,n) = \lambda, \tag{32}$$
$$U_n(c,n) = -\lambda w. \tag{33}$$

From these, the *labor wedge* (implicit tax rate) is

$$1 - T'(zn) = -\frac{U_n(c,n)}{z\, U_c(c,n)}. \tag{34}$$

5.2 Frisch (λ-constant) elasticity

The Frisch elasticity measures the labor supply response to a wage change holding the marginal utility of wealth λ constant. Interpret (32)–(33) as defining $c(\lambda, w)$ and $n(\lambda, w)$. Differentiate with respect to w:

$$U_{cc}\frac{dc}{dw} + U_{cn}\frac{dn}{dw} = 0,$$
$$U_{nc}\frac{dc}{dw} + U_{nn}\frac{dn}{dw} = -\lambda.$$

From the first equation, $dc/dw = -(U_{cn}/U_{cc})\, dn/dw$. Substituting into the second:

$$\left[U_{nn} - \frac{U_{cn}^2}{U_{cc}}\right] \frac{dn}{dw} = -\lambda.$$

Using $\lambda = -U_n/w$ from (33):

$$\frac{dn}{dw} = \frac{U_n}{w} \cdot \frac{U_{cc}}{U_{cc}U_{nn} - U_{cn}^2}.$$

The Frisch elasticity is therefore

$$\boxed{\varepsilon^f = \frac{dn}{dw}\frac{w}{n} = \frac{U_n}{n\left[U_{nn} - \dfrac{U_{cn}^2}{U_{cc}}\right]}.} \tag{35}$$

5.3 Uncompensated (Marshallian) elasticity

Now augment the budget constraint with exogenous income I: $c = wn + I$. The household's optimality requires

$$w = -\frac{U_n(wn + I,\, n)}{U_c(wn + I,\, n)}.$$

Define the implicit function

$$\mathcal{F}(w, n, I) \equiv -\frac{U_n(wn + I,\, n)}{U_c(wn + I,\, n)} - w.$$

By the implicit function theorem:

$$\left.\frac{dn}{dw}\right|_I = -\frac{\partial\mathcal{F}/\partial w}{\partial\mathcal{F}/\partial n} = \frac{U_c + wn\,U_{cc} + n\,U_{nc}}{-(U_{nn} + w^2U_{cc} + 2wU_{cn})}, \tag{36}$$

$$\left.\frac{dn}{dI}\right|_w = -\frac{\partial\mathcal{F}/\partial I}{\partial\mathcal{F}/\partial n} = \frac{wU_{cc} + U_{nc}}{-(U_{nn} + w^2U_{cc} + 2wU_{cn})}. \tag{37}$$

Using $U_n = -U_c w$ from the optimality condition:

$$\varepsilon^u = \left.\frac{dn}{dw}\right|_I \frac{w}{n} = \frac{-\frac{U_n}{n} + w^2U_{cc} + wU_{nc}}{-(U_{nn} + w^2U_{cc} + 2wU_{cn})}. \tag{38}$$

5.4 Compensated (Hicksian) elasticity

The Slutsky equation decomposes the uncompensated response into substitution and income effects:

$$\left.\frac{dn}{dw}\right|_{\bar{U}} = \left.\frac{dn}{dw}\right|_I - n\left.\frac{dn}{dI}\right|_w.$$

Therefore:

$$\varepsilon^c = \left(\left.\frac{dn}{dw}\right|_I - n\left.\frac{dn}{dI}\right|_w\right)\frac{w}{n} = \varepsilon^u - w\frac{dn}{dI} = \frac{-\frac{U_n}{n}}{-(U_{nn} + w^2U_{cc} + 2wU_{cn})}. \tag{39}$$

5.5 A key ratio

A combination that will appear repeatedly in the optimal tax formula is

$$\frac{1+\varepsilon^u}{\varepsilon^c} = 1 + n\,\frac{U_{nn} + wU_{cn}}{U_n}. \tag{40}$$

To verify: from (38) and (39), let $D \equiv -(U_{nn} + w^2U_{cc} + 2wU_{cn})$ denote the common (positive) denominator. Then

$$\frac{1+\varepsilon^u}{\varepsilon^c} = \frac{1 + \frac{-U_n/n + w^2U_{cc} + wU_{nc}}{D}}{\frac{-U_n/n}{D}} = \frac{D - U_n/n + w^2U_{cc} + wU_{nc}}{-U_n/n}$$

$$= \frac{-U_{nn} - wU_{cn} - U_n/n}{-U_n/n} = 1 + n\frac{U_{nn} + wU_{cn}}{U_n}.$$

5.6 Special cases

Case 1. Separable preferences: $U(c,n) = u(c) - v(n)$.

Since $U_{cn} = 0$:

$$\varepsilon^f = \frac{v'(n)}{n\,v''(n)}, \qquad \varepsilon^c = \frac{v'(n)/n}{v''(n) - w^2u''(c)},$$

$$\varepsilon^u = \frac{v'(n)/n + w^2u''(c)}{v''(n) - w^2u''(c)}, \qquad \frac{1+\varepsilon^u}{\varepsilon^c} = 1 + \frac{nv''(n)}{v'(n)} = \frac{1+\varepsilon^f}{\varepsilon^f}.$$

Case 2. Power utility: $U(c,n) = \frac{c^{1-\rho}}{1-\rho} - \psi\frac{n^{1+\gamma}}{1+\gamma}$.

This is a special case of separable preferences with $v'(n) = \psi n^\gamma$ and $v''(n) = \psi\gamma n^{\gamma-1}$:

$$\varepsilon^f = \frac{1}{\gamma}, \qquad \frac{1+\varepsilon^u}{\varepsilon^c} = 1 + \gamma.$$

The compensated and uncompensated elasticities depend on the endogenous allocation:

$$\varepsilon^u = \frac{1 - \psi\rho\,c^{\rho-1}n^{\gamma+1}}{\gamma + \psi\rho\,c^{\rho-1}n^{\gamma+1}}, \qquad \varepsilon^c = \frac{1}{\gamma + \psi\rho\,c^{\rho-1}n^{\gamma+1}}.$$

Case 3. Quasilinear in labor: $U(c,n) = u(c) - n$.

Here $v(n) = n$, so $v' = 1$, $v'' = 0$:

$$\varepsilon^f = \infty, \qquad \varepsilon^c = -\frac{u'(c)^2}{n\,u''(c)}, \qquad \varepsilon^u = \varepsilon^c - 1, \qquad \frac{1+\varepsilon^u}{\varepsilon^c} = 1.$$

Case 4. Quasilinear in consumption: $U(c,n) = c - v(n)$.

Here $u(c) = c$, so $u' = 1$, $u'' = 0$, and there are no income effects on labor supply:

$$\varepsilon^f = \varepsilon^u = \varepsilon^c = \frac{v'(n)}{n\,v''(n)}, \qquad w = v'(n).$$

6 Deriving the Optimal Tax Formula

6.1 Solving for the Lagrange multiplier μ

Integrating the costate equation (26) over $[z_0, \bar{z}]$ and using $\lambda(\bar{z}) - \lambda(z_0) = 0$:

$$0 = \mu \int_{z_0}^{\bar{z}} C_{\mathcal{U}}(\zeta)\, f(\zeta)\, d\zeta - \int_{z_0}^{\bar{z}} \psi(\zeta)\, f(\zeta)\, d\zeta + \int_{z_0}^{\bar{z}} \lambda(\zeta)\, U_{nc}(\zeta) \frac{y(\zeta)}{\zeta^2}\, C_{\mathcal{U}}(\zeta)\, d\zeta,$$

where we write $C_{\mathcal{U}}(\zeta) \equiv C_{\mathcal{U}}(\mathcal{U}(\zeta), y(\zeta))$, etc. Solving for μ:

$$\mu = \frac{\displaystyle\int_{z_0}^{\bar{z}} \psi(\zeta)\, f(\zeta)\, d\zeta \;-\; \int_{z_0}^{\bar{z}} \lambda(\zeta)\, U_{nc}(\zeta)\, \frac{y(\zeta)}{\zeta^2}\, C_{\mathcal{U}}(\zeta)\, d\zeta}{\displaystyle\int_{z_0}^{\bar{z}} C_{\mathcal{U}}(\zeta)\, f(\zeta)\, d\zeta}. \tag{41}$$

The multiplier μ is the *marginal social value of public funds*: it measures the welfare cost (in units of the planner's objective) of tightening the government budget constraint by one unit.

6.2 Solving for the costate $\lambda(z)$

Define the following shorthand for the two components of the costate equation:

$$H_{U,S}(z) = \left[\frac{\mu}{U_c(z)} - \psi(z)\right] f(z), \tag{42}$$

$$H_{U,N}(z) = \frac{n(z)\, U_{nc}(z)}{z\, U_c(z)}. \tag{43}$$

The subscript S stands for the "separable" part (present for all preferences), while N stands for the "non-separable" part (vanishes when $U_{cn} = 0$). The costate equation (26) becomes the first-order linear ODE

$$\lambda'(z) - H_{U,N}(z)\, \lambda(z) = H_{U,S}(z). \tag{44}$$

The integrating factor is $\exp\!\big(-\int^z H_{U,N}(\xi)\, d\xi\big)$. Multiplying through:

$$\frac{d}{dz}\left[\lambda(z) \exp\!\left(-\int_{z_0}^{z} H_{U,N}(\xi)\, d\xi\right)\right] = H_{U,S}(z) \exp\!\left(-\int_{z_0}^{z} H_{U,N}(\xi)\, d\xi\right).$$

Integrating from z to $\bar{z}$ and using $\lambda(\bar{z}) = 0$:

$$-\lambda(z) \exp\!\left(-\int_{z_0}^{z} H_{U,N}(\xi)\, d\xi\right) = \int_{z}^{\bar{z}} H_{U,S}(\zeta) \exp\!\left(-\int_{z_0}^{\zeta} H_{U,N}(\xi)\, d\xi\right) d\zeta.$$

Rearranging:

$$\boxed{\lambda(z) = -\int_{z}^{\bar{z}} H_{U,S}(\zeta) \exp\!\left(-\int_{z}^{\zeta} H_{U,N}(\xi)\, d\xi\right) d\zeta.} \tag{45}$$

6.3 The optimal tax formula (general case)

We now extract the tax formula from the control optimality condition (25). Substituting $C_y = 1 - T'(y)$ and collecting the terms involving λ, we use the identity

$$\frac{U_n}{z^2} + \left[U_{nc}\, C_y + U_{nn}\frac{1}{z}\right]\frac{y}{z^2} = -\frac{U_c\,(1-T')}{z}\cdot\frac{1+\varepsilon^u}{\varepsilon^c},$$

which follows from $C_y = (1-T') = -U_n/(zU_c)$, $w = z(1-T')$, $n = y/z$, and the identity (40). To verify, note that $U_n/z = -U_c(1-T')$ and:

$$U_{nc}\, C_y + U_{nn}\frac{1}{z} = -\frac{U_{nc}\,U_n}{z\,U_c} + \frac{U_{nn}}{z} = \frac{1}{z}\left[U_{nn} - \frac{U_{nc}\,U_n}{U_c}\right].$$

Multiplying by $y/z^2 = n/z$ and adding U_n/z^2:

$$\begin{aligned}\frac{U_n}{z^2} + \frac{n}{z^2}\left[U_{nn} - \frac{U_{nc}\,U_n}{U_c}\right] &= \frac{U_n}{z^2}\left[1 + n\,\frac{U_{nn} + wU_{nc}}{U_n}\right] \\ &= \frac{U_n}{z^2}\cdot\frac{1+\varepsilon^u}{\varepsilon^c} = -\frac{U_c(1-T')}{z}\cdot\frac{1+\varepsilon^u}{\varepsilon^c},\end{aligned}$$

where the second line uses $wU_{nc}/U_n = -U_{nc}U_n/(U_nU_c)\cdot U_c\cdot z/z$ appropriately, and the last step uses $U_n/z = -U_c(1-T')$.

Substituting back into (25):

$$\mu\, T'(y(z))\, f(z) = -\frac{\lambda(z)}{z}\, U_c(z)\,(1 - T'(y(z)))\cdot\frac{1+\varepsilon^u(z)}{\varepsilon^c(z)}.$$

Dividing both sides by $\mu\,(1-T')\,f(z)$ and rearranging:

$$\boxed{\frac{T'(y(z))}{1-T'(y(z))} = -\frac{1+\varepsilon^u(z)}{\varepsilon^c(z)}\cdot\frac{\lambda(z)}{z\,f(z)}\cdot\frac{U_c(z)}{\mu}.} \tag{46}$$

This is the *general optimal tax formula.* It expresses the tax wedge $T'/(1-T')$ as a product of three terms:

(i) The elasticity ratio $(1+\varepsilon^u)/\varepsilon^c$, capturing the distortionary cost of taxation.

(ii) The costate $\lambda(z)/(zf(z))$, capturing the social value of relaxing the incentive constraint at z.

(iii) The ratio $U_c(z)/\mu$, converting between the household's marginal utility and the social value of funds.

7 The Perturbation Approach (Saez, 2001)

The Hamiltonian derivation of Section 6 works in productivity space z and uses the mechanism-design reformulation. An alternative—and arguably more intuitive—route derives the same formula directly in *income space* by perturbing the tax schedule and balancing the mechanical revenue gain against the behavioral revenue loss.

7.1 The earnings distribution

Let $y(z) = z\,n(z)$ denote the earnings of type z under the tax schedule T. Assume that $y(\cdot)$ is strictly increasing (the single-crossing/monotonicity condition). Denote the cdf and density of earnings by H and h:

$$H(y) = F\big(z(y)\big), \qquad h(y) = \frac{f(z(y))}{y'(z(y))}, \tag{47}$$

where $z(y)$ is the inverse of $y(z)$, and $y'(z) = n(z) + z\,n'(z)$.

7.2 The elasticity of taxable income

Fix an agent of type z earning $y = y(z)$ under the tax schedule T, with $T'(y) \in (0, 1)$.

Virtual linearization. Define the **local linearization** of T at y as the affine schedule

$$T^{\text{lin}}(y'; \tau, I) \;:=\; \tau\, y' - I, \tag{48}$$

where $\tau := T'(y)$ is the local marginal rate and $I := \tau\, y - T(y)$ is the *virtual income* chosen so that $T^{\text{lin}}(y; \tau, I) = T(y)$. This ensures that the agent's budget set under T^{lin} is tangent to the nonlinear budget set at y.

Earnings under the linearized schedule. Let $\hat{y}(\tau, I, z)$ denote the optimal earnings of a type-z agent facing the linear schedule $T^{\text{lin}}(\cdot; \tau, I)$:

$$\hat{y}(\tau, I, z) \;:=\; z \cdot \arg\max_{n \geq 0}\; U\big((1-\tau)\, zn + I,\; n\big). \tag{49}$$

By construction, $\hat{y}(\tau, I, z) = y(z)$ at the status-quo (τ, I).

Compensated (Hicksian) ETI. The **compensated elasticity of taxable income** is

$$\boxed{\zeta^c(y) \;:=\; \frac{\partial \hat{y}^{\,c}}{\partial(1-\tau)} \;\cdot\; \frac{1-\tau}{y}\,,} \tag{50}$$

where $\hat{y}^{\,c}(\tau, I, z)$ denotes optimal earnings under the linearized schedule T^{lin} **compensated to hold utility constant** at the level $\bar{U} = U(y - T(y),\, y/z)$, i.e. the derivative is taken along the Hicksian (utility-constant) demand.

The **uncompensated ETI** is defined analogously without the utility compensation:

$$\zeta^u(y) \;:=\; \frac{\partial \hat{y}}{\partial(1-\tau)} \;\cdot\; \frac{1-\tau}{y}\,.$$

Remark 7.1 (Relationship to labor-supply elasticities). Since $y = z\,n$ with z fixed, ζ^c coincides with the compensated elasticity of labor supply ε^c from Section 5, and $\zeta^u = \varepsilon^u$. The income effect on labor supply is parameterized by $\eta := -(1 - T')\,\partial y/\partial I > 0$ under normality, where I is nonlabor income. The Slutsky relation gives $\zeta^c = \zeta^u + \eta$. Under quasilinear-in-consumption preferences ($U(c, n) = c - v(n)$), income effects vanish and $\zeta^c = \zeta^u = \varepsilon^f = 1/\gamma$.

7.3 The perturbation

Fix an income level $y^* > 0$. Consider a reform that increases the marginal tax rate by $d\tau > 0$ in a narrow band $[y^*, y^* + \varepsilon]$, with $\varepsilon > 0$ small. This raises the total tax paid by every earner above $y^* + \varepsilon$ by exactly $\varepsilon\, d\tau$ and increases the marginal rate faced by earners in the band. The reform has two first-order effects:

1. A **mechanical revenue gain** (net of welfare cost) from all earners above y^*.
2. A **behavioral revenue loss** from earners in the band who reduce income in response to the higher marginal rate.

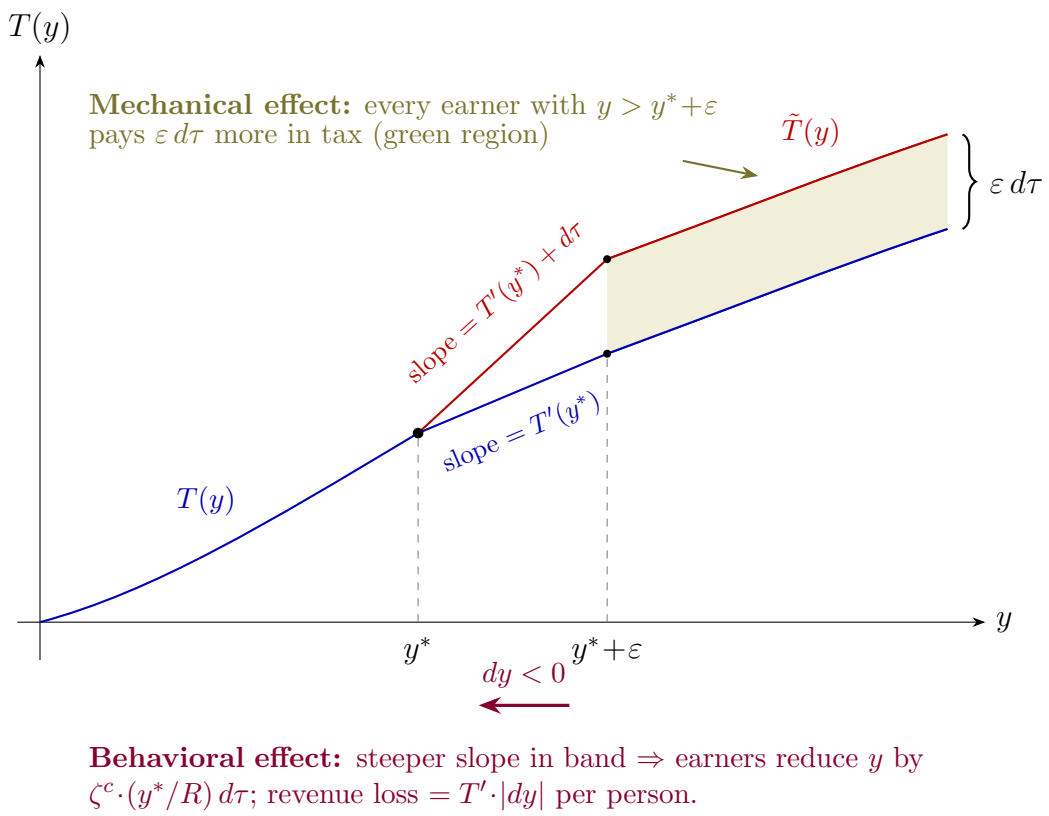

Figure 2: The Saez perturbation. The baseline schedule $T(y)$ (blue) is perturbed to $\tilde{T}(y)$ (red) by increasing the slope by $d\tau$ in $[y^*, y^* + \varepsilon]$. The two schedules coincide below y^*, diverge in the band (where $\tilde{T}$ has a steeper slope), and run parallel above it with a constant gap of $\varepsilon\, d\tau$. At the optimum the mechanical gain and behavioral loss exactly offset: $dM + dB = 0$.

7.4 Deriving dM and dB from the Lagrangian

We derive the effect of the perturbation on the planner's Lagrangian. Recall from Section 3:

$$\mathcal{L}(T) = \int_{z_0}^{\bar{z}} \psi(z)\,\mathcal{U}(z)\, f(z)\, dz \;+\; \mu \left[\int_{z_0}^{\bar{z}} T(y(z))\, f(z)\, dz \;-\; R_{\text{gov}} \right],$$

where μ is the multiplier on the government budget constraint (the marginal value of public funds, derived in (41)).

Write the perturbed schedule as $\tilde{T}(y) = T(y) + \epsilon\,\delta T(y)$, where

$$\delta T(y) = \begin{cases} 0, & y < y^*, \\ (y - y^*)\,d\tau, & y \in [y^*, y^* + \varepsilon], \\ \varepsilon\,d\tau, & y > y^* + \varepsilon, \end{cases}$$

and $\epsilon > 0$ is a perturbation parameter (set to 1 at the end). Under the perturbed schedule, each agent's earnings adjust from $y(z)$ to $y(z; \tilde{T})$. The total derivative of the Lagrangian at $\epsilon = 0$ is

$$\left.\frac{d\mathcal{L}}{d\epsilon}\right|_{\epsilon=0} = \int_{z_0}^{\bar{z}} \Bigg[\underbrace{\psi(z)\frac{d\mathcal{U}}{d\epsilon} + \mu\,\delta T(y)}_{\text{(a)}} + \underbrace{\mu\,T'(y)\frac{\partial y}{\partial \epsilon}}_{\text{(b)}}\Bigg] f(z)\,dz. \tag{51}$$

The revenue change $d[T(y)]/d\epsilon = \delta T(y) + T'(y)\,\partial y/\partial\epsilon$ has been split into direct (in (a)) and behavioral (in (b)) components.

Remark 7.2 (Connection to the Gâteaux derivative). Expression (51) is the *Gâteaux derivative* of the planner's Lagrangian $\mathcal{L}$ at the tax schedule T in the direction δT. Formally, $\mathcal{L}$ is a functional on the space of tax schedules, and

$$\left.\frac{d\mathcal{L}}{d\epsilon}\right|_{\epsilon=0} = \lim_{\epsilon\to 0} \frac{\mathcal{L}(T + \epsilon\,\delta T) - \mathcal{L}(T)}{\epsilon} \equiv D\mathcal{L}(T;\,\delta T),$$

which is the infinite-dimensional analogue of a directional derivative. At the optimum, $D\mathcal{L}(T;\,\delta T) = 0$ for all admissible directions δT—this is the necessary condition for optimality in function space. The Saez perturbation chooses a specific direction (a marginal rate increase in a narrow band), but the optimality condition must hold for *every* such direction, which is what pins down the tax schedule pointwise. See Appendix C for the general variational framework.

Envelope theorem and term (a). Each agent was already optimizing under T: the household FOC $U_c(1-T'(y)) + U_n/z = 0$ holds at the current bundle. When the schedule is perturbed, the agent re-optimizes (changing y), but by the **envelope theorem** the welfare effect of this behavioral adjustment is *exactly zero*, not merely second order. To see this, write the total change in $\mathcal{U} = U(y - T(y),\, y/z)$:

$$\frac{d\mathcal{U}}{d\epsilon} = U_c\left[-\delta T(y) + \big(1 - T'(y)\big)\frac{\partial y}{\partial \epsilon}\right] + \frac{U_n}{z}\frac{\partial y}{\partial \epsilon}.$$

The terms involving $\partial y/\partial\epsilon$ cancel by the household FOC ($U_c(1-T') + U_n/z = 0$), leaving

$$\left.\frac{d\mathcal{U}}{d\epsilon}\right|_{\epsilon=0} = -U_c\,\delta T(y).$$

Thus the only welfare change comes from the direct tax increase $\delta T(y)$, which reduces consumption dollar-for-dollar. The behavioral adjustment *does* matter for revenue (term (b)),

because the government's revenue $T(y)$ depends on the agent's choice—but it is the agent, not the government, who was at an optimum. Therefore

$$\psi(z)\frac{d\mathcal{U}}{d\epsilon} = -\psi(z)\, U_c\, \delta T(y).$$

Defining the **marginal social welfare weight**

$$g(z) := \frac{\psi(z)\, U_c(z)}{\mu}, \tag{52}$$

(so that $g(z)$ measures the social value—in dollars of public funds—of one additional dollar to a type-z agent), term (a) becomes

$$\text{(a)} = \mu\, \delta T(y)\big(1 - g(z)\big). \tag{53}$$

This is the **mechanical effect**: the net gain (revenue minus welfare cost) from the direct tax increase, with behavior held fixed.

Term (b): the behavioral revenue effect. The behavioral adjustment $\partial y/\partial\epsilon$ changes government revenue by $T'(y)\,\partial y/\partial\epsilon$ per agent. (The agent's welfare effect from re-optimization was already accounted for by the envelope theorem—it is zero.)

$$\text{(b)} = \mu\, T'(y)\frac{\partial y}{\partial\epsilon}. \tag{54}$$

This is the **behavioral effect**: the revenue change because agents adjust their earnings in response to the new marginal incentives.

Slutsky decomposition of the behavioral effect. The behavioral response $\partial y/\partial\epsilon$ in term (b) includes both substitution and income effects:

$$\frac{\partial y}{\partial\epsilon} = \underbrace{\frac{\partial y^c}{\partial\epsilon}}_{\text{substitution}} + \underbrace{\frac{\partial y^{\text{inc}}}{\partial\epsilon}}_{\text{income effect}}.$$

We now compute the income-effect contribution to revenue explicitly and show it can be absorbed into term (a), leaving only the compensated response in term (b).

Step 1: The income-effect revenue. Define the **income effect on earnings**:

$$\eta(z) := -(1 - T')\,\frac{\partial y}{\partial I} > 0,$$

where I is nonlabor income. Three ingredients enter this definition:

- $\partial y/\partial I < 0$: the raw income effect—how earnings change when nonlabor income rises by \$1. Under normality of leisure, richer agents work less.
- The minus sign: a sign convention so that $\eta > 0$ under normality, measuring how much earnings *fall* per unit increase in nonlabor income.

- The $(1 - T')$ factor: normalizes the income effect per dollar of *net-of-tax* income rather than gross income. This ensures the Slutsky relation in earnings space takes the clean form $\zeta^c = \zeta^u + \eta$, without awkward $1/(1 - T')$ factors.

The perturbation extracts $\delta T(y) = \varepsilon\, d\tau$ from each earner above y^*, which is equivalent to reducing nonlabor income by $\delta T(y)$. The income-effect behavioral response in earnings is therefore

$$dy_{\text{inc}} = \frac{\partial y}{\partial I} \cdot \delta T(y) = -\frac{\eta}{1 - T'} \cdot \delta T(y).$$

The revenue consequence of this earnings change is $T'(y) \cdot dy_{\text{inc}}$, giving

$$\text{income-effect revenue per agent} = -\mu\, T' \cdot \frac{\eta}{1 - T'} \cdot \delta T(y) = -\mu\, \frac{T'}{1 - T'}\, \eta \cdot \delta T(y).$$

The sign is positive (a revenue *bonus*): the agent is poorer, works more under normality, and pays more tax.

Step 2: Absorbing into term (a). Combine the original mechanical effect (a) $= \mu\, \delta T(y)(1 - g(z))$ with the income-effect revenue:

$$\begin{aligned}
\text{(a)} + \text{income-effect part of (b)} &= \mu\, \delta T(y) \left[(1 - g(z)) + \frac{T'}{1 - T'}\, \eta(z) \right] \\
&= \mu\, \delta T(y) \left[1 - \underbrace{\left(g(z) - \frac{T'}{1 - T'}\, \eta(z) \right)}_{=:\ \hat{g}(z)} \right].
\end{aligned}$$

This motivates defining the **adjusted welfare weight**

$$\hat{g}(z) := g(z) - \frac{T'(y(z))}{1 - T'(y(z))}\, \eta(z). \tag{55}$$

The adjustment $\frac{T'}{1-T'}\eta$ has two pieces: $1/(1 - T')$ de-normalizes η (recovering the raw $|\partial y/\partial I|$), and T' converts the earnings change into a revenue change. The adjusted weight $\hat{g}(z)$ is the net social cost of extracting a dollar from type z, *after crediting back* the income-effect revenue bonus.

Step 3: The redefined dM and dB. We can now write the total effect of the perturbation as:

$$dM := \text{(a)} + \text{income-effect part of (b)} = \mu\, \delta T(y) \big(1 - \hat{g}(z)\big), \tag{56}$$

$$dB := \text{compensated part of (b)} = \mu\, T'(y) \frac{\partial y^c}{\partial \epsilon}. \tag{57}$$

By construction, the two formulations are equivalent:

$$\underbrace{(1 - g)\, \delta T + T' \frac{\partial y}{\partial \epsilon}}_{\text{using } g \text{ and full } \partial y/\partial \epsilon} = \underbrace{(1 - \hat{g})\, \delta T + T' \frac{\partial y^c}{\partial \epsilon}}_{\text{using } \hat{g} \text{ and compensated } \partial y^c/\partial \epsilon},$$

since the $\frac{T'}{1-T'}\eta$ subtracted from g to form $\hat{g}$ exactly equals the income-effect revenue moved from (b) into (a). The Saez convention uses the second form because the compensated elasticity ζ^c is the cleaner behavioral primitive. Under quasilinear preferences ($U(c,n) = c - v(n)$), income effects vanish ($\eta = 0$), so $\hat{g} = g$ and the distinction disappears.

7.5 Evaluating the mechanical effect dM

Aggregating (56) over all agents (dividing by μ to work in units of public funds) and substituting $\delta T(y)$, the exact expression has three pieces:

(i) Earners above the band ($y > y^* + \varepsilon$) each pay the full $\varepsilon\, d\tau$, contributing

$$\varepsilon\, d\tau \int_{y^*+\varepsilon}^{\infty} \big(1 - \hat{g}(y)\big)\, h(y)\, dy.$$

(ii) Earners inside the band ($y \in [y^*, y^* + \varepsilon]$) pay only $(y - y^*)\, d\tau$, contributing

$$d\tau \int_{y^*}^{y^*+\varepsilon} (y - y^*)\big(1 - \hat{g}(y)\big)\, h(y)\, dy \;\le\; \varepsilon\, d\tau \cdot \varepsilon \sup |(1 - \hat{g})\, h| = O(\varepsilon^2\, d\tau).$$

(iii) Replacing the lower limit $y^*+\varepsilon$ by y^* in (i) changes the integral by $\int_{y^*}^{y^*+\varepsilon}(1-\hat{g})\, h\, dy = O(\varepsilon)$, so the error is $\varepsilon\, d\tau \cdot O(\varepsilon) = O(\varepsilon^2\, d\tau)$.

Combining (i)–(iii), the mechanical effect is

$$dM = \varepsilon\, d\tau \int_{y^*}^{\infty} \big(1 - \hat{g}(y)\big)\, h(y)\, dy + O(\varepsilon^2\, d\tau), \tag{58}$$

where $\hat{g}(y) \equiv \hat{g}(z(y))$. The $O(\varepsilon^2)$ remainder is negligible: the first-order condition divides by $\varepsilon\, d\tau$ and sends $\varepsilon \to 0$.

Remark 7.3 (Budget-balance normalization). The adjusted welfare weights integrate to one: $\int_0^\infty \hat{g}(y)\, h(y)\, dy = 1$. To see this, consider a uniform lump-sum tax $\delta T(y) = \varepsilon\, d\tau$ on *all* earners (i.e., set $y^* = 0$). The mechanical effect is

$$dM = \varepsilon\, d\tau \int_0^\infty (1 - \hat{g}(y))\, h(y)\, dy.$$

At the optimum, this reform must generate zero first-order welfare change (the planner has already optimized the lump-sum component of the tax). Since $\varepsilon\, d\tau > 0$, we need $\int (1 - \hat{g})\, h\, dy = 0$, i.e., $\int \hat{g}\, h\, dy = 1$. Intuitively, a uniform lump-sum transfer is a pure redistribution of the numeraire; the total social value of a dollar distributed to everyone must equal the social cost of collecting it (μ), which is normalized to one.

7.6 Evaluating the behavioral effect dB

As shown in (57), the behavioral effect captures only the compensated earnings response. An agent of type z at income y^* faces a marginal rate increase of $d\tau$ in the band, changing the local net-of-tax rate from $R(y^*) = 1 - T'(y^*)$ to $R(y^*) - d\tau$, i.e. $d(1 - \tau) = -d\tau$.

To compute the compensated response dy^c, we use the ETI from (50), which is defined via the linearized schedule T^{lin} (48). This applies to the nonlinear perturbation because the virtual linearization at y^* satisfies $T^{\text{lin}}(y^*) = T(y^*)$ and $T^{\text{lin}\prime}(y^*) = T'(y^*)$ by construction—the budget constraint under T^{lin} is *tangent* to the one under T at the current optimum. The first-order response to a marginal rate change depends only on the slope and level of the budget constraint, not on the curvature T'' (which enters only at second order), so the first-order compensated responses under T and T^{lin} coincide.

Rearranging the ETI definition:

$$\zeta^c(y) = \frac{\partial \hat{y}^c}{\partial(1-\tau)} \cdot \frac{1-\tau}{y} \qquad \Longrightarrow \qquad \frac{\partial \hat{y}^c}{\partial(1-\tau)} = \zeta^c(y)\,\frac{y}{R(y)},$$

and applying the chain rule with $d(1-\tau) = -d\tau$:

$$dy^c = \frac{\partial \hat{y}^c}{\partial(1-\tau)}\, d(1-\tau) = -\zeta^c(y^*)\,\frac{y^*}{R(y^*)}\, d\tau.$$

The sign is negative: a higher marginal rate reduces the net-of-tax rate, causing compensated earnings to fall.

The resulting revenue loss per affected agent at income y is $T'(y)\,|dy^c(y)|$, and the density of agents at y is $h(y)$. Only agents *inside* the band $[y^*, y^* + \varepsilon]$ face a changed marginal rate; agents outside the band face no marginal incentive change (they experience only a lump-sum shift, which is already captured in dM via $\hat{g}$).

Aggregating the compensated revenue loss over the band:

$$dB = \int_{y^*}^{y^*+\varepsilon} T'(y)\, dy^c(y)\, h(y)\, dy.$$

Within the band, $dy^c(y)$ and $h(y)$ are continuous, so $T'(y)\, dy^c(y)\, h(y) = T'(y^*)\, dy^c(y^*)\, h(y^*) + O(\varepsilon)$. Since the band has width ε, the integral evaluates to

$$dB = \varepsilon \cdot T'(y^*)\, dy^c(y^*)\, h(y^*) + O(\varepsilon^2\, d\tau).$$

Substituting $dy^c(y^*) = -\zeta^c(y^*)\,\frac{y^*}{R(y^*)}\, d\tau$ and $R(y^*) = 1 - T'(y^*)$:

$$dB = -\varepsilon\, d\tau\, \frac{T'(y^*)}{1 - T'(y^*)}\, \zeta^c(y^*)\, y^*\, h(y^*) + O(\varepsilon^2\, d\tau). \tag{59}$$

As with dM, the $O(\varepsilon^2\, d\tau)$ remainder vanishes after dividing by $\varepsilon\, d\tau$ and sending $\varepsilon \to 0$.

7.7 First-order condition

At the optimum, the total effect of the perturbation must be zero for every y^*:

$$dM + dB = 0.$$

Substituting (58) and (59), cancelling $\varepsilon\, d\tau > 0$, and rearranging:

$$\boxed{\frac{T'(y^*)}{1 - T'(y^*)} = \frac{1}{\zeta^c(y^*)} \cdot \frac{1}{y^*\, h(y^*)} \int_{y^*}^{\infty} \big(1 - \hat{g}(y)\big)\, h(y)\, dy.} \tag{60}$$

This is the *Saez (2001) optimal tax formula.* The tax wedge $T'/(1-T')$ at income y^* equals:

- $1/\zeta^c(y^*)$: the inverse compensated ETI (higher elasticity $\Rightarrow$ lower tax),
- $\dfrac{1-H(y^*)}{y^*\,h(y^*)}$: the inverse hazard rate of the earnings distribution (thicker tail $\Rightarrow$ higher tax),
- $\dfrac{1}{1-H(y^*)}\displaystyle\int_{y^*}^{\infty}(1-\hat{g}(y))\,h(y)\,dy$: one minus the average adjusted welfare weight above y^* (less concern for the rich $\Rightarrow$ higher tax).

7.8 Equivalence with the Hamiltonian formula

Formula (60) is equivalent to the Hamiltonian formula (46). To see this, change variables from y to z using $h(y)\,dy = f(z)\,dz$ and $\zeta^c(y) = \varepsilon^c(z)$:

$$\frac{T'(y(z))}{1-T'(y(z))} = \frac{1}{\varepsilon^c(z)} \cdot \frac{1}{y(z)\,h(y(z))} \int_z^{\bar{z}} \big(1-\hat{g}(\zeta)\big)\,f(\zeta)\,d\zeta.$$

Using $y(z)\,h(y(z)) = z\,n(z)\,f(z)/y'(z)$ and the identity (40), together with the integral representation of the costate $\lambda(z)$ from (45), one recovers exactly the three-factor structure of (46). The perturbation approach makes the economic forces transparent, while the Hamiltonian approach provides the mathematical machinery for existence and uniqueness.

7.9 Optimal top rate from the perturbation formula

The perturbation formula yields the Diamond–Saez top rate directly. Take $y^* \to \infty$ and assume:

(a) Zero welfare weight in the tail: $\hat{g}(y) \to 0$ as $y \to \infty$.

(b) Pareto tail in earnings: $1 - H(y) \propto y^{-\alpha_y}$, so that $\frac{1-H(y)}{y\,h(y)} \to \frac{1}{\alpha_y}$.

(c) Constant compensated ETI at the top: $\zeta^c(y) \to e$.

Under these conditions, $(1-\hat{g}) \to 1$ in the numerator and (60) reduces to

$$\frac{T'}{1-T'} = \frac{1}{e} \cdot \frac{1}{\alpha_y} = \frac{1}{\alpha_y\,e},$$

giving

$$T'_{\text{top}} = \frac{1}{1+\alpha_y\,e},$$

which matches (69).

8 Special Cases and the ABC Formula

8.1 Separable preferences

With $U(c,n) = u(c) - v(n)$, the non-separable term vanishes: $H_{U,N}(z) = 0$. The formulas simplify considerably.

Lagrange multiplier.

$$\mu = \frac{\displaystyle\int_{z_0}^{\bar{z}} \psi(\zeta)\, f(\zeta)\, d\zeta}{\displaystyle\int_{z_0}^{\bar{z}} \big[u'(c(\zeta))\big]^{-1} f(\zeta)\, d\zeta} = \frac{1}{\displaystyle\int_{z_0}^{\bar{z}} \big[u'(c(\zeta))\big]^{-1} f(\zeta)\, d\zeta}, \tag{61}$$

Costate. Since $H_{U,N} = 0$, the exponential in (45) equals one:

$$\lambda(z) = \int_z^{\bar{z}} \left[\psi(\zeta) - \frac{\mu}{u'(c(\zeta))}\right] f(\zeta)\, d\zeta. \tag{62}$$

Tax formula. Define the *marginal social welfare weight* of type z:

$$g(z) = \frac{u'(c(z))\, \psi(z)}{\mu}. \tag{63}$$

This measures the social value (in units of public funds) of transferring one unit of resources to type z. Then

$$\frac{T'(y(z))}{1 - T'(y(z))} = \frac{1 + \varepsilon^u(z)}{\varepsilon^c(z)} \cdot \frac{u'(c(z))}{z\, f(z)} \int_z^{\bar{z}} \frac{1 - g(\zeta)}{u'(c(\zeta))}\, f(\zeta)\, d\zeta. \tag{64}$$

Proof. Substituting (62) into (46) and using $U_c = u'(c)$:

$$\frac{T'}{1 - T'} = -\frac{1 + \varepsilon^u}{\varepsilon^c} \cdot \frac{u'(c(z))}{z\, f(z)\, \mu} \int_z^{\bar{z}} \left[\psi(\zeta) - \frac{\mu}{u'(c(\zeta))}\right] f(\zeta)\, d\zeta.$$

Distributing the minus sign and pulling $1/\mu$ inside:

$$= \frac{1 + \varepsilon^u}{\varepsilon^c} \cdot \frac{u'(c(z))}{z\, f(z)} \int_z^{\bar{z}} \frac{1}{u'(c(\zeta))} \left[1 - \frac{\psi(\zeta)\, u'(c(\zeta))}{\mu}\right] f(\zeta)\, d\zeta.$$

Recognizing the definition of g completes the derivation. □

8.2 The Diamond (1998) ABC formula

Following Diamond (1998), assume quasilinear-in-consumption preferences $U(c, n) = c - v(n)$. Define the "welfare cdf"

$$\Psi(z) = \int_{z_0}^{z} \psi(\zeta)\, f(\zeta)\, d\zeta.$$

Since $u'(c) = 1$, we have $\mu = 1$ and $g(z) = \psi(z)$. For redistributive preferences, $\psi(z)$ is decreasing in z. The tax formula (64) becomes

$$\frac{T'(y(z))}{1 - T'(y(z))} = \frac{1 + \varepsilon^u(z)}{\varepsilon^c(z)} \cdot \frac{1}{z\, f(z)} \int_z^{\bar{z}} \big[1 - \psi(\zeta)\big] f(\zeta)\, d\zeta.$$

Evaluating the integral:

$$\int_z^{\bar{z}} f(\zeta)\, d\zeta - \int_z^{\bar{z}} \psi(\zeta)\, f(\zeta)\, d\zeta = \big[1 - F(z)\big] - \big[1 - \Psi(z)\big] = \Psi(z) - F(z).$$

Hence

$$\frac{T'(y(z))}{1-T'(y(z))} = \frac{1+\varepsilon^u(z)}{\varepsilon^c(z)} \cdot \frac{\Psi(z)-F(z)}{z\,f(z)}. \tag{65}$$

For the more general separable case $U(c,n) = u(c) - v(n)$, define the inverse-marginal-utility weighted density and its cdf. Let

$$R \equiv \int_{z_0}^{\bar{z}} \frac{f(\zeta)}{u'(c(\zeta))}\, d\zeta$$

denote the total inverse-marginal-utility-weighted mass. Then define the weighted density and its cdf:

$$\omega(z) = \frac{f(z)/u'(c(z))}{R}, \qquad \Omega(z) = \int_{z_0}^{z} \omega(\zeta)\, d\zeta.$$

The tax formula generalizes to

$$\frac{T'(y(z))}{1-T'(y(z))} = \frac{1+\varepsilon^u(z)}{\varepsilon^c(z)} \cdot \frac{\Psi(z)-\Omega(z)}{z\,\omega(z)}. \tag{66}$$

When $u(c) = c$, we recover $\Omega = F$, $\omega = f$, and (65).

The ABC decomposition. Diamond (1998) decomposes this as

$$\frac{T'(y(z))}{1-T'(y(z))} = \underbrace{\frac{1+\varepsilon^u(z)}{\varepsilon^c(z)}}_{A(z)} \cdot \underbrace{\frac{\Psi(z)-\Omega(z)}{1-\Omega(z)}}_{B(z)} \cdot \underbrace{\frac{1-\Omega(z)}{z\,\omega(z)}}_{C(z)}. \tag{67}$$

- $A(z)$: **Elasticity effect.** Higher labor supply elasticity $\Rightarrow$ lower A $\Rightarrow$ lower marginal tax. Captures the efficiency cost of distorting labor supply.
- $B(z)$: **Redistribution motive.** We can write $B(z) = 1 - \frac{1-\Psi(z)}{1-\Omega(z)}$. The numerator $1-\Psi(z)$ is the planner's welfare weight on types above z; the denominator $1-\Omega(z)$ is the (inverse-marginal-utility-weighted) mass above z. When the planner puts relatively less weight on the rich ($\Psi(z) > \Omega(z)$), we have $B(z) > 0$, calling for positive taxation. When the planner is indifferent to redistribution ($\Psi = \Omega$), $B(z) = 0$ and all taxes vanish.
- $C(z)$: **Tail shape.** The ratio $(1-\Omega(z))/(z\,\omega(z))$ is the (weighted) inverse hazard rate of the type distribution. A thicker tail (more mass above z relative to density at z) raises C and calls for higher taxes, because many agents in the tail contribute revenue while only those at z are directly distorted.

8.3 Optimal top marginal tax rate

Consider the upper tail of the type distribution. Assume:

(a) Quasilinear-in-consumption preferences: $U(c,n) = c - \psi\frac{n^{1+\gamma}}{1+\gamma}$, so $\varepsilon^f = \varepsilon^u = \varepsilon^c = 1/\gamma$.

(b) No welfare weight on the tail: $\Psi(z) = 1$ for large z.

(c) The type distribution has a Pareto tail with shape parameter α: $1 - F(z) \propto z^{-\alpha}$ for large z, implying $\frac{1-F(z)}{z\,f(z)} = \frac{1}{\alpha}$.

Under these assumptions, the tax formula (65) becomes

$$\frac{T'}{1-T'} = (1+\gamma) \cdot \frac{1-F(z)}{z\,f(z)} = \frac{1+\gamma}{\alpha}.$$

Solving:

$$T' = \frac{1}{1 + \dfrac{\alpha}{1+\gamma}}. \tag{68}$$

Rewriting in terms of the earnings distribution. Since the optimal labor supply satisfies $\psi n^\gamma = (1-T')z$, with a constant marginal tax rate $\nu = 1-T'$ in the tail we have

$$y(z) = zn(z) = z\left(\frac{\nu z}{\psi}\right)^{1/\gamma} = \left(\frac{\nu}{\psi}\right)^{1/\gamma} z^{(1+\gamma)/\gamma}.$$

If z is Pareto with shape α, then earnings y are Pareto with shape

$$\alpha_y = \frac{\alpha\gamma}{1+\gamma} \qquad \Longleftrightarrow \qquad \alpha = \frac{1+\gamma}{\gamma}\,\alpha_y.$$

Proof. For $\bar{y}$ in the tail:

$$\Pr(y > \bar{y}) = \Pr\left(z > \bar{y}^{\,\gamma/(1+\gamma)}\left(\frac{\psi}{\nu}\right)^{1/(1+\gamma)}\right) \propto \left(\bar{y}^{\,\gamma/(1+\gamma)}\right)^{-\alpha} = \bar{y}^{\,-\alpha\gamma/(1+\gamma)}.$$

Hence y is Pareto with shape $\alpha_y = \alpha\gamma/(1+\gamma)$. □

Let $e = 1/\gamma$ denote the earnings elasticity with respect to the net-of-tax rate. Then

$$\frac{\alpha}{1+\gamma} = \frac{\alpha_y}{\gamma} = \alpha_y\, e.$$

Therefore, the top tax rate in terms of observables is

$$\boxed{T'_{\text{top}} = \frac{1}{1+\alpha_y\, e},} \tag{69}$$

which is the Diamond–Saez formula. With empirically plausible values—$\alpha_y \approx 1.5$ (Pareto coefficient of the U.S. earnings distribution) and $e \approx 0.25$ (compensated earnings elasticity)—this yields $T'_{\text{top}} \approx 1/(1+0.375) \approx 73\%$.

8.4 Zero tax at the bottom

At the bottom of the distribution $z = z_0$, the boundary condition $\lambda(z_0) = 0$ combined with the tax formula (46) immediately implies

$$T'(y(z_0)) = 0.$$

Intuitively, a marginal tax on the lowest type distorts that type's labor supply but generates no redistributive benefit (there is no one below to redistribute to).

Remark 8.1 (Sensitivity of bottom tax rates). The zero-at-the-bottom result depends critically on the assumption that the lower bound of the productivity distribution is strictly positive ($z_0 > 0$). If instead the support extends to zero ($z_0 = 0$), the optimal marginal tax rate at the bottom converges to 100%, because the marginal disutility of producing the first dollar of income vanishes for any $z > 0$, making the indifference curve locally flat at zero income. In the baseline calibration with $z_0 > 0$, the zero-rate region is vanishingly small and marginal rates rise rapidly from the bottom.

More broadly, Heathcote and Tsujiyama (2026) show that optimal tax rates at low income levels are surprisingly sensitive to how one models zero-earning individuals. A mass of zero-productivity types (a "disability" interpretation) generates a U-shaped marginal tax profile with high rates over a wide range at the bottom. A fixed utility cost of labor force participation (a "costly participation" interpretation) instead delivers low, increasing rates at the bottom. These findings highlight that the classic zero-at-the-bottom result, while theoretically correct for the standard model with $z_0 > 0$, offers limited practical guidance for designing transfers at the bottom of the income distribution.

8.5 Zero tax at the top (with bounded support)

Similarly, at $z = \bar{z}$ with $\lambda(\bar{z}) = 0$:

$$T'(y(\bar{z})) = 0.$$

The reasoning is symmetric: a marginal tax increase on the highest type generates no revenue from types above (there are none), so it is purely distortionary. With an unbounded Pareto tail, this result does not apply and the limiting tax rate is given by (69).

8.6 Summary of forces

The optimal tax schedule balances three forces at every income level:

1. **Equity:** Higher taxes enable redistribution from richer to poorer households. The redistribution motive is captured by $B(z)$ and the social welfare weights $g(z)$.

2. **Efficiency:** Higher taxes distort labor supply. The efficiency cost is governed by the elasticities ε^u and ε^c, summarized in $A(z)$.

3. **Revenue:** The shape of the income distribution determines how many taxpayers are affected by the tax at each point. A thicker tail (lower hazard rate) means more revenue per unit of distortion, summarized in $C(z)$.

9 Computational Approaches

Computing optimal tax schedules requires solving a challenging fixed-point problem: the tax schedule determines behavior, which in turn determines the optimal tax schedule. This section presents three computational approaches, following Heathcote and Tsujiyama (2026). Throughout, we work with the separable preference specification

$$u(c, y; z) = \log c - \frac{1}{1+\sigma}\left(\frac{y}{z}\right)^{1+\sigma},$$

where $1/\sigma$ is the Frisch elasticity of labor supply. We discretize the productivity distribution onto a grid $\{z_1, \ldots, z_N\}$ with probabilities $\{\pi_1, \ldots, \pi_N\}$, evenly spaced in logs. The *coarseness* of the grid is $\kappa = z_{i+1}/z_i = (z_N/z_1)^{1/(N-1)}$.

9.1 Mirrlees approach

The *Mirrlees approach* directly solves for the constrained-efficient allocation by choosing type-specific consumption and income bundles. With our preference class, the local incentive compatibility constraints are necessary and sufficient for global IC. These consist of downward incentive constraints (DIC) and upward incentive constraints (UIC):

$$U(z_i, z_i) \geq U(z_i, z_{i-1}), \quad i = 2, \ldots, N, \tag{DIC}$$

$$U(z_{i-1}, z_{i-1}) \geq U(z_{i-1}, z_i), \quad i = 2, \ldots, N. \tag{UIC}$$

We now argue that, at the optimum, the DICs are binding and the UICs reduce to a monotonicity condition.

DICs are binding. Suppose for contradiction that the DIC for type i is slack:

$$\log(c_i) - \frac{(y_i/z_i)^{1+\sigma}}{1+\sigma} > \log(c_{i-1}) - \frac{(y_{i-1}/z_i)^{1+\sigma}}{1+\sigma}.$$

Then the planner can reduce c_i by ϵ and increase c_{i-1} by $\epsilon\, \pi_i/\pi_{i-1}$, keeping the resource constraint satisfied. The welfare change is

$$\pi_i\left(-\frac{1}{c_i}\right) + \pi_{i-1}\left(\frac{1}{c_{i-1}}\right)\frac{\pi_i}{\pi_{i-1}} = \pi_i\left(\frac{1}{c_{i-1}} - \frac{1}{c_i}\right) > 0,$$

where the inequality holds because $c_{i-1} < c_i$ (lower types consume less at any IC allocation). It remains to verify that, for small ϵ, no IC constraint is violated by this perturbation:

- **DIC for type** i (the slack one): Both $\log(c_i)$ decreases and $\log(c_{i-1})$ increases, tightening the constraint from both sides. But since it was strictly slack, continuity guarantees it remains satisfied for small enough ϵ.
- **DIC for type** $i+1$: This requires $\mathcal{U}_{i+1} \geq U(z_{i+1}, z_i)$, and the RHS involves $\log(c_i)$, which decreases. The constraint is *relaxed*.
- **DIC for type** $i-1$ (if $i \geq 3$): This requires $\mathcal{U}_{i-1} \geq U(z_{i-1}, z_{i-2})$, and the LHS involves $\log(c_{i-1})$, which increases. The constraint is *relaxed*.
- **UIC for type** $i-1$: This requires $\mathcal{U}_{i-1} \geq U(z_{i-1}, z_i)$. The LHS increases (higher c_{i-1}) and the RHS decreases (lower c_i), so the constraint is *relaxed*.
- **UIC for type** i: This requires $\mathcal{U}_i \geq U(z_i, z_{i+1})$. The LHS decreases (lower c_i), tightening the constraint. But the UIC was satisfied (by monotonicity), so it remains satisfied for small ϵ.

All other constraints are unaffected (only c_i and c_{i-1} change). Thus the perturbation is feasible and welfare-improving, contradicting optimality.

UICs hold if and only if $y_i \geq y_{i-1}$. This is the discrete single-crossing argument. With every DIC binding, type $i+1$ is *indifferent* between its own bundle (c_{i+1}, y_{i+1}) and type i's bundle (c_i, y_i):

$$\log(c_{i+1}) - \frac{(y_{i+1}/z_{i+1})^{1+\sigma}}{1+\sigma} = \log(c_i) - \frac{(y_i/z_{i+1})^{1+\sigma}}{1+\sigma}.$$

The UIC for type i requires that i weakly prefers its own bundle to $i+1$'s:

$$\log(c_i) - \frac{(y_i/z_i)^{1+\sigma}}{1+\sigma} \geq \log(c_{i+1}) - \frac{(y_{i+1}/z_i)^{1+\sigma}}{1+\sigma}.$$

Subtracting the binding DIC from the UIC and rearranging:

$$\frac{y_{i+1}^{1+\sigma}}{1+\sigma}\left(\frac{1}{z_i^{1+\sigma}} - \frac{1}{z_{i+1}^{1+\sigma}}\right) \geq \frac{y_i^{1+\sigma}}{1+\sigma}\left(\frac{1}{z_i^{1+\sigma}} - \frac{1}{z_{i+1}^{1+\sigma}}\right).$$

Since $z_i < z_{i+1}$, the factor $1/z_i^{1+\sigma} - 1/z_{i+1}^{1+\sigma} > 0$, and dividing through gives $y_{i+1}^{1+\sigma} \geq y_i^{1+\sigma}$, i.e., $y_{i+1} \geq y_i$.

We now show the converse direction holds as well, establishing the full equivalence. Suppose $y_{i+1} \geq y_i$. We must show the UIC for type i holds: $U(c_i, y_i/z_i) \geq U(c_{i+1}, y_{i+1}/z_i)$, i.e.,

$$\log(c_i) - \frac{(y_i/z_i)^{1+\sigma}}{1+\sigma} \geq \log(c_{i+1}) - \frac{(y_{i+1}/z_i)^{1+\sigma}}{1+\sigma}.$$

From the binding DIC for type $i+1$:

$$\log(c_{i+1}) - \log(c_i) = \frac{(y_{i+1}/z_{i+1})^{1+\sigma} - (y_i/z_{i+1})^{1+\sigma}}{1+\sigma}.$$

Substituting into the UIC requirement and rearranging:

$$\frac{y_{i+1}^{1+\sigma} - y_i^{1+\sigma}}{1+\sigma} \cdot \frac{1}{z_i^{1+\sigma}} \geq \frac{y_{i+1}^{1+\sigma} - y_i^{1+\sigma}}{1+\sigma} \cdot \frac{1}{z_{i+1}^{1+\sigma}}.$$

Since $y_{i+1} \geq y_i$, the factor $y_{i+1}^{1+\sigma} - y_i^{1+\sigma} \geq 0$. Since $z_i < z_{i+1}$, we have $1/z_i^{1+\sigma} > 1/z_{i+1}^{1+\sigma}$. The product of a nonnegative and a positive factor is nonnegative, so the inequality holds. (It is strict whenever $y_{i+1} > y_i$.)

Conversely, if $y_{i+1} < y_i$, the factor $y_{i+1}^{1+\sigma} - y_i^{1+\sigma} < 0$ reverses the inequality, violating the UIC.

Remark 9.1 (Connection to single crossing). The algebraic argument above is the discrete incarnation of the continuous single-crossing proof in Appendix A. The factor $1/z_i^{1+\sigma} - 1/z_{i+1}^{1+\sigma} > 0$ is the discrete analogue of the slope difference $S(c, y, z_i) - S(c, y, z_{i+1}) > 0$ (Lemma 3.1): producing a given output costs more labor disutility for the less productive type. The monotonicity condition $y_{i+1} \geq y_i$ ensures that the allocation does not ask the less productive type to produce more, which would create an incentive to swap bundles.

The planner's problem is therefore:

$$\max_{\{c_i,y_i\}_{i=1}^N} \sum_{i=1}^N \pi_i \left[\log(c_i) - \frac{1}{1+\sigma}\left(\frac{y_i}{z_i}\right)^{1+\sigma}\right] \tag{70}$$

$$\text{s.t.} \quad \log(c_i) - \frac{(y_i/z_i)^{1+\sigma}}{1+\sigma} = \log(c_{i-1}) - \frac{(y_{i-1}/z_i)^{1+\sigma}}{1+\sigma}, \quad i = 2,\dots,N, \tag{71}$$

$$y_i \geq y_{i-1}, \quad i = 2,\dots,N, \tag{72}$$

$$\sum_{i=1}^N \pi_i y_i \geq \sum_{i=1}^N \pi_i c_i + G. \tag{73}$$

Denoting by δ_i the multiplier on the DIC for type i and by μ the multiplier on the resource constraint, the first-order conditions with respect to c_i and y_i are:

$$c_i^{-1}(\pi_i + \delta_i - \delta_{i+1}) = \mu\pi_i, \tag{74}$$

$$\frac{y_i^\sigma}{z_i^{1+\sigma}}(\pi_i + \delta_i) - \frac{y_i^\sigma}{z_{i+1}^{1+\sigma}}\delta_{i+1} = -\mu\pi_i, \tag{75}$$

with $\delta_1 = 0$ and $\delta_{N+1} = 0$. Rather than solving these FOCs directly, we exploit the binding-DIC structure to eliminate c and solve an unconstrained problem in y (see below).

Given the optimal allocation $\{y_i^*, c_i^*\}$, the implied marginal tax rates follow from the household's first-order condition in the decentralized economy:

$$T_i^{*\prime} = 1 - \frac{c_i^*}{z_i}\left(\frac{y_i^*}{z_i}\right)^\sigma. \tag{76}$$

Eliminating consumption via binding DIC. Because every DIC binds at the optimum, consumption is pinned by income alone (up to a single level c_1). Writing $n_i = y_i/z_i$, the binding DIC for type i gives

$$\log c_i = \log c_{i-1} + \underbrace{\frac{n_i^{1+\sigma} - (y_{i-1}/z_i)^{1+\sigma}}{1+\sigma}}_{S_i - S_{i-1}}, \tag{77}$$

so $c_i = c_1 \exp(S_i)$ where $S_1 = 0$ and $S_i = \sum_{j=2}^i [n_j^{1+\sigma} - (y_{j-1}/z_j)^{1+\sigma}]/(1+\sigma)$. The level c_1 is then determined by the resource constraint:

$$c_1 = \frac{\sum_i \pi_i y_i - G}{\sum_i \pi_i \exp(S_i)}. \tag{78}$$

Thus the planner's problem reduces to choosing the income vector y alone.

Reparametrization for monotonicity. The UIC requires $y_i \geq y_{i-1}$. We enforce this with a change of variables:

$$x = \big(\log y_1,\ \delta_2, \dots, \delta_N\big), \qquad \delta_i = \log(y_i/y_{i-1}) \geq 0.$$

Income is recovered as $\log y_i = \sum_{j=1}^i x_j$. The constraint $\delta_i \geq 0$ is a simple box constraint, handled directly by L-BFGS-B.

Algorithm 1 Mirrlees Approach

Require: Grid $\{z_i, \pi_i\}_{i=1}^N$, government spending G, Frisch parameter σ
Ensure: Optimal allocation $\{c_i^*, y_i^*\}_{i=1}^N$ and marginal tax rates $\{T_i^{*\prime}\}_{i=1}^N$

Initialize: Set $x_0 = (\log y_1^{\text{HSV}}, \delta_2^{\text{HSV}}, \ldots)$ from the baseline HSV allocation; for $N > 1000$, warm-start by interpolating from a coarser solution

Solve $\min_x -W(x)$ subject to x_1 free, $x_i \geq 0$ for $i \geq 2$, using L-BFGS-B:
At each evaluation of x:
(a) Recover y from $\log y_i = \sum_{j=1}^i x_j$
(b) Compute c from (77)–(78)
(c) Evaluate $W = \sum_i \pi_i[\log c_i - n_i^{1+\sigma}/(1+\sigma)]$ and its analytic gradient

Recover tax rates:
for $i = 1, \ldots, N$ **do**
$T_i^{*\prime} \leftarrow 1 - \dfrac{c_i^*}{z_i}\left(\dfrac{y_i^*}{z_i}\right)^\sigma$ ▷ marginal rate, from (76)
$T_i^*/y_i^* \leftarrow (y_i^* - c_i^*)/y_i^*$ ▷ average rate
end for

Remark 9.2 (Grid sensitivity). When the grid is coarse (κ large), only a small number of IC constraints bind, and the planner can achieve substantial redistribution with low marginal rates—exploiting "unused" regions of income space between grid points to set high marginal rates that raise revenue without distorting equilibrium labor supply. With $N = 10$ grid points, the income-weighted average marginal tax rate is only about 5%, and GDP is essentially at its first-best level; with a fine grid ($N = 10{,}000$), the average marginal rate rises to around 46% and GDP falls by roughly 14% relative to the no-tax baseline. This is because a coarse grid artificially relaxes IC constraints, making the allocation closer to the first best. The Mirrlees approach is therefore most informative on a fine grid, where the binding DIC constraints actually restrict the planner.

Remark 9.3 (Implementation details). The analytic gradient uses reverse cumulative sums and the chain rule through the DIC chain, making each evaluation $O(N)$. For $N = 10{,}000$, the optimizer typically converges in a few thousand L-BFGS-B iterations. Warm-starting from a coarser solution (e.g., $N = 1{,}000$) via log-linear interpolation of h substantially reduces solve time. The binding DIC can be verified ex post by checking that $U(z_i, z_i) - U(z_i, z_{i-1})$ is zero to machine precision.

9.2 Tax formula approach

The *tax formula approach* iterates on the Diamond–Saez formula to jointly find the optimal tax schedule and the corresponding equilibrium allocation. We first derive the discrete-state version of the DS formula via a perturbation argument, then describe the iterative algorithm.

9.2.1 Derivation of the discrete Diamond–Saez formula

We derive the discrete DS formula directly from the Mirrlees FOCs (74)–(75). The derivation has three steps: solve for the DIC multipliers δ_{i+1} from the consumption FOC,

substitute into the income FOC, and use the household optimality condition to express the result in terms of marginal tax rates.

Step 1: Solving for δ_{i+1} from the consumption FOC. The consumption FOC (74) gives

$$\delta_i - \delta_{i+1} = \pi_i(\mu c_i - 1).$$

Summing from $j = i + 1$ to $j = N$ and using $\delta_{N+1} = 0$:

$$\delta_{i+1} = \sum_{s=i+1}^{N} \pi_s(\mu c_s - 1). \tag{79}$$

The boundary condition $\delta_1 = 0$ pins down μ: setting $i = 0$ gives $0 = \sum_{s=1}^{N} \pi_s(\mu c_s - 1)$, hence

$$\mu = \frac{1}{\mathbb{E}[c^*]}. \tag{80}$$

Substituting back:

$$\delta_{i+1} = \sum_{s=i+1}^{N} \pi_s \left(\frac{c_s^*}{\mathbb{E}[c^*]} - 1 \right). \tag{81}$$

This is the discrete analogue of the costate $\lambda(z)$: it measures the cumulative "redistribution pressure" above type i, with high-consumption types ($c_s > \mathbb{E}[c]$) contributing positively and low-consumption types contributing negatively.

Step 2: Combining the consumption and income FOCs. The income FOC (75) involves both δ_i and δ_{i+1}. On the log-spaced grid with $z_{i+1}/z_i = \kappa$, we have $z_{i+1}^{1+\sigma} = \kappa^{1+\sigma} z_i^{1+\sigma}$, so (75) can be written as

$$\frac{n_i^\sigma}{z_i} \left[(\pi_i + \delta_i) - \frac{\delta_{i+1}}{\kappa^{1+\sigma}} \right] = \mu \pi_i,$$

where $n_i = y_i/z_i$. Now substitute $(\pi_i + \delta_i) = \mu \pi_i c_i + \delta_{i+1}$ from the consumption FOC:

$$\frac{n_i^\sigma}{z_i} \left[\mu \pi_i c_i + \delta_{i+1} \big(1 - \kappa^{-(1+\sigma)}\big) \right] = \mu \pi_i.$$

The household FOC in the decentralized economy gives $c_i n_i^\sigma / z_i = 1 - T_i'$ (equation (76)), so the first term becomes $\mu \pi_i (1 - T_i')$. Rearranging:

$$\mu \pi_i T_i' = \big(1 - \kappa^{-(1+\sigma)}\big) \frac{n_i^\sigma}{z_i} \delta_{i+1}. \tag{82}$$

The left-hand side is the marginal revenue gain from type i's tax; the right-hand side is the incentive cost, proportional to the DIC multiplier δ_{i+1} and the discrete elasticity factor $1 - \kappa^{-(1+\sigma)}$.

Step 3: Expressing in terms of tax rates. Divide (82) by $\mu\pi_i(1-T_i') = \mu\pi_i c_i n_i^\sigma / z_i$:

$$\frac{T_i'}{1-T_i'} = \frac{(1-\kappa^{-(1+\sigma)})}{\mu\pi_i c_i}\,\delta_{i+1}.$$

Substituting $\mu = 1/\mathbb{E}[c^*]$ and (81):

$$\frac{T_i^{*\prime}}{1-T_i^{*\prime}} = \left[1-\kappa^{-(1+\sigma)}\right]\frac{1}{\pi_i}\sum_{s=i+1}^{N}\pi_s\left(1-\frac{\mathbb{E}[c^*]}{c_s^*}\right)\frac{c_s^*}{c_i^*}, \tag{83}$$

where the three factors have the same interpretation as in the continuous ABC formula:

- $A_i^{\text{disc}} = 1-\kappa^{-(1+\sigma)}$: the discrete elasticity factor, capturing the efficiency cost of a discrete IC-binding deviation on the grid. As $\kappa \to 1$, $1-\kappa^{-(1+\sigma)} \approx (1+\sigma)\log\kappa \to 0$ at the same rate as $\pi_i \approx f(z_i)\,z_i\log\kappa \to 0$, so the ratio A_i^{disc}/π_i converges to $(1+\sigma)/(z_i f(z_i))$, recovering the continuous formula.
- $1-\mathbb{E}[c^*]/c_s^*$: one minus the welfare weight of type s, measuring the redistribution motive.
- c_s^*/c_i^*: the ratio of marginal utilities $U_c(c_i)/U_c(c_s) = c_s/c_i$, converting between type s's utils and type i's utils. This factor is unity under quasilinear preferences (constant U_c) but not under log consumption.

Remark 9.4 (Origin of the $\kappa^{-(1+\sigma)}$ factor). The factor $1-\kappa^{-(1+\sigma)}$ arises from the discrete grid structure. In the income FOC, the DIC multiplier δ_{i+1} enters twice: once through type i's own DIC (with coefficient $1/z_i^{1+\sigma}$) and once through type $i+1$'s DIC (with coefficient $1/z_{i+1}^{1+\sigma}$). The difference between these two coefficients produces the factor $1-(z_i/z_{i+1})^{1+\sigma} = 1-\kappa^{-(1+\sigma)}$. On a coarse grid ($\kappa$ large), this factor is close to one and the formula differs substantially from the continuous version. On a fine grid ($\kappa \to 1$), $1-\kappa^{-(1+\sigma)} \approx (1+\sigma)\log\kappa$, which cancels with the $\pi_i \approx f(z_i)\,z_i\,\log\kappa$ in the denominator to recover the continuous elasticity ratio $(1+\sigma)/(z_i f(z_i))$.

Remark 9.5 (Continuous limit). As the grid becomes fine ($\kappa \to 1$, $N \to \infty$), the discrete formula (83) converges to the continuous ABC formula. The sum $\sum_{s>i}\pi_s(\cdot)$ converges to the integral $\int_z^{\bar{z}}(\cdot)\,f(\zeta)\,d\zeta$, the welfare weight $\mathbb{E}[c^*]/c_s^*$ converges to the continuous $g(z)$, and the c_s^*/c_i^* factor converges to unity for adjacent types.

9.2.2 The iterative algorithm

The tax function $\tilde{T}$ is approximated as piecewise linear in *wages*, characterized by type-specific marginal rates $\tilde{T}_i'$ and a common lump-sum transfer Tr. Since the optimal income allocation y^* is not known ex ante, it is convenient to think of $\tilde{T}$ as varying by wage rather than income—this also naturally accommodates bunching (multiple types choosing the same income). Given $(\tilde{T}_i', Tr)$, each type z_i faces a constant marginal rate: the budget constraint is $c_i = (1-\tilde{T}_i')y_i + Tr$ and the household FOC for labour supply $h_i = y_i/z_i$ is

$$(1-\tilde{T}_i')\,z_i\left(n_i^{-\sigma} - n_i\right) = Tr. \tag{84}$$

This is a single nonlinear equation in n_i (given $\tilde{T}_i'$, z_i, and Tr), solved by Newton's method or bisection.

Algorithm 2 Tax Formula Approach

Require: Grid $\{z_i, \pi_i\}_{i=1}^N$, government spending G, Frisch parameter σ, tolerance ε
Ensure: Optimal tax rates $\{T_i^{*\prime}\}_{i=1}^N$ and allocation $\{c_i^*, y_i^*\}_{i=1}^N$

Initialize: Guess marginal tax rates $\{\tilde{T}_i'\}_{i=1}^N$ (e.g., from the current tax system)

repeat
  Step 1: Compute equilibrium allocation given $\tilde{T}$
  Find the transfer Tr such that the government budget balances:
    $\sum_{i=1}^N \pi_i \big[y_i(Tr) - c_i(Tr)\big] = G$
   (use Newton's method on Tr; at each trial Tr, solve (84) for all n_i)
  Record $y_i = z_i n_i$ and $c_i = (1 - \tilde{T}_i')y_i + Tr$ for each type

  Step 2: Update tax rates using discrete DS formula
  for $i = 1, \ldots, N-1$ **do**
    $$\frac{T_i^{\text{new}\prime}}{1 - T_i^{\text{new}\prime}} \leftarrow \left[1 - \kappa^{-(1+\sigma)}\right] \frac{1}{\pi_i} \sum_{s=i+1}^{N} \pi_s \left(1 - \frac{\mathbb{E}[c]}{c_s}\right) \frac{c_s}{c_i}$$
  end for
  Set $T_N^{\text{new}\prime} = 0$ ▷ zero tax at top
  Damped update: $\tilde{T}_i' \leftarrow \alpha\, T_i^{\text{new}\prime} + (1-\alpha)\, \tilde{T}_i'$ for all i ▷ $\alpha \in (0.04, 0.3)$
until $\max_i |T_i^{\text{new}\prime} - \tilde{T}_i'| < \varepsilon$

Remark 9.6 (Failure with coarse grids). When the grid is coarse, the piecewise linear tax function is a poor approximation to the true optimal schedule, which is highly nonlinear between grid points. The candidate allocation may violate global IC constraints: the indifference curve for type z_i may pass *above* the income-consumption pair (y_{i-1}^*, c_{i-1}^*) rather than through it, meaning the local downward IC does not bind and welfare can be improved.

The severity depends on the grid: with $N = 10$ ($\kappa = 2.25$, $\tilde{A} = 0.91$), the average IC slack is 0.46—far from zero—and the income-weighted average marginal rate is only 35%, well below the 46% obtained with fine grids. With $N = 100$ ($\kappa = 1.08$), the average IC slack drops to 0.01 and the tax schedule is much closer to the ODE solution. By $N = 10{,}000$ ($\kappa \approx 1.001$), the tax formula approach converges to the same schedule as the ODE method.

Remark 9.7 (Correct discrete formula). It is important to use the correct discrete DS formula (83), which includes the coarseness parameter κ, rather than a naive discretization of the continuous formula. Most implementations in the literature start from the continuous-productivity formula and discretize for numerical purposes. This does not deliver the correct optimal tax formula for a discrete-productivity environment.

Remark 9.8 (Implementation details). Two practical details matter for convergence. First, the damping parameter α in the tax-rate update should start moderately (e.g., $\alpha = 0.3$) and decrease over iterations (to $\alpha \approx 0.04$ after 2000 iterations); undamped updates ($\alpha = 1$) typically oscillate and fail to converge. Second, the discrete probabilities π_i are constructed by evaluating the wage density on the log-spaced grid and normalizing; the underlying variance parameter σ_θ^2 should be adjusted so that $\mathrm{Var}(\log\theta)$ on the discrete grid matches its target value (0.348 in the HT2026 calibration), following Footnote 11 of the paper.

9.3 Differential equations approach

The *differential equations approach* works directly with the continuous productivity distribution, avoiding discretization of the economic problem. The key idea is to cast the planner's optimality conditions—the envelope condition (18), the costate equation (26), and the tax formula (46) derived in Sections 4–6—as a system of ODEs in z that can be solved with a standard numerical integrator.

Specializing to our separable preferences $u = \log c - \frac{1}{1+\sigma}(y/z)^{1+\sigma}$, the general framework simplifies considerably. Since $U_{cn} = 0$, the non-separable term in the costate equation (26) vanishes. Moreover, $U_c = 1/c$, so $C_{\mathcal{U}} = c$, and the key elasticity ratio becomes $(1+\varepsilon^u)/\varepsilon^c = 1+\sigma$. Define utility $U(z)$, labor supply $n(z) = y(z)/z$, consumption $c(z)$, and the marginal tax ratio $Q(z) = T'(y(z))/(1-T'(y(z)))$.

Envelope condition. The general envelope condition (18), $\mathcal{U}'(z) = -U_n\, y/z^2$, becomes (with $U_n = -n^\sigma$ and $n = y/z$):

$$U'(z) = \frac{1}{z}\, n(z)^{1+\sigma}. \tag{85}$$

Consumption. From $U(z) = \log c(z) - \frac{1}{1+\sigma} n(z)^{1+\sigma}$, consumption and its derivative are:

$$c(z) = \exp\left\{U(z) + \frac{1}{1+\sigma}\, n(z)^{1+\sigma}\right\}, \quad c'(z) = c(z)[U'(z) + n(z)^\sigma\, n'(z)]\,. \tag{86}$$

Optimality ODE. The general tax formula (46) can be written in terms of the tax ratio as $Q(z) = -(1+\sigma)\frac{\lambda(z)}{zf(z)} \cdot \frac{1}{\mu c(z)}$. Differentiating this expression with respect to z and substituting the costate equation (26) (with $U_{cn} = 0$, so $\lambda'(z) = [\mu/c - \psi(z)]f$) yields:

$$Q'(z) = -Q(z)\left[\frac{1}{z} + \frac{f'(z)}{f(z)} + \frac{c'(z)}{c(z)}\right] + \frac{1+\sigma}{z}\left(\frac{1}{\mu\, c(z)} - 1\right), \tag{87}$$

where $\mu = [\int c(s)\, dF(s)]^{-1}$ is the multiplier on the resource constraint.

The difficulty with the Q formulation. The ODE (87) involves $c'(z)/c(z)$. From (86), $c'(z)/c(z) = U'(z) + n(z)^\sigma\, n'(z)$. The envelope condition gives $U'(z) = n^{1+\sigma}/z$, but $n'(z)$—the derivative of labor supply with respect to type—must also be computed. This requires differentiating the household FOC $c\, n^\sigma/z = 1/(1+Q)$ with respect to z:

$$\frac{c'n^\sigma + c\sigma n^{\sigma-1}n'}{z} - \frac{cn^\sigma}{z^2} = -\frac{Q'}{(1+Q)^2}.$$

Substituting $c' = c[n^{1+\sigma}/z + n^\sigma n']$ and collecting terms in n':

$$n'(z) = \frac{1}{c\, n^{\sigma-1}(n+\sigma)/z}\left[\frac{cn^\sigma}{z^2} - \frac{cn^{2\sigma+1}}{z^2} - \frac{Q'}{(1+Q)^2}\right]. \tag{88}$$

The problem is immediate: $n'(z)$ depends on $Q'(z)$, and $Q'(z)$ depends on $c'(z)/c(z)$, which depends on $n'(z)$. The system is *implicitly coupled*. To use the Q formulation, one must either:

(a) Promote n to a third state variable and solve the resulting 3×3 implicit ODE system (Q, U, n) at each step, or

(b) Algebraically eliminate n' from (88) and (87) simultaneously, yielding a single (complicated) explicit expression for Q' in terms of (Q, U, z).

Both approaches are feasible but cumbersome. Option (a) requires solving a 3×3 linear system at every RK4 evaluation to disentangle Q', n', and c'. Option (b) produces a long explicit formula that is error-prone to implement. Algorithm 3 shows the pseudocode for option (a).

Algorithm 3 ODE Approach Using Q Directly (3-State System)

Require: Density $f(z)$ on $[z_0, \bar{z}]$, government spending G, Frisch parameter σ, ODE grid size N, tolerance ε

Ensure: Optimal allocation $\{c(z), y(z)\}$ and marginal tax rates $\{T'(y(z))\}$

Initialize: Guess (μ, n_1) via grid search over candidate values

repeat

  Step 1: Set initial values at $z = z_0$

    $Q(z_0) \leftarrow 0, \quad c(z_0) \leftarrow z_0/n_1^{\sigma}, \quad U(z_0) \leftarrow \log c(z_0) - n_1^{1+\sigma}/(1+\sigma)$

    $R \leftarrow 0$ ▷ resource integral accumulator

  Step 2: Integrate (Q, U) **forward via RK4 from** z_0 **to** $\bar{z}$

  for $i = 1, \ldots, N-1$ **do**

    Compute $\mathbf{k}_1, \ldots, \mathbf{k}_4$ at z_i, $z_m = \sqrt{z_i z_{i+1}}$, z_{i+1}:

      At each of the 4 evaluation points (z, Q, U):

      (a) Recover n by solving $c\, n^{\sigma}/z = 1/(1+Q)$ with $c = e^{U + n^{1+\sigma}/(1+\sigma)}$

      (b) Compute $U' = n^{1+\sigma}/z$ ▷ envelope, no coupling

      (c) Solve for (Q', n') jointly from the coupled system:

        $Q' = -Q[1/z + f'/f + n^{1+\sigma}/z + n^{\sigma} n'] + (1+\sigma)(1/(\mu c) - 1)/z$

        $n' = \left[cn^{\sigma}/z^2 - cn^{2\sigma+1}/z^2 - Q'/(1+Q)^2\right] / \left[cn^{\sigma-1}(n+\sigma)/z\right]$

        (substitute the second into the first, solve the resulting linear equation for n',

        then back out Q')

      (d) Return $\mathbf{F} = (Q', U')$

    $(Q^{i+1}, U^{i+1}) \leftarrow (Q^i, U^i) + \frac{\Delta z}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$

    Recover n^{i+1} from (a), set $c^{i+1} = \exp\{U^{i+1} + (n^{i+1})^{1+\sigma}/(1+\sigma)\}$

    if $z_{i+1} n^{i+1} < z_i n^i$ **then** ▷ bunching: y non-monotone

      $n^{i+1} \leftarrow z_i n^i / z_{i+1}$; recompute c^{i+1} ▷ hold y flat

    end if

    Accumulate: $R \mathrel{+}= \frac{\Delta z}{2}[(y^i - c^i) f(z_i) + (y^{i+1} - c^{i+1}) f(z_{i+1})]$

  end for

  Step 3: Evaluate residuals

    $r_1 \leftarrow Q(\bar{z}) \cdot \bar{z}\, f(\bar{z})$ ▷ transversality

    $r_2 \leftarrow R - G$ ▷ resource constraint

  Step 4: Newton update

    $(\mu, n_1) \leftarrow (\mu, n_1) - J^{-1}(r_1, r_2)^{\top}$ ▷ $J = \partial(r_1, r_2)/\partial(\mu, n_1)$ via autodiff

until $|r_1| + |r_2| < \varepsilon$

Recover tax rates:

for each grid point z_i **do**

  $T'(y_i) \leftarrow Q_i/(1+Q_i)$

end for

Step (c) is the bottleneck: at every one of the $4N$ RK4 function evaluations, one must solve a coupled linear system to disentangle (Q', n'). This is not difficult in principle, but

it is an unnecessary complication. The Q_M formulation below eliminates step (c) entirely.

The Mirrlees Q_M formulation. The key insight, due to Mirrlees (1971) and used by Heathcote and Tsujiyama (2026), is that a simple rescaling of Q absorbs the $c'(z)/c(z)$ term, yielding an ODE that does not involve $n'(z)$. We now derive this formulation.

Start from the planner's optimality condition. With separable preferences, the costate equation (26) simplifies (since $U_{cn} = 0$) to $\lambda'(z) = [\mu\, c(z) - \psi(z)]f(z)$. With utilitarian welfare ($\psi(z) = 1$), integrating from z to $\bar{z}$ using $\lambda(\bar{z}) = 0$ gives

$$\lambda(z) = \int_z^{\bar{z}} [1 - \mu\, c(s)]\, dF(s).$$

Substituting into the tax formula (46) (with $(1 + \varepsilon^u)/\varepsilon^c = 1 + \sigma$ and $U_c = 1/c$), and noting that $Q = -(1 + \sigma)\lambda/(z f \mu c)$:

$$Q(z) = (1 + \sigma)\, \frac{1}{z\, f(z)\, c(z)} \int_z^{\bar{z}} \left[c(s) - \frac{1}{\mu}\right] dF(s). \tag{89}$$

Now define

$$Q_M(z) \equiv \frac{1 - \frac{1}{z}\, c(z)\, n(z)^\sigma}{(1 + \sigma)\, n(z)^\sigma}. \tag{90}$$

Note that Q_M is a simple rescaling of Q: from the household FOC, $c(z)\, n(z)^\sigma/z = 1 - T'(y(z))$, so $Q_M(z) = T'(y(z))/[(1 + \sigma)\, n(z)^\sigma]$, which differs from $Q = T'/(1 - T')$ by a factor involving n and the elasticity. Specifically, $Q = (1 + \sigma)\, z\, Q_M/c$.

To derive the ODE for Q_M, substitute $Q = (1 + \sigma)\, z\, Q_M/c$ into (89) to obtain the integral representation

$$Q_M(z) = \frac{1}{z^2 f(z)} \int_z^{\bar{z}} \left[c(s) - \frac{1}{\mu}\right] dF(s).$$

Note that $Q_M(z_0) = 0$ automatically encodes $\mu = 1/\mathbb{E}[c]$. Differentiating with respect to z:

$$\begin{aligned} Q_M'(z) &= \frac{d}{dz}\left[\frac{1}{z^2 f(z)}\right] \int_z^{\bar{z}} \left[c(s) - \tfrac{1}{\mu}\right] dF(s) + \frac{1}{z^2 f(z)} \cdot \left(-\left[c(z) - \tfrac{1}{\mu}\right] f(z)\right) \\ &= -\frac{Q_M(z)}{z}\left[\frac{z\, f'(z)}{f(z)} + 2\right] - \frac{c(z) - 1/\mu}{z^2}, \end{aligned}$$

where the first term uses $\int_z^{\bar{z}} [c(s) - 1/\mu]\, dF(s) = z^2 f(z)\, Q_M(z)$. Thus:

$$\boxed{Q_M'(z) = -\frac{Q_M(z)}{z}\left[\frac{z\, f'(z)}{f(z)} + 2\right] - \frac{c(z) - 1/\mu}{z^2}.} \tag{91}$$

Unlike (87), this formulation does not involve $c'(z)$ or $n'(z)$, but it does depend on $c(z)$ (which is recovered from the state variables at each integration step).

The original marginal tax ratio is recovered from

$$Q(z) = \frac{z - c(z)\, n(z)^\sigma}{c(z)\, n(z)^\sigma}. \tag{92}$$

Boundary conditions. The two boundary conditions follow from the transversality conditions $\lambda(z_0) = 0$ and $\lambda(\bar{z}) = 0$ via the tax formula $Q(z) = -(1+\sigma)\,\lambda(z)/(z\,f(z)\,\mu\,c(z))$.

At the bottom ($z = z_0$): $\lambda(z_0) = 0$ directly gives $Q(z_0) = 0$.

At the top ($z = \bar{z}$): When $\bar{z} < \infty$, $\lambda(\bar{z}) = 0$ gives $Q(\bar{z}) = 0$ directly. When $\bar{z} = \infty$ (unbounded support, e.g., a Pareto tail), we cannot evaluate Q at $\bar{z}$. The transversality condition becomes $\lim_{z\to\infty} \lambda(z) = 0$. Inverting the tax formula, $\lambda(z) = -Q(z)\,z\,f(z)\,\mu\,c(z)/(1+\sigma)$. Since μ and $c(z)$ remain bounded and positive, $\lambda(z) \to 0$ is equivalent to

$$Q(z)\,z\,f(z) \to 0 \quad \text{as } z \to \infty.$$

This condition does *not* require Q itself to vanish—only that Q does not grow faster than $1/(z\,f(z))$. For a Pareto tail with $f(z) \sim z^{-(\alpha+1)}$, the product $z\,f(z) \sim z^{-\alpha} \to 0$, so Q can converge to a positive constant (the Diamond–Saez rate $Q_\infty = (1+\gamma)/\alpha$) and the condition is still satisfied. In the robust case (Section 10), $Q(z) \to 0$ itself, so the condition holds a fortiori.

These two conditions, together with the resource constraint, pin down the two free parameters in the shooting algorithm below.

Numerical solution: a shooting method. The complete system to be solved consists of two ODEs, one implicit constraint, two boundary conditions, and one integral (resource) constraint. We collect them here for reference:

ODEs (integrated forward from z_0 to $\bar{z}$):

$$U'(z) = \frac{1}{z}\,n(z)^{1+\sigma}, \tag{Envelope}$$

$$Q_M'(z) = -\frac{Q_M(z)}{z}\left[\frac{z\,f'(z)}{f(z)} + 2\right] - \frac{c(z) - 1/\mu}{z^2}. \tag{Optimality}$$

Implicit constraint (solved at each evaluation point):

$$(1+\sigma)\,n(z)^\sigma\,Q_M(z) + \frac{c(z)\,n(z)^\sigma}{z} = 1, \qquad c(z) = \exp\left\{U(z) + \frac{n(z)^{1+\sigma}}{1+\sigma}\right\}. \tag{n-solve}$$

Boundary conditions:

$$Q_M(z_0) = 0, \qquad Q(\bar{z})\,\bar{z}\,f(\bar{z}) = 0. \tag{BC}$$

Resource constraint:

$$\int_{z_0}^{\bar{z}} \big[z\,h(z) - c(z)\big]\,f(z)\,dz = G. \tag{RC}$$

The system has two unknowns not pinned by initial conditions: $\mu = 1/\mathbb{E}[c]$ (the resource multiplier) and $h(z_0)$ (labor supply at the bottom). Once these are guessed, the ODEs can be integrated forward as an initial value problem. The bottom boundary condition gives $Q_M(z_0) = 0$ and $U'(z_0) = h(z_0)^{1+\sigma}/z_0$, but two quantities remain unknown:

(a) μ, the multiplier on the resource constraint (equivalently, $1/\mathbb{E}[c]$), and

(b) $n(z_0)$, the labor supply of the least productive type.

We guess these two values, integrate the ODE forward from z_0 to $\bar{z}$, and check whether the terminal and feasibility conditions are satisfied. This gives a two-dimensional root-finding problem.

Write the 2-state system as

$$\frac{d}{dz}\begin{pmatrix} Q_M \\ U \end{pmatrix} = \begin{pmatrix} D_{Q_M}(z, Q_M, U, n) \\ D_U(z, n) \end{pmatrix} \equiv \mathbf{F}(z, Q_M, U),$$

where the right-hand sides are:

$$\begin{aligned} D_U(z, n) &= \frac{1}{z}\, n^{1+\sigma}, \\ D_{Q_M}(z, Q_M, U, n) &= -\frac{Q_M}{z}\left[\frac{z f'(z)}{f(z)} + 2\right] - \frac{c(z) - 1/\mu}{z^2}, \end{aligned}$$

with $c = \exp\{U + h^{1+\sigma}/(1+\sigma)\}$ recovered from the utility definition. Rather than integrating n as a third state variable (which requires the numerically unstable derivative $n'(z)$), we recover n at each RK4 evaluation point by solving the constraint equation

$$(1+\sigma)\, n^\sigma\, Q_M + \frac{c\, n^\sigma}{z} = 1, \tag{93}$$

where $c = \exp\{U + h^{1+\sigma}/(1+\sigma)\}$. This is a single nonlinear equation in h (given the current Q_M, U, and z), solved by Newton's method. The constraint follows from the definition of Q_M in (90): $Q_M = [1 - c\, n^\sigma/z]/[(1+\sigma)\, n^\sigma]$, rearranged to $(1+\sigma)\, n^\sigma\, Q_M = 1 - c\, n^\sigma/z$.

Algorithm 4 Differential Equations (Shooting) Approach

Require: Density $f(z)$ on $[z_0, \bar{z}]$, government spending G, Frisch parameter σ, ODE grid size N, tolerance ε

Ensure: Optimal allocation $\{c(z), y(z)\}$ and marginal tax rates $\{T'(y(z))\}$

Initialize: Guess (μ, n_1) via grid search over candidate values

repeat

  Step 1: Set initial values at $z = z_0$

    $Q_M(z_0) \leftarrow 0, \quad c(z_0) \leftarrow z_0/n_1^\sigma, \quad U(z_0) \leftarrow \log c(z_0) - n_1^{1+\sigma}/(1+\sigma)$

  Step 2: Integrate (Q_M, U) **forward via RK4 from** z_0 **to** $\bar{z}$

  for $i = 1, \ldots, N-1$ **do**

    Compute $\mathbf{k}_1, \ldots, \mathbf{k}_4$ using $\mathbf{F}$ at z_i, $z_m = \sqrt{z_i z_{i+1}}$, z_{i+1}

      (at each evaluation, recover n by solving (93))

    $(Q_M^{i+1}, U^{i+1}) \leftarrow (Q_M^i, U^i) + \frac{\Delta z}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$

    Recover n^{i+1} from (93), set $c^{i+1} = \exp\{U^{i+1} + (n^{i+1})^{1+\sigma}/(1+\sigma)\}$

    if $z_{i+1} n^{i+1} < z_i n^i$ **then** ▷ bunching: y non-monotone

      $n^{i+1} \leftarrow z_i n^i / z_{i+1}$; recompute c^{i+1} ▷ hold y flat

    end if

    Accumulate resource integral: $R \mathrel{+}= \frac{\Delta z}{2}[(y^i - c^i) f(z_i) + (y^{i+1} - c^{i+1}) f(z_{i+1})]$

  end for

  Step 3: Evaluate residuals

    $r_1 \leftarrow Q(\bar{z}) \cdot \bar{z}\, f(\bar{z})$ ▷ transversality

    $r_2 \leftarrow R - G$ ▷ resource constraint

  Step 4: Newton update

    $(\mu, n_1) \leftarrow (\mu, n_1) - J^{-1}(r_1, r_2)^\top$ ▷ $J = \partial(r_1, r_2)/\partial(\mu, n_1)$ via autodiff

until $|r_1| + |r_2| < \varepsilon$

Recover tax rates:

for each grid point z_i **do**

  $Q_i \leftarrow (z_i - c_i n_i^\sigma)/(c_i n_i^\sigma)$

  $T'(y_i) \leftarrow Q_i/(1 + Q_i)$

end for

The RK4 stages in Step 2 use geometric midpoints $z_m = \sqrt{z_i \cdot z_{i+1}}$ (since the grid is log-spaced) and achieve $O((\Delta z)^5)$ local truncation error per step. Explicitly:

$$\mathbf{k}_1 = \mathbf{F}(z_i,\ Q_M^i, U^i), \qquad \hat{n}_1 = n^i,$$

$$\mathbf{k}_2 = \mathbf{F}\Big(z_m,\ Q_M^i + \tfrac{\Delta z}{2} k_1^{Q_M},\ U^i + \tfrac{\Delta z}{2} k_1^U\Big), \qquad \hat{n}_2 \text{ from (93)},$$

$$\mathbf{k}_3 = \mathbf{F}\Big(z_m,\ Q_M^i + \tfrac{\Delta z}{2} k_2^{Q_M},\ U^i + \tfrac{\Delta z}{2} k_2^U\Big), \qquad \hat{n}_3 \text{ from (93)},$$

$$\mathbf{k}_4 = \mathbf{F}\Big(z_{i+1},\ Q_M^i + \Delta z\, k_3^{Q_M},\ U^i + \Delta z\, k_3^U\Big), \qquad \hat{n}_4 \text{ from (93)}.$$

Unlike the Mirrlees and tax-formula approaches, the grid $\{z_1, \ldots, z_N\}$ here controls only the *step size* of the numerical integrator—the underlying economic model remains

continuous. This is why the method is far less sensitive to N: increasing N refines the numerical approximation to the ODE, but does not change the set of incentive constraints in the economic problem.

Remark 9.9 (Why Q_M instead of Q?). The Q_M formulation, eq. (91), is preferred for computation because, unlike the Q formulation (87), it does not involve the derivative $c'(z)$. The Q_M ODE depends on Q_M, z, the density f, and $c(z)$ (recovered from the state variables at each step), but avoids the need to compute $n'(z)$. Combined with the implicit recovery of h from the constraint equation (93), this yields a 2-state system (Q_M, U) that is simpler and more numerically stable than integrating n explicitly. Heathcote and Tsujiyama (2026) verify that using the alternative formulations (eq. (87) or the Saez variant) delivers identical results.

Remark 9.10 (Robustness to discretization). Because the ODEs are derived from the continuous-type formulation *before* any numerical approximation, the method is far less sensitive to the number of grid points N used in the ODE solver. The solution with $N = 100$ Runge–Kutta steps is already indistinguishable from the fine-grid ($N = 10{,}000$) Mirrlees solution, whereas the Mirrlees and tax-formula approaches require $N \geq 1{,}000$ for comparable accuracy. The reason is that the grid controls only the *step size* of the numerical integrator, not the number of incentive constraints in the economic problem.

Remark 9.11 (Comparison of approaches). Each method has strengths. The Mirrlees approach solves the constrained planner's problem directly and is conceptually transparent, but is sensitive to grid coarseness. The tax formula approach is easy to implement and connects directly to the Diamond–Saez formula, but inherits the piecewise-linear approximation and can fail on coarse grids. The ODE approach is the most robust to discretization, but requires more sophisticated numerical methods (shooting, RK4) and careful treatment of boundary conditions.

10 Robust Optimal Taxation

The analysis in the preceding sections takes the type distribution $F(z)$ as known. In practice, the shape of the distribution—especially in the tails—is subject to substantial uncertainty, and the optimal tax schedule can be highly sensitive to the assumed distribution. This section introduces model misspecification concerns following Bhandari, Borovička, and Yao (2024), and shows that even small concerns about the correctness of F can dramatically alter the optimal tax schedule at the top of the income distribution.

10.1 The tax problem with misspecification concerns

The government contemplates a set of alternative type distributions $\widetilde{F}(z)$ that are statistically close to the benchmark distribution $F(z)$. Let $m(z) = d\widetilde{F}/dF$ denote the likelihood ratio (Radon–Nikodým derivative) between the alternative and benchmark distributions, so that $\mathbb{E}[m] = 1$ by construction. For any integrable function $X(z)$,

$$\widetilde{\mathbb{E}}[X] = \int_{\mathcal{Z}} X(z)\, m(z)\, dF(z) = \mathbb{E}[mX].$$

The statistical distance between F and $\widetilde{F}$ is measured by a divergence function $\mathcal{E}(m)$. A natural choice is the relative entropy (Kullback–Leibler divergence):

$$\mathcal{E}_0(m) = \mathbb{E}[m \log m] = \int_{\mathcal{Z}} m(z) \log m(z)\, f(z)\, dz. \tag{94}$$

The set of plausible alternative distributions is bounded by a divergence constraint $\mathcal{E}(m) \leq \kappa$, forming a "divergence ball" around the benchmark:

$$\mathcal{F}(F, \kappa) = \big\{\widetilde{F} : \mathcal{E}(F, \widetilde{F}) \leq \kappa\big\}.$$

The robust government chooses a tax function to maximize welfare under the worst-case distribution:

$$\max_{T} \min_{\widetilde{F} \in \mathcal{F}(F,\kappa)} \quad \widetilde{\mathbb{E}}[\psi\, \mathcal{U}] + V\big(\widetilde{\mathbb{E}}[T(\mathcal{Y})]\big). \tag{95}$$

The interpretation is a two-player game: the government commits to a tax schedule T, and then a malevolent nature chooses the distribution $\widetilde{F}$ within the ambiguity set to minimize the government's objective. The government, anticipating this, chooses T to maximize its worst-case welfare. The solution $\widetilde{F}$ is called the *worst-case distribution*.

It is convenient to work with an equivalent penalized formulation. Introducing the Lagrange multiplier θ on the divergence constraint, the problem becomes

$$\max_{T} \min_{\substack{m>0 \\ \mathbb{E}[m]=1}} \quad \mathbb{E}[m\psi\, \mathcal{U}] + V\big(\mathbb{E}[m\, T(\mathcal{Y})]\big) + \theta\, \mathcal{E}_0(m). \tag{96}$$

The parameter θ controls the degree of misspecification concerns. As $\theta \to \infty$, the penalty becomes prohibitive, the worst-case distribution converges to the benchmark ($m \equiv 1$), and we recover the standard problem. As θ decreases, the government entertains a larger set of alternative distributions and the tax schedule adjusts more aggressively.

Remark 10.1 (Welfare and budgetary channels). The malevolent nature exploits two channels. First, it underweighs types with high weighted utility $\psi(z)\mathcal{U}(z)$ to lower $\mathbb{E}[m\psi\mathcal{U}]$ (welfare channel). Second, it underweighs net taxpayers and overweighs net recipients to reduce $\mathbb{E}[m\, T(\mathcal{Y})]$ (budgetary channel). Both channels are penalized by the divergence term $\theta\, \mathcal{E}_0(m)$.

Remark 10.2 (Minimax theorem and the order of optimization). Problem (95) is a max-min problem: the government commits to T, then nature minimizes by choosing $\widetilde{F}$. This is the standard formulation in the ambiguity aversion literature—the decision maker (government) acts first, and is cautious because it anticipates that nature will exploit any vulnerability in the chosen policy.

For the analysis below, it is convenient to exchange the order of optimization and work with the equivalent min-max problem $\min_m \max_T$, where nature commits to a distribution and the government best-responds. By the max-min inequality, $\max_T \min_m \leq \min_m \max_T$ in general. Bhandari, Borovička, and Yao (2024) verify in the discrete-type case that the conditions of Sion's minimax theorem (1958) are satisfied: the objective in (96) is strictly concave in the allocation variables (after a change of variables to $u_i = u(c_i)$ and $v_i = v(y_i/z_i)$) and strictly convex in m (due to the entropy term $\theta\, m \log m$), and the constraint sets can be made compact and convex. The minimax theorem then guarantees equality: $\max_T \min_m = \min_m \max_T$, and the saddle point (T^*, m^*) is well defined.

This equivalence is important because it means the optimality conditions derived below—which characterize the saddle point by optimizing over T for fixed m and over m for fixed T—are valid regardless of the timing protocol.

Remark 10.3 (Ex-post Bayesian interpretation). The minimax equivalence enables an *ex-post Bayesian* interpretation of the robust tax problem. At the saddle point, fix the worst-case distortion m^*. The inner maximization over T is then a *standard* (non-robust) tax problem under the distorted distribution $\widetilde{f}(z) = m^*(z)\, f(z)$. All the techniques from Sections 3–6—the taxation principle, the envelope condition, the Hamiltonian, the ABC formula—apply directly, with $\widetilde{f}$ replacing f.

This decomposition is analytically powerful: it separates the robust problem into two logical components:

(i) **What shapes the worst-case distribution?** The distortion $m^*(z)$ is determined by the first-order condition for the minimization over m (equation (99) below), which tilts the benchmark distribution exponentially based on the tax-and-welfare profile.

(ii) **How does the tax schedule respond?** Given $\widetilde{f}$, the optimal tax formula is the standard ABC formula evaluated at the worst-case distribution (equation (100) below).

The solution is a fixed point: m^* depends on T^* (through the exponential tilting), and T^* depends on m^* (through the ABC formula). This fixed-point structure is what makes the problem non-trivial despite the ex-post Bayesian reduction.

10.2 Mechanism design formulation

Since the incentive-compatibility constraint is type-by-type and does not depend on the underlying distribution, misspecification concerns do not alter its structure. The local IC (envelope) condition

$$\frac{d\mathcal{U}}{dz} = -U_n\left(c(z), \frac{y(z)}{z}\right)\frac{y(z)}{z^2}$$

remains the same as in Section 3. We can therefore replace the inner maximization over the tax function T by a mechanism design problem for each given alternative distribution $\widetilde{f}(z) = m(z)f(z)$.

Using the entropy penalty (94) and the minimax equivalence (Remark 10.2), we write the problem in the analytically convenient min-max form:

$$\min_{\substack{m>0\\ \mathbb{E}[m]=1}} \max_{c,y} \int_{z_0}^{\bar{z}} \psi(z)\,\mathcal{U}(z)\, m(z)\, f(z)\, dz + V(G) + \theta \int_{z_0}^{\bar{z}} m(z)\log m(z)\, f(z)\, dz \tag{97}$$

subject to the envelope condition and the budget constraint

$$G = \int_{z_0}^{\bar{z}} \big[y(z) - c(z)\big]\, m(z)\, f(z)\, dz.$$

For a fixed $m(z)$, the inner problem is a standard Mirrlees problem under distribution $\widetilde{f}(z) = m(z)f(z)$.

10.3 Hamiltonian and the robust tax formula

Treat $\mathcal{U}$ as the state variable, λ as its costate, and y and m as control variables. The Hamiltonian is

$$\begin{aligned} H(\mathcal{U}, y, m, \lambda) = \psi(z)\,\mathcal{U}(z)\,m(z)\,f(z) + \theta\,m(z)\log m(z)\,f(z) - \chi\,m(z)\,f(z) \\ - \lambda(z)\,U_n\left(c(z), \frac{y(z)}{z}\right)\frac{y(z)}{z^2} + \mu\big[y(z) - c(z)\big]\,m(z)\,f(z), \end{aligned} \tag{98}$$

where μ is the multiplier on the budget constraint, χ is the multiplier on $\mathbb{E}[m] = 1$, and $c(z) = C(\mathcal{U}(z), y(z))$ is defined implicitly from the utility function.

Optimal worst-case distortion. The first-order condition with respect to $m(z)$ yields

$$\psi(z)\,\mathcal{U}(z) + \theta(1 + \log m(z)) - \chi + \mu\big[y(z) - c(z)\big] = 0.$$

Solving for $m(z)$ and using the normalization $\mathbb{E}[m] = 1$ to absorb constants into $\bar{m}$:

$$m(z) = \bar{m}\exp\left(-\frac{1}{\theta}\big[\psi(z)\,\mathcal{U}(z) + \mu\,T(y(z))\big]\right), \tag{99}$$

where $T(y(z)) = y(z) - c(z)$ is the effective tax on type z, and $\bar{m}$ is the normalization constant ensuring $\mathbb{E}[m] = 1$. The worst-case distortion is an *exponential tilting* of the benchmark distribution: it underweighs types with high weighted utility or high tax contributions, and overweighs those where the planner is most vulnerable.

Robust marginal tax formula. The remaining optimality conditions (with respect to y and $\mathcal{U}$) yield a formula for the marginal tax rate that takes the same ABC structure as in Section 6, but with all distributional quantities evaluated under the worst-case distribution $\widetilde{f}(z) = m(z)f(z)$.

In the general separable case $U(c, n) = u(c) - v(n)$, one replaces Ω and ω from (66) with their worst-case counterparts $\widetilde{\Omega}$ and $\widetilde{\omega}$, constructed using $\widetilde{f}$ in place of f. For the remainder of this section, we specialize to the analytically tractable case that also delivers the strongest results.

Assumption 10.1 (Quasilinear isoelastic preferences)**.** Workers' preferences are $U(c, n) = c - n^{1+\gamma}/(1 + \gamma)$, and $V(G)$ satisfies the Inada conditions $\lim_{G \searrow \underline{G}} V'(G) = \infty$ and $\lim_{G \nearrow \infty} V'(G) = 0$.

Under quasilinear preferences, $u'(c) \equiv 1$, so $\widetilde{\Omega} = \widetilde{F}$, $\widetilde{\omega} = \widetilde{f}$, and $(1 + \varepsilon^u)/\varepsilon^c = 1 + \gamma$. The tax formula becomes

$$\boxed{\frac{T'(y(z))}{1 - T'(y(z))} = (1 + \gamma) \cdot \frac{\widetilde{\Psi}(z) - \widetilde{F}(z)}{1 - \widetilde{F}(z)} \cdot \frac{1 - \widetilde{F}(z)}{z\,\widetilde{f}(z)}} \tag{100}$$

where $\widetilde{F}(z)$ and $\widetilde{f}(z) = m(z)f(z)$ are the CDF and density of the worst-case distribution, and the cumulative welfare weight under the worst case is

$$\widetilde{\Psi}(z) = \int_{z_0}^{z} \frac{\psi(\zeta)\,\widetilde{f}(\zeta)}{\int_{z_0}^{\bar{z}} \psi(\xi)\,\widetilde{f}(\xi)\,d\xi}\,d\zeta.$$

The three factors retain their standard ABC interpretations: $(1+\gamma) = A$ captures the distortionary cost of taxation on labor supply; $(\widetilde{\Psi}(z) - \widetilde{F}(z))/(1 - \widetilde{F}(z)) = B$ measures the desire for redistribution, bounded above by one; and $(1 - \widetilde{F}(z))/(z\,\widetilde{f}(z)) = C$ is the inverse hazard rate of the worst-case distribution, capturing the tradeoff between the benefit of raising lump-sum revenue from types above z and the distortionary cost at z. The crucial difference from the standard formula is that B and C are now evaluated at the *endogenous* worst-case distribution $\widetilde{f}(z) = m(z)f(z)$.

Remark 10.4 (Joint determination). The worst-case distortion $m(z)$ in (99) and the tax function $T(y(z))$ in (100) are determined jointly as the saddle point of the minimax problem (97). Given the tax function, m delivers the lowest penalized objective; given m, the tax function maximizes welfare.

Remark 10.5 (Effect of θ on the worst-case distribution). A smaller θ (stronger misspecification concerns) produces more aggressive exponential tilting in (99), yielding a worst-case density $\widetilde{f}$ that is more different from the benchmark f. In the limit $\theta \to \infty$, the worst-case density converges to f and we recover the standard Diamond–Saez formula.

10.4 Top tax rates: the quasilinear–Rawlsian–Pareto case

To sharpen the analysis of top marginal tax rates, we specialize to the case that generates the highest taxes in the standard model and is therefore the most informative about the effects of robustness concerns.

Under the quasilinear preferences of Assumption 10.1, the income function is $y(z) = (1 - T'(y(z)))^{1/\gamma}\, z^{(1+\gamma)/\gamma}$. We further assume a Rawlsian–Pareto welfare weight: there exists $\hat{z}$ such that $\psi(z) = 0$ for all $z \geq \hat{z}$. This eliminates welfare concerns in the right tail, so that $\widetilde{\Psi}(z) = 1$ for $z \geq \hat{z}$, and the B factor equals one. The tax formula (100) becomes

$$\frac{T'(y(z))}{1 - T'(y(z))} = (1+\gamma)\,\frac{1 - \widetilde{F}(z)}{z\,\widetilde{f}(z)}, \qquad z \geq \hat{z}. \tag{101}$$

The worst-case distortion in the tail simplifies to

$$m(z) = \bar{m}\exp\Big(-\frac{\mu}{\theta}\,T(y(z))\Big), \qquad z \geq \hat{z}, \tag{102}$$

reflecting purely budgetary concerns—the robust planner worries that there are fewer high-productivity types contributing tax revenue.

Assumption 10.2 (Density regularity)**.** There exists $\hat{z}$ such that the type density $f(z)$ is continuously differentiable on $[\hat{z}, \bar{z})$ and $zf(z)$ is strictly decreasing on $[\hat{z}, \bar{z})$, with

$$\lim_{z\to\bar{z}} \frac{d\log f(z)}{d\log z} < -1,$$

where the limit may be $-\infty$.

This assumption is used in two places in the proof of Proposition 10.1:

- The condition $\lim_{z\to\bar{z}} \frac{d\log f}{d\log z} < -1$ ensures that $z\,f(z) \to 0$ as $z \to \infty$. This is needed so that, in the *standard* (non-robust) model, the inverse hazard rate $(1 - F)/(zf)$ converges to a finite positive limit (the Pareto case) rather than diverging. Without it, the standard top tax rate could be ill-defined.

- The continuous differentiability of f on $[\hat{z}, \bar{z})$ ensures that the L'Hôpital argument in the proof is valid: we differentiate $1 - \widetilde{F}(z)$ and $z\,\widetilde{f}(z)$ and need both to be smooth.

The assumption is compatible with Pareto tails ($f(z) \propto z^{-\alpha-1}$, giving $d\log f/d\log z = -\alpha - 1 < -1$ for $\alpha > 0$), lognormal tails (where $d\log f/d\log z \to -\infty$), and any mixture thereof.

10.4.1 Zero marginal taxes at the top

In the standard model without misspecification concerns ($m \equiv 1$), the limiting marginal tax rate depends on the tail of $f(z)$. For a Pareto tail with shape parameter α, the ratio $(1 - F(z))/(zf(z)) = 1/\alpha$, yielding the well-known positive top tax rate $T'_\infty = (1+\gamma)/(1+\gamma+\alpha)$. This has led to widely cited policy prescriptions: for instance, Diamond and Saez (2011) find a top marginal tax of 73% based on $\gamma^{-1} = 0.25$ and $\alpha = 1.875$.

With misspecification concerns, the picture changes dramatically.

Proposition 10.1 (Zero top marginal tax rate)**.** *Under Assumptions 10.1 and 10.2 with $\bar{z} = \infty$ and $\theta < \infty$, the marginal tax rate vanishes at the top:*

$$\lim_{z\to\infty} T'(y(z)) = 0.$$

This holds irrespective of the shape of the benchmark distribution $f(z)$.

Proof sketch. Suppose for contradiction that $\lim_{z\to\infty} T'(y(z)) = \tau_\infty > 0$. Then $y(z) \approx (1-\tau_\infty)^{1/\gamma} z^{(1+\gamma)/\gamma}$, and the tax revenue $T(y(z))$ grows without bound. The worst-case distortion becomes

$$m(z) \approx \bar{m} \exp\Big(-\frac{\mu}{\theta}\,\tau_\infty\,(1-\tau_\infty)^{1/\gamma}\,z^{(1+\gamma)/\gamma}\Big),$$

which decays super-exponentially. The tax formula (101) says $T'/(1-T') = (1+\gamma)(1-\widetilde{F}(z))/(z\,\widetilde{f}(z))$, so we need to evaluate $\lim_{z\to\infty}(1-\widetilde{F}(z))/(z\,\widetilde{f}(z))$. We proceed in three steps.

Step 1: The ratio is a 0/0 form. As $z \to \infty$, the numerator $1-\widetilde{F}(z) = \int_z^\infty \widetilde{f}(s)\,ds \to 0$ (the tail probability vanishes). The denominator $z\,\widetilde{f}(z) \to 0$ as well: $\widetilde{f}(z) = m(z)\,f(z)$, and $m(z)$ decays super-exponentially (faster than any power of z), while $z\,f(z)$ grows at most polynomially. This is a 0/0 indeterminate form, so L'Hôpital's rule applies.

Step 2: Apply L'Hôpital. Differentiate numerator and denominator with respect to z:

$$\begin{aligned}
\text{Numerator:}\quad & \frac{d}{dz}\big[1-\widetilde{F}(z)\big] = -\widetilde{f}(z),\\
\text{Denominator:}\quad & \frac{d}{dz}\big[z\,\widetilde{f}(z)\big] = \widetilde{f}(z) + z\,\widetilde{f}'(z) = \widetilde{f}(z)\left[1 + z\,\frac{d\log\widetilde{f}(z)}{dz}\right].
\end{aligned}$$

Their ratio gives:

$$\lim_{z\to\infty}\frac{1-\widetilde{F}(z)}{z\,\widetilde{f}(z)} = \lim_{z\to\infty}\frac{-\widetilde{f}(z)}{\widetilde{f}(z)\left[1 + z\,\frac{d\log\widetilde{f}}{dz}\right]} = \lim_{z\to\infty}\frac{-1}{1 + z\,\frac{d\log\widetilde{f}}{dz}}.$$

Step 3: Show the denominator diverges. Decompose the log-derivative of $\widetilde{f} = m \cdot f$:

$$\frac{d \log \widetilde{f}}{dz} = \frac{d \log f}{dz} + \frac{d \log m}{dz}.$$

From $m(z) = \bar{m} \exp(-(\mu/\theta)\, T(y(z)))$, we have $\frac{d \log m}{dz} = -\frac{\mu}{\theta}\, T'(y(z))\, y'(z)$. Multiplying through by z:

$$z\, \frac{d \log \widetilde{f}}{dz} = \underbrace{z\, \frac{d \log f}{dz}}_{\text{bounded (Assumption 10.2)}} - \underbrace{\frac{\mu}{\theta}\, z\, T'(y)\, y'(z)}_{\to \infty}.$$

The second term diverges because, under the hypothesis $T' \to \tau_\infty > 0$, we have $y(z) \approx (1-\tau_\infty)^{1/\gamma} z^{(1+\gamma)/\gamma}$, so $y'(z) \approx \frac{1+\gamma}{\gamma} z^{1/\gamma}$ and

$$\frac{\mu}{\theta}\, z\, \tau_\infty\, y'(z) \;\approx\; \frac{\mu\, \tau_\infty (1+\gamma)}{\theta\, \gamma}\, z^{1+1/\gamma} \;\to\; \infty.$$

The first term $z\, d \log f / dz$ converges to a finite constant by Assumption 10.2 (e.g., $-(\alpha+1)$ for a Pareto tail), so it is dominated. Therefore $z\, \frac{d \log \widetilde{f}}{dz} \to -\infty$, the denominator in the L'Hôpital ratio diverges to $-\infty$, and

$$\lim_{z \to \infty} \frac{1 - \widetilde{F}(z)}{z\, \widetilde{f}(z)} = 0.$$

By the tax formula (101), this gives $\lim T'(y) = 0$, contradicting $\tau_\infty > 0$. □

The key intuition is that the robust planner is concerned that there are fewer high-productivity workers who can be taxed. The worst-case distortion (102) is more severe for higher types because they contribute more tax revenue $T(y(z))$. This effectively generates a thinner tail in the worst-case distribution $\widetilde{f}$, and the optimal response is a lower and ultimately vanishing marginal tax rate.

Importantly, the conclusion holds for *any* finite θ: even vanishingly small misspecification concerns overturn the positive top marginal tax result.

Remark 10.6 (Comparison with the standard model). The single-crossing property implies $y(z)$ is strictly increasing and, since marginal taxes are positive, $m(z)$ is strictly decreasing. Therefore, for any $z \geq \hat{z}$,

$$\frac{1 - \widetilde{F}(z)}{z\, \widetilde{f}(z)} < \frac{1 - F(z)}{z\, f(z)},$$

so the robust marginal tax rate is *everywhere lower* than the standard one, not just at the limit.

10.4.2 Rate of decay of the marginal tax rate

Proposition 10.1 establishes that the marginal tax converges to zero but does not pin down the rate. For quantitative applications the rate of convergence matters, as it determines the practical importance of the asymptotic result for finite incomes.

Proposition 10.2 (Decay rate of the marginal tax)**.** *Under the conditions of Proposition 10.1, if additionally the right tail of $f(z)$ is Pareto with shape parameter α, then:*

$$\lim_{y\to\infty} \frac{\mu}{\theta}\,[T'(y)]^2\, y = \gamma. \tag{103}$$

In particular, the limiting elasticity of the marginal tax with respect to income is

$$\lim_{y\to\infty} \frac{d\log T'(y)}{d\log y} = -\frac{1}{2}. \tag{104}$$

Expression (103) implies that the marginal tax rate asymptotically behaves as

$$T'(y) \approx \sqrt{\frac{\gamma\theta}{\mu}} \cdot y^{-1/2}.$$

A tenfold increase in income therefore cuts the marginal tax rate by approximately two-thirds ($10^{-1/2} \approx 0.316$).

Heuristic derivation. Suppose the marginal tax rate is well approximated by a constant-elasticity function $T'(y) \propto y^{\nu}$ for large y. Then $T(y) \approx \text{const} + y^{1+\nu}/(1+\nu)$ and the worst-case distortion behaves as

$$m(z) \approx \widetilde{m} \exp\Big(-\frac{\mu}{\theta}\, y(z)^{1+\nu}\Big)\,.$$

Since $T'(y) \to 0$, the worker's FOC $y(z) = (1 - T'(y))^{1/\gamma}\, z^{(1+\gamma)/\gamma}$ gives $(1 - T')^{1/\gamma} \to 1$, so $y(z) \approx z^{(1+\gamma)/\gamma}$ (the undistorted labor supply). To evaluate the right-hand side of (101), note that as $z \to \infty$ both the numerator $1 - \widetilde{F}(z) \to 0$ and the denominator $z\,\widetilde{f}(z) \to 0$ (the worst-case density decays super-exponentially due to the $e^{-(\mu/\theta)T(y)}$ factor in $m(z)$, which overwhelms the polynomial $z\,f(z)$). This is a 0/0 indeterminate form, so we apply L'Hôpital's rule: To see this, apply L'Hôpital to $(1 - \widetilde{F}(z))/(z\,\widetilde{f}(z))$. Differentiating the numerator: $\frac{d}{dz}[1 - \widetilde{F}(z)] = -\widetilde{f}(z)$. Differentiating the denominator: $\frac{d}{dz}[z\,\widetilde{f}(z)] = \widetilde{f}(z) + z\,\widetilde{f}'(z) = \widetilde{f}(z)[1 + z\,\frac{d\log\widetilde{f}}{dz}]$. Their ratio is

$$\frac{1 - \widetilde{F}(z)}{z\,\widetilde{f}(z)} \;\to\; \frac{-1}{1 + z\,\frac{d\log\widetilde{f}}{dz}}.$$

Now expand $\frac{d\log\widetilde{f}}{dz} = \frac{d\log f}{dz} + \frac{d\log m}{dz}$. Since $m(z) = \bar{m}\exp(-\frac{\mu}{\theta}T(y(z)))$, we have $\frac{d\log m}{dz} = -\frac{\mu}{\theta}T'(y)\,y'(z)$. Substituting and multiplying through by z:

$$z\,\frac{d\log\widetilde{f}}{dz} = \frac{d\log f}{d\log z} - \frac{\mu}{\theta}\,z\,T'(y)\,y'(z).$$

Therefore

$$\lim_{z\to\infty} \frac{T'(y(z))}{1 - T'(y(z))} = (1+\gamma)\lim_{z\to\infty} \frac{1}{\frac{\mu}{\theta}\,z\,T'(y)\,y'(z) - \frac{d\log f}{d\log z} - 1}.$$

We now compute the growth rates of each term in the denominator under the ansatz $T'(y) \propto y^{\nu}$.

Growth rate of $z\,T'(y)\,y'(z)$. Since $T' \to 0$, the undistorted labor supply gives $y(z) \approx z^{(1+\gamma)/\gamma}$, so $y'(z) \approx \frac{1+\gamma}{\gamma} z^{1/\gamma}$. With $T'(y) \propto y^\nu \propto z^{(1+\gamma)\nu/\gamma}$, the product is

$$z \cdot T'(y) \cdot y'(z) \;\propto\; z \cdot z^{(1+\gamma)\nu/\gamma} \cdot z^{1/\gamma} = z^{1+(1+\gamma)\nu/\gamma+1/\gamma} = z^{(1+\gamma)(1+\nu)/\gamma}.$$

Since $\nu < 0$ but $\nu > -1$ (the tax rate decays but revenue still grows), the exponent $(1+\gamma)(1+\nu)/\gamma > 0$, so this term diverges.

Growth rate of $d\log f/d\log z$. For any Pareto or sub-exponential density, $d\log f/d\log z$ grows at most polynomially (for Pareto, it is the constant $-(\alpha+1)$). This is dominated by the exponentially growing first term.

The left-hand side. With $T'(y) \propto y^\nu$ and $T' \to 0$, $T'/(1-T') \approx T' \propto y^\nu \propto z^{(1+\gamma)\nu/\gamma}$.

Matching exponents. Since the denominator is dominated by the first term, the right-hand side behaves as $z^{-(1+\gamma)(1+\nu)/\gamma}$. The left-hand side behaves as $z^{(1+\gamma)\nu/\gamma}$. For these to be consistent:

$$\frac{1+\gamma}{\gamma}\,\nu = -\frac{1+\gamma}{\gamma}(1+\nu) \quad\Longrightarrow\quad 2\nu = -1 \quad\Longrightarrow\quad \nu = -\frac{1}{2}.$$

Remark 10.7 (Universality of the $-1/2$ elasticity). The elasticity $-1/2$ does not depend on the Pareto tail parameter α, the labor supply elasticity γ, or the robustness parameter θ. It reflects a fundamental balance: if the tax rate decayed faster (more negative ν), tax revenue would grow more slowly, the worst-case distortion would be milder, and the optimal tax formula would call for slower decay—and vice versa. The elasticity $-1/2$ is the unique fixed point of this feedback loop, identifying the saddle point of the min-max problem.

10.4.3 Derivation of the ODE for the marginal tax

The decay result can be formalized through a differential equation for $T'(y)$. The derivation requires three preliminary results. Throughout, assume the benchmark density is Pareto with shape parameter α in the right tail: $f(z) = \alpha z^{-\alpha-1}$ (normalized on $[1,\infty)$).

Lemma 10.1 (Income derivative)**.** *Under quasilinear preferences (Assumption 10.1), optimal output satisfies*

$$y'(z) = \frac{(1+\gamma)\,\frac{y(z)}{z}}{\gamma + \frac{T''(y(z))}{1-T'(y(z))}\,y(z)}. \tag{105}$$

Proof. The worker FOC gives $y(z) = (1-T'(y(z)))^{1/\gamma}\,z^{(1+\gamma)/\gamma}$. Differentiating with respect to z:

$$y'(z) = -\frac{1}{\gamma}\,(1-T')^{(1-\gamma)/\gamma}\,z^{(1+\gamma)/\gamma}\,T''(y)\,y'(z) + \frac{1+\gamma}{\gamma}\,(1-T')^{1/\gamma}\,z^{1/\gamma}.$$

Using $y(z) = (1-T')^{1/\gamma} z^{(1+\gamma)/\gamma}$ to simplify, the first term becomes $-\frac{1}{\gamma}\frac{T''}{1-T'}\,y\,y'(z)$ and the second becomes $\frac{1+\gamma}{\gamma}\frac{y}{z}$. Collecting $y'(z)$ on the left and solving gives (105). □

Lemma 10.2 (Derivative of the inverse hazard rate)**.** *Define* $\widetilde{\phi}(z) = \frac{1-\widetilde{F}(z)}{z\,\widetilde{f}(z)} = \frac{1}{1+\gamma}\frac{T'(y(z))}{1-T'(y(z))}$, *where the second equality follows from the tax formula* (101)*. Then*

$$\widetilde{\phi}'(z) = -\frac{1}{z} - \widetilde{\phi}(z)\left[\frac{1}{z} - \frac{\mu}{\theta}\,T'(y(z))\,y'(z) + \frac{f'(z)}{f(z)}\right]. \tag{106}$$

Proof. By the quotient rule,

$$\frac{d}{dz}\frac{1-\widetilde{F}}{z\widetilde{f}} = \frac{-\widetilde{f}\cdot z\widetilde{f} - (1-\widetilde{F})(\widetilde{f}+z\widetilde{f}')}{(z\widetilde{f})^2} = -\frac{1}{z} - \frac{1-\widetilde{F}}{z\widetilde{f}}\left[\frac{1}{z} + \frac{\widetilde{f}'}{\widetilde{f}}\right].$$

Since $\widetilde{f}(z) = m(z)f(z)$ with $m(z) = \bar{m}\exp(-\mu T(y(z))/\theta)$, we have

$$\frac{d}{dz}\log\widetilde{f}(z) = \frac{d\log m}{dz} + \frac{d\log f}{dz} = -\frac{\mu}{\theta}\,T'(y)\,y'(z) + \frac{f'(z)}{f(z)}.$$

Substituting yields (106). □

Proposition 10.3 (ODE for $T'(y)$)**.** *When the benchmark type distribution is Pareto with shape parameter* α*, the optimal marginal tax satisfies the differential equation*

$$T''(y) = \frac{1-T'(y)}{y}\left[2 - \frac{1+\gamma+\alpha}{1+\gamma}\,T'(y)\right]^{-1}\left[\frac{\mu}{\theta}\,[T'(y)]^2\,y - \gamma + \gamma\,\frac{1+\gamma+\alpha}{1+\gamma}\,T'(y)\right]. \tag{107}$$

Proof. Differentiate the tax formula $T'/(1-T') = (1+\gamma)\,\widetilde{\phi}(z)$ with respect to z:

$$\frac{T''(y(z))\,y'(z)}{(1-T'(y(z)))^2} = (1+\gamma)\,\widetilde{\phi}'(z). \tag{108}$$

Substitute the expression for $\widetilde{\phi}'(z)$ from Lemma 10.2:

$$\frac{T''\,y'}{(1-T')^2} = (1+\gamma)\left[-\frac{1}{z} - \widetilde{\phi}\left(\frac{1}{z} - \frac{\mu}{\theta}T'\,y' + \frac{f'}{f}\right)\right].$$

Collect terms involving $y'(z)$. Note $\widetilde{\phi} = T'/((1+\gamma)(1-T'))$:

$$\left[\frac{T''}{T'(1-T')} - \frac{\mu}{\theta}\,T'\right]y'(z)\,z = -\widetilde{\phi}^{-1} - 1 - z\frac{f'(z)}{f(z)}. \tag{109}$$

Now substitute $y'(z)$ from Lemma 10.1. After multiplying by the denominator $\gamma + T''y/(1-T')$ and using $\widetilde{\phi}^{-1} = (1+\gamma)(1-T')/T'$, the terms rearrange to

$$\left[2 - \frac{\gamma - zf'/f}{1+\gamma}\,T'\right]\frac{T''\,y}{1-T'} = \frac{\mu}{\theta}(T')^2 y - \gamma + \gamma\,\frac{\gamma - zf'/f}{1+\gamma}\,T'.$$

For the Pareto density, $-zf'(z)/f(z) = 1+\alpha$. After this substitution the equation depends on z only through $y(z)$, and since $y(z)$ is strictly monotonic we can rewrite it as an ODE in y, yielding (107). □

Remark 10.8 (Recovering Proposition 10.2 from the ODE)*.* The ODE directly implies the decay rate. Since $\lim_{y\to\infty} T'(y) = 0$, we have $\lim_{y\to\infty}(1-T') = 1$, so $T''y/(1-T') \to 0$. For this to hold, the last bracket in (107) must vanish, which requires $\lim_{y\to\infty}\frac{\mu}{\theta}[T'(y)]^2 y = \gamma$, confirming (103).

This is a first-order ODE for $T'(y)$ with terminal condition $\lim_{y\to\infty} T'(y) = 0$. The unique strictly positive solution satisfying the terminal condition is a saddle path: starting from any other initial condition, trajectories either converge to one or become negative. This saddle-path structure also suggests a stable numerical algorithm that starts from a terminal condition in the tail and integrates backward.

Remark 10.9 (Exactness of the ODE). An important feature of the ODE (107) is that it holds *exactly* for all z in the region where the benchmark density is Pareto, not merely as an asymptotic approximation. This is because the Pareto substitution $-zf'(z)/f(z) = 1+\alpha$ is an exact identity for all z in the Pareto range, not a limiting result. For distributions that are only *asymptotically* Pareto (i.e., $-zf'(z)/f(z) \to 1+\alpha$ as $z \to \infty$), the ODE becomes an approximation that grows increasingly accurate in the tail.

10.4.4 Numerical implementation

The saddle-path structure of the ODE (107) suggests a natural backward-integration algorithm. We describe the procedure in detail and illustrate its output for the baseline calibration $\alpha = 1.5$, $\gamma = 2$, $\mu = 1$.

The ODE as a dynamical system. Write $\tau(y) = T'(y)$ for the marginal tax rate. Then (107) takes the form

$$\frac{d\tau}{dy} = \underbrace{\frac{1-\tau}{y}}_{\text{scaling}} \cdot \underbrace{\frac{1}{2-\beta\tau}}_{\text{denominator}} \cdot \underbrace{\left[\frac{\mu}{\theta}\tau^2 y - \gamma + \gamma\beta\tau\right]}_{\equiv B(\tau,y)\ \text{(bracket)}}, \tag{110}$$

where $\beta = (1+\gamma+\alpha)/(1+\gamma)$. This is a first-order non-autonomous ODE in τ with y as the independent variable (non-autonomous because the right-hand side depends explicitly on y). Its qualitative behavior is governed by:

- The **isocline** $B(\tau, y) = 0$, on which $d\tau/dy = 0$. Below the isocline ($B < 0$) the tax rate is decreasing; above ($B > 0$) it is increasing.
- The **terminal condition** $\tau(y) \to 0$ as $y \to \infty$, which selects the saddle path.

Why backward integration is stable. Consider a small perturbation ε to the saddle-path value at some y_0. Integrating *forward* (increasing y), the linearized dynamics around the saddle path amplify ε: trajectories starting above the saddle diverge toward $\tau = 1$, while those starting below diverge toward $\tau < 0$ (see the phase diagram in Figure 4). This is precisely the *instability* that makes the saddle path unique.

Reversing the direction—integrating from large y toward small y—inverts the stability. The saddle path becomes the *attracting* trajectory: perturbations to the terminal condition shrink as integration proceeds backward. This is the key insight: by starting far in the tail where we have a sharp asymptotic approximation and integrating toward lower incomes, numerical errors are damped at every step.

Discretization. We use a logarithmically spaced grid, which concentrates resolution where τ varies most rapidly (moderate incomes) while efficiently covering the long tail:

$$y_i = \exp\Big(\log\bar{y} - (i-1)\,\Delta\Big), \quad i = 1, \ldots, N, \qquad \Delta = \frac{\log\bar{y} - \log\underline{y}}{N-1},$$

so $y_1 = \bar{y}$ (the tail) and $y_N = \underline{y}$ (the lower boundary). In our implementation we use $\bar{y} = 10^7$, $\underline{y} = 0.5$, and $N = 500{,}000$ grid points.

RK4 scheme. At each step we advance from (y_i, τ_i) to (y_{i+1}, τ_{i+1}) with step size $h_i = y_{i+1} - y_i < 0$ (backward):

$$\begin{aligned} k_1 &= f(y_i,\ \tau_i), \\ k_2 &= f\big(y_i + \tfrac{h_i}{2},\ \tau_i + \tfrac{h_i}{2}\, k_1\big)\,, \\ k_3 &= f\big(y_i + \tfrac{h_i}{2},\ \tau_i + \tfrac{h_i}{2}\, k_2\big)\,, \\ k_4 &= f(y_i + h_i,\ \tau_i + h_i\, k_3), \end{aligned}$$

$$\tau_{i+1} = \tau_i + \frac{h_i}{6}\,(k_1 + 2k_2 + 2k_3 + k_4)\,,$$

where $f(y, \tau) = d\tau/dy$ is the right-hand side of (110). The log-spacing means $|h_i|$ is small when y_i is small (where the solution changes rapidly) and large when y_i is large (where τ is nearly flat), yielding roughly uniform accuracy across the domain.

Initial condition from asymptotics. The terminal condition $\tau \to 0$ is imposed approximately. From Proposition 10.2, $\tau(y)^2 y \to \gamma\theta/\mu$, so we initialize:

$$\tau_1 = \tau(\bar{y}) = \sqrt{\frac{\gamma\theta}{\mu\bar{y}}}.$$

For $\bar{y} = 10^7$ and $\theta = 1$, this gives $\tau_1 \approx 4.5 \times 10^{-4}$—deep enough in the tail that higher-order corrections are negligible. We verify this *a posteriori* by checking $\tau(y)^2 y$ at an interior point (e.g. $y = 10^5$); the results in the table below show agreement with $\gamma\theta/\mu$ to within a few percent for moderate θ (the approximation is less tight for large θ, where the asymptotic regime sets in at higher incomes).

θ	$T'(10^2)$	$T'(10^3)$	$T'(10^4)$	$[T']^2 y\big\vert_{y=10^5}$	$\gamma\theta/\mu$
0.5	0.091	0.031	0.010	0.99	1.0
1.0	0.123	0.043	0.014	1.98	2.0
2.0	0.165	0.059	0.020	3.95	4.0
5.0	0.236	0.091	0.031	9.80	10.0
10.0	0.299	0.123	0.043	19.4	20.0
100.0	0.514	0.299	0.123	183.0	200.0

Results. Figure 3 displays the saddle-path solution $T'(y)$ for several values of the robustness parameter θ. For all finite θ, the marginal tax rate decays monotonically to zero, in contrast to the constant Diamond–Saez rate $T' = (1+\gamma)/(1+\gamma+\alpha) \approx 0.667$ obtained in the non-robust limit $\theta = \infty$. Stronger robustness concerns (smaller θ) pull tax rates down uniformly. The dotted gray curve shows the asymptotic envelope $\sqrt{\gamma/y}$ for $\theta = 1$, which the numerical solution tracks closely in the tail.

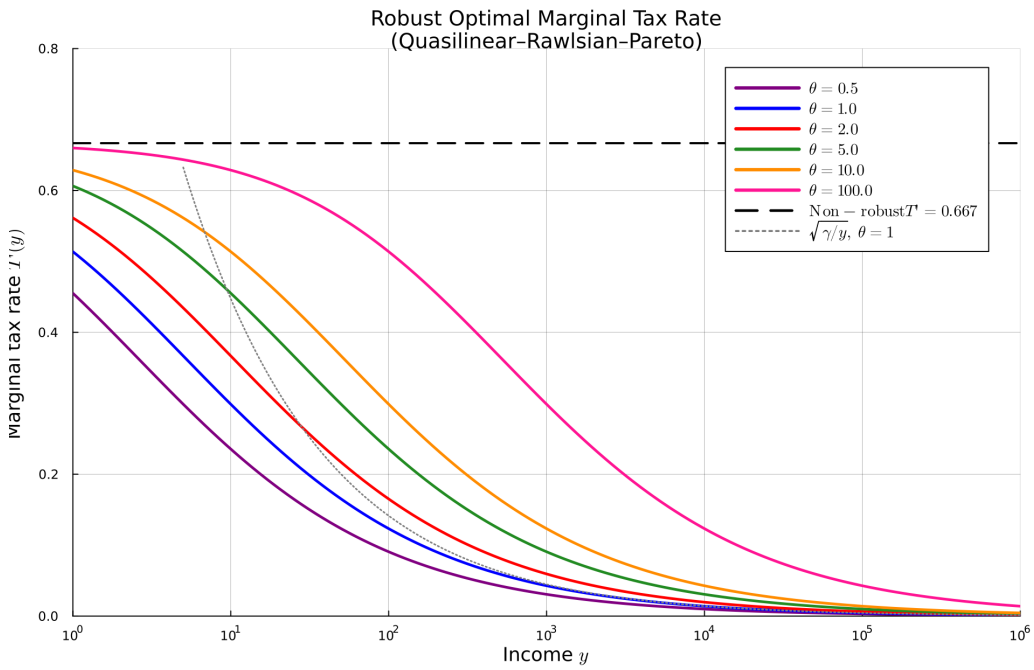

Figure 3: Robust optimal marginal tax rate $T'(y)$ for the quasilinear–Rawlsian–Pareto case. Parameters: $\alpha = 1.5$, $\gamma = 2$, $\mu = 1$. The dashed line is the non-robust Diamond–Saez top rate.

Figure 4 shows the phase diagram for $\theta = 1$. The magenta curve is the saddle path—the unique trajectory satisfying $T'(y) \to 0$. Gray curves are off-saddle trajectories initialized at perturbations $T'(y_0) = T'_{\text{saddle}}(y_0) \pm \delta$ for various δ: those starting above the saddle path diverge toward $T' = 1$, while those starting below eventually become negative. The dashed curve is the $T'' = 0$ isocline, obtained by setting the last bracket in (107) to zero:

$$\frac{\mu}{\theta}\,[T']^2\,y - \gamma + \gamma\,\frac{1+\gamma+\alpha}{1+\gamma}\,T' = 0 \quad \Longrightarrow \quad T'_{\text{iso}}(y) = \frac{-\gamma\beta + \sqrt{\gamma^2\beta^2 + 4\gamma\mu y/\theta}}{2\mu y/\theta},$$

where $\beta = (1+\gamma+\alpha)/(1+\gamma)$. For large y this isocline decays as $\sqrt{\gamma\theta/(\mu y)}$, matching the saddle path. For small y it approaches $1/\beta$, the Diamond–Saez rate. The saddle path lies just below the isocline (where $T'' < 0$), confirming that $T'(y)$ is strictly decreasing.

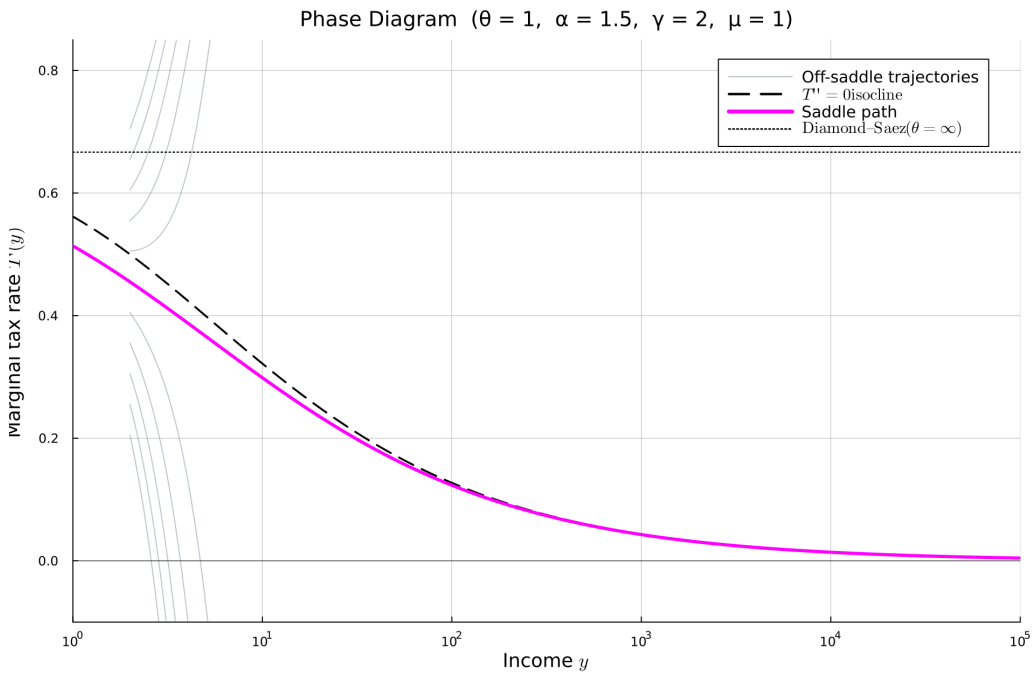

Figure 4: Phase diagram for $\theta = 1$, $\alpha = 1.5$, $\gamma = 2$, $\mu = 1$. The saddle path (magenta) is the unique trajectory with $T'(y) \to 0$. Off-saddle trajectories (gray) diverge.

Remark 10.10 (Forward vs. backward integration: comparing the two ODE approaches). It is instructive to contrast the numerical strategies for the non-robust ODE (Section 9.3) and the robust ODE derived here.

Non-robust case (Section 9.3): forward shooting. The non-robust ODE system (85)–(91) is a two-state system (Q_M, U) in the type z, supplemented by an implicit equation for labor supply h at each evaluation point. The system has two known boundary conditions—$Q_M(z_0) = 0$ at the bottom and $Q(\bar{z})\,\bar{z}\,f(\bar{z}) = 0$ at the top—and two unknown parameters (ζ, the resource multiplier, and $n_1 = n(z_0)$, the labor supply at the bottom). The solution proceeds *forward*, integrating from z_0 to $\bar{z}$:

(i) Guess (μ, n_1).

(ii) Integrate the 2-state ODE forward via RK4 from $z = z_0$ to $z = \bar{z}$, recovering h implicitly at each evaluation via Newton on the constraint (93).

(iii) Check two residuals: transversality ($Q(\bar{z})\,\bar{z}\,f(\bar{z}) = 0$) and feasibility ($\int (y - c)\,dF = G$).

(iv) Update (μ, n_1) via 2D Newton with autodiff Jacobians; repeat.

This is a *boundary value problem* solved by shooting: the initial conditions at z_0 are not fully known, so we guess and iterate. Forward integration is natural here because the ODE is *stable* in the forward direction—the costate $\lambda(z)$ is a smooth integral from z to $\bar{z}$, and the singularity-free Q_M formulation keeps the system well-conditioned.

Robust case: backward integration from the tail. The robust ODE (107) is fundamentally different in structure. It is a *single* first-order equation for $\tau = T'(y)$ as a function of income y (not type z), and the Pareto density assumption makes it depend only on (τ, y) (no separate dependence on z). There is only one boundary condition: the terminal condition $\tau(y) \to 0$ as $y \to \infty$. This means:

- There is *no shooting*: the asymptotic decay rate (Proposition 10.2) provides the initial condition $\tau(\bar{y}) = \sqrt{\gamma\theta/(\mu\bar{y})}$ directly. No unknown parameters need to be guessed or iterated over.
- Integration proceeds *backward* (from $\bar{y}$ toward $\underline{y}$) because the saddle path is *unstable* in the forward direction. Forward integration from any starting point would amplify small errors exponentially: trajectories above the saddle path shoot up to $\tau = 1$, and those below become negative (Figure 4). Backward integration reverses the stability, turning the saddle path into an attractor—errors shrink at every step.
- No auxiliary equations are needed: there is no implicit h-solve, no resource constraint to satisfy, and no shooting loop. The entire computation is a single backward pass of RK4 on a scalar ODE.

11 The Perturbation Approach for Robust Taxation

Section 7 derived the optimal tax formula via the Saez perturbation. Section 10 derived the robust tax formula via the Hamiltonian. This section re-derives the robust formula using the perturbation approach.

The section is organized in two parts:

Part I. The formula (§11.1–11.7). Assuming T^* exists, we derive the robust Saez formula in income space. The derivation uses only nature's unique best response, the envelope theorem, and the standard Saez perturbation. It holds for any preferences satisfying Assumptions 1.1–1.2. No minimax theorem is needed.

Part II. The quasilinear isoelastic case (§11.8–§11.9). We specialize to $U(c, n) = c - n^{1+\gamma}/(1+\gamma)$, derive the ABC formula in type space, and prove that the optimal allocation exists and is unique.

11.1 The government's value function

The robust planner solves

$$\max_T \; V(T), \qquad V(T) = \min_{\substack{m>0 \\ \mathbb{E}[m]=1}} \mathcal{L}(T, m), \tag{111}$$

where the Lagrangian is

$$\mathcal{L}(T, m) = \int_{\mathcal{Z}} \big[\psi(z)\,\mathcal{U}(z;T) + \mu\,T(y(z;T))\big]\, m(z)\, f(z)\, dz + \theta \int_{\mathcal{Z}} m(z) \log m(z)\, f(z)\, dz. \tag{112}$$

Here $y(z;T)$ is the earnings of type z (from the household FOC under T), $\mathcal{U}(z;T) = U(y - T(y),\, y/z)$ is the resulting indirect utility, $\mu > 0$ is the marginal value of public funds, and $\theta > 0$ is the robustness parameter. Both y and $\mathcal{U}$ depend on T through the household's optimization; we suppress this dependence and write $y(z)$, $\mathcal{U}(z)$ when T is held fixed.

11.2 Nature's best response: the exponential twist

Proposition 11.1 (Nature's unique best response)**.** *For any tax schedule T and $\theta > 0$, the problem $\min_{m>0,\,\mathbb{E}[m]=1} \mathcal{L}(T, m)$ has a unique solution:*

$$\boxed{m^*(z;T) = \bar{m}\,\exp\!\left(-\frac{1}{\theta}\big[\psi(z)\,\mathcal{U}(z) + \mu\,T(y(z))\big]\right),} \tag{113}$$

where $\bar{m}$ ensures $\mathbb{E}[m^*] = 1$.

Proof. Define $g(z) = \psi(z)\mathcal{U}(z) + \mu\,T(y(z))$. For fixed T, $\mathcal{L}(T, m) = \int g\,m\,f\,dz + \theta\int m\log m\,f\,dz$. The first term is linear in m; the second is strictly convex ($t\log t$ is strictly convex for $t > 0$). Therefore $\mathcal{L}(T, \cdot)$ is strictly convex on $\{m > 0 : \mathbb{E}[m] = 1\}$ and has at most one minimizer.

The Lagrangian for the constrained minimization (with multiplier χ for $\mathbb{E}[m] = 1$) gives the pointwise FOC $g(z) + \theta(1 + \log m(z)) - \chi = 0$. Solving: $\log m(z) = -g(z)/\theta + (\chi-\theta)/\theta$. Exponentiating gives (113) with $\bar{m} = \exp((\chi-\theta)/\theta)$, determined by $\mathbb{E}[m^*] = 1$. The strict convexity of the objective confirms this is the unique global minimum. □

Remark 11.1 (No assumptions on preferences). Proposition 11.1 holds for any utility function. Only strict convexity of $t\log t$ and $\theta > 0$ are used. On unbounded support ($\bar{z} = \infty$), the formula requires $\int e^{-g(z)/\theta} f(z)\,dz < \infty$; this holds whenever g is bounded below, which is ensured by the allocation bounds on bounded support.

11.3 The envelope theorem for the value function

Proposition 11.2 (Envelope theorem for the value function)**.** *Define the perturbation path $T_\alpha = T + \alpha\,\delta T$ for $\alpha \in \mathbb{R}$, and let $y_\alpha(z)$ solve the household's problem under T_α. Write*

$$g_\alpha(z) = \psi(z)\,\mathcal{U}_\alpha(z) + \mu\,T_\alpha(y_\alpha(z)) \tag{114}$$

for the exponential-twist argument, and $V(\alpha) = V(T_\alpha)$ for the government's value along the path. Suppose:

(i) for f-a.e. $z \in \mathcal{Z}$, $\alpha \mapsto g_\alpha(z)$ is differentiable on $(-\alpha_0, \alpha_0)$;

(ii) $Z(\alpha) = \int_{\mathcal{Z}} e^{-g_\alpha(z)/\theta}\,f(z)\,dz < \infty$ for all $|\alpha| < \alpha_0$;

(iii) there exists $H \in L^1(f\,dz)$ such that $|\partial_\alpha g_\alpha(z)|\,e^{-g_\alpha(z)/\theta} \le H(z)$ for all $|\alpha| < \alpha_0$ and f-a.e. z.

Then V is differentiable at $\alpha = 0$ with

$$V'(0) = D_T\mathcal{L}(T,\, m^*(T))[\delta T]. \tag{115}$$

That is, the derivative of the government's value equals the derivative of the Lagrangian holding m fixed at nature's optimal response.

Proof. By Proposition 11.1, the unique minimizer has the explicit exponential form, so substituting back gives the entropic dual:

$$V(\alpha) = -\theta\log Z(\alpha), \qquad Z(\alpha) = \int_{\mathcal{Z}} e^{-g_\alpha(z)/\theta} f(z)\,dz.$$

Hypothesis (ii) ensures $0 < Z(\alpha) < \infty$.

Step 1 (Differentiate Z under the integral). For f-a.e. z, the difference quotient $(e^{-g_\alpha/\theta} - e^{-g_0/\theta})/\alpha$ converges pointwise to $-\frac{1}{\theta}\,\partial_\alpha g_0(z)\,e^{-g_0(z)/\theta}$ as $\alpha \to 0$ (hypothesis (i)). By the mean value theorem, it equals $\partial_\alpha e^{-g_{\tilde\alpha}/\theta}$ for some $\tilde\alpha$ between 0 and α, which is dominated by $H(z)/\theta$ (hypothesis (iii)). By DCT (Lemma E.3):

$$Z'(0) = -\frac{1}{\theta}\int_{\mathcal{Z}} \partial_\alpha g_0(z)\,e^{-g_0(z)/\theta}\,f(z)\,dz.$$

Step 2 (Conclude). Since $V(\alpha) = -\theta \log Z(\alpha)$ and $Z(0) > 0$:

$$\begin{aligned} V'(0) &= -\theta\,\frac{Z'(0)}{Z(0)} = \frac{\int \partial_\alpha g_0(z)\,e^{-g_0(z)/\theta}\,f(z)\,dz}{\int e^{-g_0(z)/\theta}\,f(z)\,dz} \\ &= \int \partial_\alpha g_0(z)\,m^*(0;z)\,f(z)\,dz = D_T\mathcal{L}(T, m^*(T))[\delta T]. \end{aligned}$$

The last equality is the definition of $D_T\mathcal{L}(T, m_0^*)[\delta T]$: at fixed $m = m_0^*$, the entropy term does not depend on α, so $D_T\mathcal{L} = \int \partial_\alpha g_0\, m_0^*\, f\, dz$. □

Remark 11.2. The proof uses only: the closed-form m^* (from strict convexity of $t\log t$), the entropic dual formula, and DCT. No saddle-point, minimax equality, compactness of the constraint set, or differentiation of the argmin map $m^*(\alpha)$ is invoked.

Remark 11.3 (Verification of hypotheses (i)–(iii)). Hypotheses (i)–(iii) are traced back to primitive conditions on U, T, δT, ψ, and $\mathcal{Z}$ in Appendix E.7. On bounded support with $U \in C^2$, admissibility ($0 \le T' < 1$), bounded ψ, and bounded behavioral derivatives, all three hold with H constant. On unbounded support, (ii) and (iii) are genuine tail integrability requirements. For the Saez band perturbation, differentiability (i) can fail at the measure-zero set of types with base earnings exactly at a kink; this does not affect the integral (see Appendix E.7).

11.4 The worst-case earnings distribution

Definition 11.1 (Worst-case distribution)**.** Given T and $m^*(T)$ from (113), define $\widetilde{f}(z) = m^*(z;T)\,f(z)$, $\widetilde{F}(z) = \int_{z_0}^{z} \widetilde{f}\,d\zeta$. The worst-case earnings density and cdf are $\widetilde{h}(y) = \widetilde{f}(z(y))/y'(z(y))$ and $\widetilde{H}(y) = \widetilde{F}(z(y))$.

11.5 The Saez perturbation

By Proposition 11.2, the first-order effect of δT on V is $\partial\mathcal{L}/\partial\epsilon$ evaluated at $m = m^*(T)$. With m fixed, the Lagrangian has the same form as in Section 7.4 with f replaced by $\widetilde{f}$.

Fix $y^* > 0$, raise the marginal rate by $d\tau > 0$ in $[y^*, y^* + \varepsilon]$ as in Section 7.3. The household envelope theorem gives $d\mathcal{U}/d\epsilon = -U_c\,\delta T(y)$, so term (a) is $\mu\,\delta T(y)(1 - g(z))$ with $g(z) = \psi U_c/\mu$. The Slutsky decomposition (Section 7.4) yields the adjusted weight $\hat g(z) = g(z) - \frac{T'}{1-T'}\eta(z)$ and the redefined effects:

$$d\widetilde{M} = \mu\,\delta T(y)\,(1 - \hat g(z)), \tag{116}$$

$$d\widetilde{B} = \mu\,T'(y)\,\partial y^c/\partial\epsilon. \tag{117}$$

Remark 11.4 (Welfare weights are independent of m). $g(z) = \psi U_c/\mu$ is a per-capita quantity: it measures the social value of a dollar to type z. The distortion m affects the integration weight $\widetilde{f}$, not the per-capita value.

Aggregation. The mechanical and behavioral effects, aggregated under $\widetilde{f}$, are:

$$d\widetilde{M} = \varepsilon\, d\tau \int_{y^*}^{\infty} (1-\hat{g})\, \widetilde{h}\, dy + O(\varepsilon^2\, d\tau), \tag{118}$$

$$d\widetilde{B} = -\varepsilon\, d\tau\, \frac{T'(y^*)}{1-T'(y^*)}\, \zeta^c(y^*)\, y^*\, \widetilde{h}(y^*) + O(\varepsilon^2\, d\tau), \tag{119}$$

by the same calculation as in Sections 7.5–7.6, with h replaced by $\widetilde{h}$ throughout. (The per-capita responses δT, dy^c are unchanged; only the density of agents at each income level changes.)

11.6 The robust optimal tax formula

Setting $d\widetilde{M} + d\widetilde{B} = 0$:

Proposition 11.3 (Robust Saez formula)**.** *Under Assumptions 1.1–1.2:*

$$\boxed{\frac{T'(y^*)}{1-T'(y^*)} = \frac{1}{\zeta^c(y^*)} \cdot \frac{1}{y^*\, \widetilde{h}(y^*)} \int_{y^*}^{\infty} (1-\hat{g}(y))\, \widetilde{h}(y)\, dy.} \tag{120}$$

Remark 11.5 (Generality). This holds for any preferences satisfying the regularity conditions. The three ingredients: (a) unique m^* (entropy convexity), (b) envelope theorem (Proposition 11.2), (c) Saez perturbation (Section 7). No minimax theorem is used.

11.7 Three-factor decomposition

Formula (120) decomposes as in Section 7.7:

$$\frac{T'(y^*)}{1-T'(y^*)} = \underbrace{\frac{1}{\zeta^c(y^*)}}_{\text{inverse ETI}} \cdot \underbrace{\frac{1-\widetilde{H}(y^*)}{y^*\, \widetilde{h}(y^*)}}_{\substack{\text{inverse hazard}\\ \text{(worst case)}}} \cdot \underbrace{\frac{1}{1-\widetilde{H}(y^*)} \int_{y^*}^{\infty} (1-\hat{g})\, \widetilde{h}\, dy}_{\substack{\text{avg. net welfare}\\ \text{cost above } y^*}}.$$

11.8 The ABC formula (quasilinear isoelastic)

We specialize to $U(c,n) = c - n^{1+\gamma}/(1+\gamma)$ (Assumption 10.1) for the rest of this section. Under these preferences: $u(c) = c$, $v(n) = n^{1+\gamma}/(1+\gamma)$, income effects vanish ($\eta = 0$, $\hat{g} = g = \psi/\mu$, $U_c \equiv 1$), and $\zeta^c = 1/\gamma$.

The income-space Saez formula (120) is the perturbation result. The following type-space equivalent is obtained via the Hamiltonian formula (Section 10.3), not by a separate perturbation argument.

Corollary 11.1 (Robust ABC formula)**.** *Under quasilinear isoelastic preferences:*

$$\frac{T'(y(z))}{1-T'(y(z))} = (1+\gamma) \cdot \frac{\widetilde{\Psi}(z) - \widetilde{F}(z)}{1-\widetilde{F}(z)} \cdot \frac{1-\widetilde{F}(z)}{z\, \widetilde{f}(z)}, \tag{121}$$

where $\widetilde{F}(z) = \int_{z_0}^{z} \widetilde{f}\, d\zeta$ *and* $\widetilde{\Psi}$ *is the* ***normalized*** *welfare cdf*

$$\widetilde{\Psi}(z) = \frac{\int_{z_0}^{z} \psi(\zeta)\, \widetilde{f}(\zeta)\, d\zeta}{\int_{z_0}^{\bar{z}} \psi(\xi)\, \widetilde{f}(\xi)\, d\xi}. \tag{122}$$

This normalization absorbs μ: the budget-balance condition (from a lump-sum perturbation at $y^* = 0$ in (120)) gives $\mu = \int \psi \widetilde{f}\, dz$, so dividing by μ yields $\widetilde{\Psi}(\bar{z}) = 1$.

Proof. The income-space formula (120) and the type-space formula (121) are related by the same change of variables as in the non-robust case (Section 7.8 of the main notes, with $f \to \widetilde{f}$). The Hamiltonian formula (Section 10.3) provides a direct derivation in type space. □

11.9 Existence and uniqueness for the fixed-μ robust problem

The formulas above characterize T^* *if it exists.* We now prove existence and uniqueness of the optimal allocation for the fixed-μ Lagrangian problem. The connection to the primal problem (welfare maximization subject to a government budget constraint) requires a separate choice of μ so the budget binds; this step is not carried out here (Remark 11.10).

Assumption 11.1 (Bounded support)**.** $\mathcal{Z} = [z_0, \bar{z}]$ with $0 < z_0 < \bar{z} < \infty$.

Assumption 11.2 (Admissibility)**.** Tax schedules satisfy $0 \le T'(y) \le 1$ (nonnegative marginal rates, no confiscatory taxation).

Remark 11.6. Assumption 11.1 is needed only for compactness. The lower bound $T' \ge 0$ in Assumption 11.2 is needed for the output bound (Step 3); without it, marginal subsidies ($T' < 0$) allow unbounded labor supply. The formulas (120)–(121) and the exponential twist (113) are valid without these assumptions.

11.9.1 Step 1: From tax schedules to allocations

The government's problem is $\max_T V(T)$ with $V(T) = \min_{m>0,\, \mathbb{E}[m]=1} \mathcal{L}(T, m)$. By the **taxation principle** (Lemma E.5), maximizing over implementable tax schedules is equivalent to maximizing over IC allocations. Under single crossing, IC is equivalent to the envelope condition plus monotonicity of y (Lemma E.6). Letting $\mathcal{A}_{IC}$ denote the set of IC allocations:

$$\max_T V(T) = \max_{a \in \mathcal{A}_{IC}} W(a), \qquad W(a) \equiv \min_{m>0,\, \mathbb{E}[m]=1} \mathcal{L}(a, m), \tag{123}$$

where $W(a)$ is the government's reduced value as a function of the allocation a. All subsequent steps work with W.

11.9.2 Step 2: The q-parameterization

Define

$$q(z) = y(z)^{1+\gamma}. \tag{124}$$

Since $t \mapsto t^{1+\gamma}$ is strictly increasing on $\mathbb{R}_+$:

$$y(z) \text{ nondecreasing} \quad \Longleftrightarrow \quad q(z) \text{ nondecreasing}. \tag{125}$$

This is the key property: the IC monotonicity constraint becomes a *linear order constraint* in q.

In terms of q:

$$y(z) = q(z)^{1/(1+\gamma)}, \tag{126}$$

$$\mathcal{U}(z) = u_0 + \int_{z_0}^{z} \frac{q(s)}{s^{2+\gamma}}\, ds, \tag{127}$$

$$c(z) = \mathcal{U}(z) + \frac{q(z)}{(1+\gamma)\, z^{1+\gamma}}, \tag{128}$$

$$T(y(z)) = q(z)^{1/(1+\gamma)} - u_0 - \int_{z_0}^{z} \frac{q(s)}{s^{2+\gamma}}\, ds - \frac{q(z)}{(1+\gamma)\, z^{1+\gamma}}. \tag{129}$$

The envelope ODE in q follows from $\mathcal{U}'(z) = v'(n)\, n/z = n^{1+\gamma}/z = (y/z)^{1+\gamma}/z = q/z^{2+\gamma}$. In (u_0, q)-space: $\mathcal{U}$ is **affine**, c is **affine**, the linear part of T is **affine**, and the only nonlinear piece is $\mu\, q^{1/(1+\gamma)}$ (strictly concave).

The objective in (u_0, q)**.** Substituting (126)–(129) into $g_a(z) = \psi(z)\,\mathcal{U}(z) + \mu\, T(y(z))$:

$$g_a(z) = (\psi(z) - \mu)\left[u_0 + \int_{z_0}^{z} \frac{q(s)}{s^{2+\gamma}}\, ds\right] + \mu\left[q(z)^{1/(1+\gamma)} - \frac{q(z)}{(1+\gamma)\, z^{1+\gamma}}\right]. \tag{130}$$

The government's reduced value is $W(u_0, q) = -\theta \log \int_{\mathcal{Z}} e^{-g_a(z)/\theta}\, f(z)\, dz$ (the entropic dual, verified in §11.9 below). The first bracket is affine in (u_0, q); the second contains the only nonlinearity, $\mu\, q^{1/(1+\gamma)}$.

11.9.3 Step 3: The feasible set

Admissibility ($0 \le T' \le 1$) gives $n^\gamma = z(1 - T') \le z$, hence $y(z) \le z^{(1+\gamma)/\gamma}$ and

$$0 \le q(z) \le \bar{q} = \bar{z}^{\,(1+\gamma)^2/\gamma}. \tag{131}$$

Definition 11.2 (Output set and allocation set)**.**

$$\mathcal{Q} = \big\{q : \mathcal{Z} \to [0, \bar{q}] \;:\; q \text{ nondecreasing}\big\}.$$

For $M > 0$, define $\mathcal{A}_q(M) = [-M, M] \times \mathcal{Q}$ and $\mathcal{A}_q = \mathbb{R} \times \mathcal{Q}$.

Lemma 11.1 (Equivalence of parameterizations)**.** *Under quasilinear isoelastic preferences and Assumptions 11.1–11.2:*

(i) The map $(u_0, q) \mapsto (c, y)$ defined by (126)–(128) is a bijection from $\mathcal{A}_q$ onto the set of admissible IC allocations $\mathcal{A}_{IC}^{\text{adm}}$.

(ii) Under this bijection, $W(u_0, q) = W(c, y)$.

Therefore $\max_{\mathcal{A}_q} W = \max_{\mathcal{A}_{IC}^{\text{adm}}} W$.

Proof. Forward: $(u_0, q) \in \mathcal{A}_q \Rightarrow$ admissible IC allocation. Given (u_0, q) with q nondecreasing and $q \le \bar{q}$: $y = q^{1/(1+\gamma)}$ is nondecreasing (monotonicity); $\mathcal{U}$ satisfies the envelope condition by construction of (127); the allocation is IC by Lemma E.6. For admissibility: $T'(y(z)) = 1 - (y(z)/z)^\gamma/z$ (Proposition 11.6). The bound $T' \le 1$ is immediate ($y/z \ge 0$). For $T' \ge 0$: $q \le \bar{q} = \bar{z}^{(1+\gamma)^2/\gamma}$ gives $y \le z^{(1+\gamma)/\gamma}$, hence $(y/z)^\gamma \le z$ and $T' \ge 0$.

Reverse: admissible IC allocation $\Rightarrow (u_0, q) \in \mathcal{A}_q$. Given an admissible IC allocation (c, y): set $u_0 = \mathcal{U}(z_0)$, $q = y^{1+\gamma}$. Monotonicity of y gives q nondecreasing. Admissibility $(0 \le T' \le 1)$ gives $n^\gamma = z(1 - T') \le z$ hence $q \le \bar{q}$. Equations (126)–(128) reproduce (c, y).

Part (ii). W depends on the allocation only through $\mathcal{U}$, $T(y)$, and y (via the Lagrangian $\mathcal{L}$), all of which are determined by (u_0, q) through (126)–(129). □

Remark 11.7 (Admissibility). The existence and uniqueness results below are for the admissible problem $\max_{\mathcal{A}_q} W$, which restricts to $0 \le T' \le 1$. The upper bound $T' \le 1$ is never binding at an interior optimum (confiscatory rates yield zero revenue at the margin). The lower bound $T' \ge 0$ is known to hold at the optimum for quasilinear isoelastic preferences with decreasing welfare weights, which is the case studied here. Formally verifying that the solution of the FOC satisfies these bounds is a separate step that we do not carry out.

Lemma 11.2 (Compactness of $\mathcal{Q}$)**.** *$\mathcal{Q}$ is convex and sequentially compact in the pointwise topology. Hence $\mathcal{A}_q(M)$ is sequentially compact for every $M < \infty$.*

Proof. Convexity is immediate (convex combinations of nondecreasing functions are nondecreasing). Sequential compactness of $\mathcal{Q}$: given $\{q_n\} \subset \mathcal{Q}$, Helly's theorem (Lemma E.2) yields a subsequence $q_{n_k}(z) \to q(z)$ pointwise for every $z \in \mathcal{Z}$, with q nondecreasing and $0 \le q \le \bar{q}$. Hence $q \in \mathcal{Q}$. For $\mathcal{A}_q(M) = [-M, M] \times \mathcal{Q}$: given $(u_{0,n}, q_n)$, first extract a subsequence with $q_{n_k} \to q$ in $\mathcal{Q}$ (Helly), then a further subsequence with $u_{0,n_{k_j}} \to u_0 \in [-M, M]$ (Bolzano–Weierstrass). □

Remark 11.8 (Right-continuous representatives). The pointwise limit of nondecreasing functions is nondecreasing but need not be right-continuous. Since $W(a)$ depends on q only through Lebesgue integrals (equations (127)–(129)), modifying q on a countable set does not change W. After finding the optimal q^*, we may therefore replace it by its right-continuous modification $q^{*+}(z) = \lim_{s \downarrow z} q^*(s)$ without affecting the value $W(u_0^*, q^*)$ or the on-path tax schedule.

The output schedule q lives in the compact set $\mathcal{Q}$, but $u_0 \in \mathbb{R}$ is unbounded. To apply Weierstrass, we need to show that the optimal u_0^* lies in a bounded interval. We establish this through a *coercivity argument*: $W(u_0, q) \to -\infty$ as $|u_0| \to \infty$, so extreme values of u_0 cannot be optimal.

To state the argument precisely, we first decompose the government's objective. For any allocation $a = (u_0, q)$, define

$$g_a(z) = \psi(z)\,\mathcal{U}_a(z) + \mu\, T_a(y_a(z)), \tag{132}$$

where $\mathcal{U}_a$ and T_a are given by (127)–(129). Substituting, g_a decomposes as

$$g_a(z) = \underbrace{r_q(z)}_{\text{depends on } q \text{ only}} + (\psi(z) - \mu)\, u_0, \tag{133}$$

where

$$r_q(z) = (\psi(z) - \mu) \int_{z_0}^{z} \frac{q(s)}{s^{2+\gamma}}\, ds + \mu \left[q(z)^{1/(1+\gamma)} - \frac{q(z)}{(1+\gamma)\, z^{1+\gamma}} \right].$$

The key observation: g_a is *affine in* u_0 with slope $\psi(z) - \mu$, which varies across types.

The government's reduced value function is

$$W(a) = \inf_{m>0,\,\mathbb{E}[m]=1} \mathcal{L}(a,m) = -\theta \log\left(\int_{\mathcal{Z}} e^{-g_a(z)/\theta} f(z)\,dz\right). \tag{134}$$

The second equality is the **entropic dual**: substituting nature's optimal response $m^* = e^{-g_a/\theta}/\int e^{-g_a/\theta} f$ into $\mathcal{L}$ collapses the Lagrangian to $-\theta \log Z$, where $Z = \int e^{-g_a/\theta} f$.

Verification of the entropic dual

Nature's unique best response (Proposition 11.1) is

$$m_a^*(z) = \frac{\exp(-g_a(z)/\theta)}{\int_{\mathcal{Z}} \exp(-g_a(\zeta)/\theta)\, f(\zeta)\,d\zeta}. \tag{135}$$

Verification of (134). Set $Z = \int e^{-g_a/\theta} f$ and substitute $m_a^* = e^{-g_a/\theta}/Z$:

$$\begin{aligned}
\mathcal{L}(a, m_a^*) &= \int g_a \frac{e^{-g_a/\theta}}{Z} f + \theta \int \frac{e^{-g_a/\theta}}{Z} \log \frac{e^{-g_a/\theta}}{Z} f \\
&= \frac{1}{Z}\int g_a\, e^{-g_a/\theta} f + \frac{\theta}{Z}\int e^{-g_a/\theta}(-g_a/\theta - \log Z)\, f \\
&= \frac{1}{Z}\int g_a\, e^{-g_a/\theta} f - \frac{1}{Z}\int g_a\, e^{-g_a/\theta} f - \theta \log Z \\
&= -\theta \log Z.
\end{aligned}$$

□

Lemma 11.3 (Uniform bound on r_q)**.** *$|r_q(z)| \le C_r$ for all $q \in \mathcal{Q}$ and $z \in \mathcal{Z}$, where*

$$C_r = (\|\psi\|_\infty + \mu)\,\bar{q} \int_{z_0}^{\bar{z}} \frac{ds}{s^{2+\gamma}} + \mu\,\bar{q}^{1/(1+\gamma)} + \frac{\mu\,\bar{q}}{(1+\gamma)\,z_0^{1+\gamma}}. \tag{136}$$

Proof. Each term in r_q is bounded using $0 \le q \le \bar{q}$ and $|\psi - \mu| \le \|\psi\|_\infty + \mu$. □

Assumption 11.3 (Coercive welfare weights)**.** $\operatorname{ess\,inf}_{z\in\mathcal{Z}} \psi(z) < \mu < \operatorname{ess\,sup}_{z\in\mathcal{Z}} \psi(z)$.

This says the multiplier μ lies strictly between the lowest and highest welfare weights. The economic content: there exist types z with $\psi(z) > \mu$ (society values their utility more than the fiscal cost) and types with $\psi(z) < \mu$ (the reverse). This is the generic case for any redistributive planner.

Lemma 11.4 (Coercivity in u_0)**.** *Under Assumption 11.3:* $\sup_{q\in\mathcal{Q}} W(u_0, q) \to -\infty$ *as* $u_0 \to \pm\infty$.

The intuition: u_0 is the base utility level. When the government sets u_0 very high, nature responds by putting more weight on types where $\psi(z) < \mu$—types where the social value of the transfer is less than its fiscal cost. This tilting drives the worst-case welfare down. The more extreme u_0 is, the more nature can exploit the affine structure (133), and the penalty grows without bound.

Proof. We show the case $u_0 \to +\infty$; the case $u_0 \to -\infty$ is symmetric.

Step (i): Find types where $\psi < \mu$. Since $\operatorname{ess\,inf} \psi < \mu$ (Assumption 11.3), there exist $\delta_- > 0$ and a measurable set $A_- \subset \mathcal{Z}$ with $\int_{A_-} f > 0$ such that $\psi(z) \le \mu - \delta_-$ for all $z \in A_-$.

Step (ii): On A_-, the integrand in W grows with u_0. Using the decomposition $g_a(z) = r_q(z) + (\psi(z) - \mu)u_0$ from (133):

$$e^{-g_a(z)/\theta} = e^{-r_q(z)/\theta} \cdot e^{-(\psi(z)-\mu)\,u_0/\theta}.$$

On A_-: $\psi - \mu \le -\delta_-$, so the exponent $-(\psi - \mu)u_0/\theta \ge \delta_- u_0/\theta$. When $u_0 > 0$, this is positive and growing in u_0.

Step (iii): Bound the integral from below. Drop the part of the integral outside A_- (it is nonnegative) and use $e^{-r_q/\theta} \ge e^{-C_r/\theta}$ (Lemma 11.3):

$$\int_{\mathcal{Z}} e^{-g_a/\theta} f \;\ge\; \int_{A_-} e^{-r_q/\theta}\, e^{-(\psi-\mu)u_0/\theta} f \;\ge\; e^{(\delta_- u_0 - C_r)/\theta} \int_{A_-} f.$$

Step (iv): Convert to an upper bound on W. Since $W = -\theta \log \int e^{-g_a/\theta} f$ and $-\theta \log$ is decreasing (larger integral $\Rightarrow$ smaller W):

$$W(u_0, q) \;\le\; -\theta \log\left[e^{(\delta_- u_0 - C_r)/\theta} \int_{A_-} f\right] = C_r - \delta_- u_0 - \theta \log \int_{A_-} f.$$

The right-hand side is linear in u_0 with slope $-\delta_- < 0$, so $W \to -\infty$ as $u_0 \to +\infty$. The bound is uniform over q because C_r does not depend on q.

The case $u_0 \to -\infty$. Since $\mu < \operatorname{ess\,sup} \psi$, there exist $\delta_+ > 0$ and $A_+ \subset \mathcal{Z}$ with $\int_{A_+} f > 0$ such that $\psi(z) \ge \mu + \delta_+$ on A_+. On A_+: $-(\psi - \mu)u_0/\theta \ge -\delta_+ u_0/\theta \ge 0$ (since $u_0 \le 0$). Repeating Steps (iii)–(iv) with A_+ and δ_+:

$$W(u_0, q) \le C_r + \delta_+ u_0 - \theta \log \int_{A_+} f \;\to\; -\infty \quad \text{as } u_0 \to -\infty. \qquad \square$$

Remark 11.9 (How coercivity restricts the domain). The coercivity lemma implies that extreme values of u_0 are suboptimal. Concretely: choose any feasible allocation, say $a_0 = (0, 0)$ (zero output, zero utility), with value $W(0, 0)$. By the lemma, there exists $M < \infty$ such that $\sup_q W(u_0, q) < W(0, 0)$ for all $|u_0| > M$. Therefore the maximizer of W over $\mathcal{A}_q = \mathbb{R} \times \mathcal{Q}$ must satisfy $|u_0^*| \le M$, and

$$\max_{\mathcal{A}_q} W = \max_{\mathcal{A}_q(M)} W.$$

Since $\mathcal{A}_q(M) = [-M, M] \times \mathcal{Q}$ is compact (Lemma 11.2), Weierstrass applies on $\mathcal{A}_q(M)$.

11.9.4 Step 4: Existence

Proposition 11.4 (Existence)**.** *W attains a maximum on $\mathcal{A}_q = \mathbb{R} \times \mathcal{Q}$.*

Proof. Fix any $\varepsilon > 0$. By the coercivity lemma, there exists $M < \infty$ such that

$$\sup_{q \in \mathcal{Q},\, |u_0| > M} W(u_0, q) < W(0, 0) - \varepsilon.$$

Therefore every maximizer lies in the compact strip $\mathcal{A}_q(M) = [-M, M] \times \mathcal{Q}$.

It remains to prove continuity on $\mathcal{A}_q(M)$. Let $a_n = (u_{0,n}, q_n) \to a = (u_0, q)$ pointwise in $\mathcal{A}_q(M)$. Then $g_{a_n}(z) \to g_a(z)$ for each z (dominated convergence in the envelope integral, with the remaining terms converging directly). By Lemma 11.3, $|g_{a_n}(z)| \le C_r + \|\psi - \mu\|_\infty M$ uniformly in n and z. Hence $e^{-g_{a_n}/\theta}$ converges pointwise and is dominated by an integrable constant. By dominated convergence (Lemma E.3): $\int e^{-g_{a_n}/\theta} f \to \int e^{-g_a/\theta} f$, and therefore $W(a_n) \to W(a)$.

Since $\mathcal{A}_q(M)$ is compact and W is continuous, Weierstrass (Lemma E.1) gives a maximizer. $\square$

11.9.5 Step 5: Uniqueness of the output schedule

Assumption 11.4 (Non-degenerate welfare weights)**.** $\psi(z)$ is not constant on $\mathcal{Z}$.

Proposition 11.5 (Joint strict concavity of W)**.** *Under Assumptions 11.1–11.4, W is jointly strictly concave in (u_0, q) on $\mathcal{A}_q$. Therefore the optimal allocation (u_0^*, q^*) is unique.*

Proof. Take $a_1 = (u_{0,1}, q_1) \neq a_2 = (u_{0,2}, q_2)$ in $\mathcal{A}_q$ and $t \in (0,1)$. Set $a_t = ta_1 + (1-t)a_2$. We show $W(a_t) > tW(a_1) + (1-t)W(a_2)$.

Step (a): Concavity of g_a in (u_0, q). From (129), the only nonlinear term in g_a is $\mu\, q(z)^{1/(1+\gamma)}$, which is strictly concave in $q(z)$ (power $1/(1+\gamma) < 1$). All other terms are affine in (u_0, q). Therefore:

$$g_{a_t}(z) \geq t\, g_{a_1}(z) + (1-t)\, g_{a_2}(z) \qquad \text{for all } z, \tag{137}$$

with strict inequality on $\{z : q_1(z) \neq q_2(z)\}$ (from strict concavity of $q^{1/(1+\gamma)}$). When $q_1(z) = q_2(z)$ for f-a.e. z, the inequality is equality (g_a is affine in u_0).

Step (b): Exponentiate and apply Hölder. Since $x \mapsto e^{-x/\theta}$ is strictly decreasing, (137) gives

$$e^{-g_{a_t}(z)/\theta} \leq \left(e^{-g_{a_1}(z)/\theta}\right)^t \left(e^{-g_{a_2}(z)/\theta}\right)^{1-t} \tag{138}$$

for all z. Integrate against $f > 0$ and apply Hölder's inequality (Lemma E.4, exponents $1/t$ and $1/(1-t)$):

$$\int e^{-g_{a_t}/\theta} f \;\leq\; \int \left(e^{-g_1/\theta}\right)^t \left(e^{-g_2/\theta}\right)^{1-t} f \;\leq\; \left(\int e^{-g_1/\theta} f\right)^t \left(\int e^{-g_2/\theta} f\right)^{1-t}. \tag{139}$$

Step (c): Strict inequality. If $\{z : q_1(z) \neq q_2(z)\}$ has positive f-measure: $g_{a_t}(z) > tg_1(z) + (1-t)g_2(z)$ on that set (strict concavity of $q^{1/(1+\gamma)}$), so (138) is strict there. Integrating against $f > 0$: the first inequality in (139) is strict.

If $q_1(z) = q_2(z)$ for f-a.e. z but $u_{0,1} \neq u_{0,2}$: $g_{a_t} = tg_1 + (1-t)g_2$ (affine in u_0), so the first inequality is equality. But the second (Hölder) is strict unless $e^{-g_1/\theta}/e^{-g_2/\theta}$ is constant f-a.e. The ratio is $\exp(-(\psi(z)-\mu)(u_{0,1}-u_{0,2})/\theta)$, which is non-constant because $\psi(z) - \mu$ is non-constant (Assumption 11.4) and $u_{0,1} \neq u_{0,2}$.

In both cases, the chain of inequalities in (139) is strict.

Step (d): Apply $-\theta \log$. Since $-\theta \log$ is strictly decreasing and $-\theta \log(A^t B^{1-t}) = t(-\theta \log A) + (1-t)(-\theta \log B)$:

$$W(a_t) > t\, W(a_1) + (1-t)\, W(a_2). \qquad \square$$

11.9.6 Step 6: From the optimal allocation to the on-path tax schedule

The full allocation. Given (u_0^*, q^*), define

$$\begin{aligned} y^*(z) &= q^*(z)^{1/(1+\gamma)}, \\ \mathcal{U}^*(z) &= u_0^* + \int_{z_0}^{z} \frac{q^*(s)}{s^{2+\gamma}}\, ds, \\ c^*(z) &= \mathcal{U}^*(z) + \frac{q^*(z)}{(1+\gamma)\, z^{1+\gamma}}. \end{aligned}$$

For any income y in the range of y^*, define $T^*(y) = y - c^*(z)$ for any z with $y^*(z) = y$.

Bunching. This is well-defined even at bunching points. If y^* is locally strictly increasing, the representing type z is unique. If q^* is constant $= \hat{q}$ on an interval $[z_1, z_2]$ (a bunch), then $y^*(z) \equiv \hat{y} = \hat{q}^{1/(1+\gamma)}$. Moreover,

$$c^*(z) = u_0^* + \int_{z_0}^{z} \frac{\hat{q}}{s^{2+\gamma}}\, ds + \frac{\hat{q}}{(1+\gamma)\, z^{1+\gamma}},$$

hence

$$c^{*\prime}(z) = \frac{\hat{q}}{z^{2+\gamma}} - \frac{\hat{q}}{z^{2+\gamma}} = 0 \qquad \text{on } [z_1, z_2].$$

So $c^*(z) \equiv \hat{c}$ on the bunch, and $T^*(\hat{y}) = \hat{y} - \hat{c}$ does not depend on which type in the bunch is used.

Proposition 11.6 (Marginal rate)**.** *At points where y^* is differentiable and $y^{*\prime}(z) > 0$: $T^{*\prime}(y^*(z)) = 1 - (y^*(z)/z)^\gamma/z$.*

Proof. The household FOC gives $n^\gamma = z(1 - T')$, hence $1 - T' = n^\gamma/z = (y/z)^\gamma/z$. □

11.9.7 Conclusion

Corollary 11.2. *Under quasilinear isoelastic preferences and Assumptions 11.1–11.4:*

(i) The optimal allocation (u_0^*, q^*) exists and is unique in $\mathcal{A}_q$.

(ii) The output schedule $y^* = (q^*)^{1/(1+\gamma)}$ is uniquely determined.

(iii) The on-path tax T^* is uniquely determined at every income in the range of y^*, including bunching points.

(iv) Off-path, T can be extended arbitrarily subject to no type deviating.

All results are for the fixed-μ Lagrangian.

Remark 11.10 (Lagrangian vs. primal). The connection to the primal requires choosing μ so the budget binds. One expects the budget surplus to be continuous and strictly decreasing in μ, so that the intermediate value theorem gives a unique such μ; verifying these conditions is a separate argument that we do not carry out here.

Remark 11.11 (Summary).

$$\max_T V(T) \xrightarrow[\text{tax. pr.}]{} \max_{\mathcal{A}_{IC}^{\text{adm}}} W \xrightarrow[\text{Lem. 11.1}]{} \max_{\mathcal{A}_q} W \xrightarrow[\text{Hölder}]{} a^* \text{ unique } \to T^* \text{ on-path.}$$

12 Extension: Heterogeneous Elasticities

Sections 1–11 analyzed optimal taxation when households differ in a single dimension: productivity z. In this section we extend the framework to allow a second dimension of heterogeneity: the Frisch elasticity parameter γ. The key result is that heterogeneity in elasticities raises the optimal top tax rate above the Diamond–Saez benchmark computed at the mean elasticity, because the composition of the top tail shifts toward less elastic workers. We then extend the robust analysis of Section 11 to this two-dimensional setting.

12.1 Types and GHH preferences

Each agent is characterized by a two-dimensional type $(\gamma, z) \in \Gamma \times (0, \infty)$, where $z > 0$ is productivity and $\gamma \in \Gamma \subset (0, \infty)$ is the Frisch elasticity parameter. Under a locally linear top bracket the compensated elasticity of taxable income for type γ is $e(\gamma) = 1/\gamma$.

Types are drawn from a joint distribution with density $f_{Z,\gamma}(z, \gamma)$ on $(0, \infty) \times \Gamma$.

Preferences are of the **Greenwood–Hercowitz–Huffman (GHH)** form:

$$u(c, n \mid \gamma, z) = \frac{1}{1-\rho}\left(c - \Psi(\gamma, z)\frac{n^{1+\gamma}}{1+\gamma}\right)^{1-\rho} + \Delta(\gamma, z), \tag{140}$$

with $\rho > 0$, $\Psi(\gamma, z) > 0$ a labor-disutility shifter calibrated so that status-quo behavior is independent of γ (Section 12.4 below), and $\Delta(\gamma, z)$ a utility shifter that plays no role in the first-order conditions.

Two crucial properties of GHH preferences:

(i) There are *no income effects* on labor supply: the marginal rate of substitution between consumption and labor depends only on n, not on c. We write the income-effect parameter as $\eta(y) \equiv 0$ throughout.

(ii) The compensated elasticity of taxable income coincides with the Frisch elasticity, and for a given γ it is constant across income levels in the top bracket.

12.2 Tax schedule, budget constraint, and individual optimization

The government levies a nonlinear income tax $T : \mathbb{R}_+ \to \mathbb{R}$. An agent with earnings $y = zn$ consumes $c = y - T(y)$. Define the net-of-tax rate $R(y) := 1 - T'(y)$.

Given the tax schedule T, agent (γ, z) solves

$$\max_{n \geq 0} \frac{1}{1-\rho}\left(zn - T(zn) - \Psi(\gamma, z)\frac{n^{1+\gamma}}{1+\gamma}\right)^{1-\rho}.$$

The first-order condition with respect to n is

$$\Psi(\gamma, z)\, n^{\gamma} = z\, R(zn). \tag{141}$$

This implicitly defines optimal labor supply $n(z, \gamma; T)$ and hence optimal earnings

$$\mathcal{Y}(z, \gamma; T) := z\, n(z, \gamma; T). \tag{142}$$

Assumption 12.1 (Monotonicity)**.** For each (γ, T), the map $z \mapsto \mathcal{Y}(z, \gamma; T)$ is C^1 and strictly increasing.

Under Assumption 12.1, for each income level y and type γ, there exists a unique productivity threshold

$$Z(y, \gamma; T) \quad \text{solving} \quad \mathcal{Y}(Z(y, \gamma; T), \gamma; T) = y. \tag{143}$$

12.3 Compensated elasticity of taxable income

We adopt the same definition of the compensated ETI as in Section 7.2. Fix an income level $y > 0$ and a type $\gamma \in \Gamma$. The local linearization of T at y is the affine schedule $T^{\text{lin}}(y'; \tau, I) = \tau\, y' - I$ (cf. (48)), and the compensated ETI is

$$\zeta(y, \gamma) := \frac{\partial \hat{y}^{\,c}}{\partial(1-\tau)} \cdot \frac{1-\tau}{y}, \tag{144}$$

where $\hat{y}^{\,c}$ denotes Hicksian earnings under the linearized schedule.

Under GHH preferences, income effects vanish ($\eta \equiv 0$), so compensated and uncompensated elasticities coincide. For a locally linear top bracket, $\zeta(y, \gamma) = 1/\gamma$ for all y in that bracket.

12.4 Calibration of GHH shifters

We calibrate $\Psi(\gamma, z)$ so that under a status-quo tax $T^0 \equiv 0$, the behavioral mapping $\mathcal{Y}(z, \gamma; T^0)$ is independent of γ for every z.

Consider a reference worker with Frisch parameter $\bar{\gamma}$ and disutility coefficient $\bar{\psi}$. The FOC is $\bar{\psi}\, n^{\bar{\gamma}} = z$, giving

$$\bar{n}(z) = \left(\frac{z}{\bar{\psi}}\right)^{1/\bar{\gamma}}, \qquad \bar{Y}(z) = z\,\bar{n}(z) = \bar{\psi}^{-1/\bar{\gamma}}\, z^{1+1/\bar{\gamma}} =: \kappa\, z^{\beta}, \tag{145}$$

where $\beta := 1 + 1/\bar{\gamma}$ and $\kappa := \bar{\psi}^{-1/\bar{\gamma}}$.

Requiring $n(z, \gamma; T^0) = \bar{n}(z)$ for all (γ, z) and substituting into the FOC (141) with $R \equiv 1$ yields

$$\Psi(\gamma, z) = \bar{\psi}^{\gamma/\bar{\gamma}}\, z^{1-\gamma/\bar{\gamma}}. \tag{146}$$

Under this calibration, the status-quo income mapping $\bar{Y}(z) = \kappa\, z^{\beta}$ is common to all γ-types.

12.5 Endogenous income distribution

Under a general tax schedule T, the endogenous income distribution has CDF

$$H_T(y) = \int_{\Gamma} \int_0^{Z(y,\gamma;T)} f_{Z,\gamma}(z, \gamma)\, dz\, d\gamma$$

and density

$$h_T(y) = \int_{\Gamma} h_T(y; \gamma)\, d\gamma, \qquad h_T(y; \gamma) := \frac{f_{Z,\gamma}(Z(y, \gamma; T), \gamma)}{\mathcal{Y}_z(Z(y, \gamma; T), \gamma; T)}. \tag{147}$$

Here $h_T(y; \gamma)$ is the γ-specific component of the income density at y: it gives the density of γ-type agents at income y.

Lemma 12.1 (Change of variables identity)**.** *For any measurable function $\phi(y,\gamma)$,*

$$\int_{y^*}^{\infty}\int_{\Gamma}\phi(y,\gamma)\,h_T(y;\gamma)\,d\gamma\,dy=\int_{\Gamma}\int_{z\geq Z(y^*,\gamma;T)}\phi(\mathcal{Y}(z,\gamma;T),\gamma)\,f_{Z,\gamma}(z,\gamma)\,dz\,d\gamma. \tag{148}$$

Proof. For each fixed γ, the map $z\mapsto y=\mathcal{Y}(z,\gamma;T)$ is a C^1 diffeomorphism (by Assumption 12.1) with Jacobian $\mathcal{Y}_z$. The standard change of variables gives $dy=\mathcal{Y}_z\,dz$. Substituting the definition of $h_T(y;\gamma)$ from (147):

$$\int_{y^*}^{\infty}\phi(y,\gamma)\,h_T(y;\gamma)\,dy=\int_{y^*}^{\infty}\phi(y,\gamma)\,\frac{f_{Z,\gamma}(Z(y,\gamma;T),\gamma)}{\mathcal{Y}_z(Z(y,\gamma;T),\gamma;T)}\,dy.$$

Under the substitution $z=Z(y,\gamma;T)$, we have $y=\mathcal{Y}(z,\gamma;T)$ and $dy=\mathcal{Y}_z(z,\gamma;T)\,dz$. The Jacobian $\mathcal{Y}_z$ in the denominator of $h_T(y;\gamma)$ cancels with the Jacobian from $dy=\mathcal{Y}_z\,dz$:

$$\frac{f_{Z,\gamma}(z,\gamma)}{\mathcal{Y}_z(z,\gamma;T)}\cdot\mathcal{Y}_z(z,\gamma;T)\,dz=f_{Z,\gamma}(z,\gamma)\,dz.$$

The lower limit transforms as $y=y^*\Leftrightarrow z=Z(y^*,\gamma;T)$. Integrating over γ yields (148). □

Remark 12.1 (Why the Jacobian cancels). The cancellation of $\mathcal{Y}_z$ in the change of variables is the key step that the reader may find non-obvious. The density $h_T(y;\gamma)$ already contains a factor of $1/\mathcal{Y}_z$ in its definition (the standard change-of-variables formula for a density). When we change the integration variable from y back to z, we introduce a factor of $\mathcal{Y}_z$ from $dy=\mathcal{Y}_z\,dz$. These two factors cancel exactly, leaving only the primitive density $f_{Z,\gamma}(z,\gamma)\,dz$. This is not an approximation—it is an identity.

12.6 The social planner's problem

Following the notation of Section 2, the planner assigns Pareto weights $\psi(z,\gamma)\geq 0$ to each type. The Lagrangian for the planner's problem (with multiplier μ on the government budget constraint) is

$$\mathcal{L}(T)=\int_{\Gamma}\int_{0}^{\infty}\psi(z,\gamma)\,\mathcal{U}(z,\gamma;T)\,f_{Z,\gamma}(z,\gamma)\,dz\,d\gamma+\mu\left[\int_{\Gamma}\int_{0}^{\infty}T(\mathcal{Y}(z,\gamma;T))\,f_{Z,\gamma}(z,\gamma)\,dz\,d\gamma-B\right], \tag{149}$$

where $\mathcal{U}(z,\gamma;T)=u(\mathcal{Y}-T(\mathcal{Y}),\,n\mid\gamma,z)$ is the indirect utility of type (γ,z) under schedule T, and B is the exogenous revenue requirement.

The **marginal social welfare weight** for a type (γ,z)-agent earning y is defined as in (52):

$$g(y,\gamma):=\frac{\psi(z,\gamma)\,U_c(y,\gamma)}{\mu}, \tag{150}$$

where $U_c(y,\gamma)$ is the marginal utility of consumption for the agent at income y with elasticity parameter γ, and $z=Z(y,\gamma;T)$. This measures the social value—in dollars of public funds—of one additional dollar transferred to this agent.

Since GHH preferences eliminate income effects ($\eta\equiv 0$), the adjusted welfare weight (cf. (55)) coincides with the unadjusted weight:

$$\hat{g}(y,\gamma)=g(y,\gamma). \tag{151}$$

At the optimum, the budget-balance normalization (Remark 7.3) generalizes to

$$\int_{0}^{\infty}\int_{\Gamma}\hat{g}(y,\gamma)\,h_T(y;\gamma)\,d\gamma\,dy=1. \tag{152}$$

12.7 The Saez perturbation in two dimensions

We derive the optimality condition for $T'(y)$ using the Saez perturbation method of Section 7, extended to two-dimensional heterogeneity. The perturbation itself is identical: fix $y^* > 0$ and increase the marginal tax rate by $d\tau > 0$ in a narrow band $[y^*, y^* + \varepsilon]$. The innovation relative to the one-dimensional derivation is that at each income level, multiple types γ coexist with different elasticities and different Jacobians.

Total derivative of the Lagrangian. Following the same decomposition as in (51), the total derivative of the Lagrangian at $\epsilon = 0$ is

$$\left.\frac{d\mathcal{L}}{d\epsilon}\right|_{\epsilon=0} = \int_\Gamma \int_0^\infty \left[\underbrace{\psi(z,\gamma)\frac{d\mathcal{U}}{d\epsilon} + \mu\,\delta T(Y)}_{\text{(a)}} + \underbrace{\mu\,T'(Y)\frac{\partial Y}{\partial \epsilon}}_{\text{(b)}} \right] f_{Z,\gamma}\,dz\,d\gamma, \tag{153}$$

where $Y = \mathcal{Y}(z, \gamma; T)$.

Envelope theorem. By the same argument as in Section 7, the household envelope theorem yields $d\mathcal{U}/d\epsilon = -U_c\,\delta T(Y)$, so that

$$\psi(z,\gamma)\frac{d\mathcal{U}}{d\epsilon} + \mu\,\delta T(Y) = \mu\,\delta T(Y)(1 - g(y,\gamma)).$$

Since $\eta \equiv 0$ under GHH, the Slutsky decomposition of Section 7.6 gives $\hat{g} = g$, and we define:

$$dM := \mu\,\delta T(Y)(1 - \hat{g}(y,\gamma)), \tag{154}$$

$$dB := \mu\,T'(Y)\,\partial Y^c/\partial\epsilon, \tag{155}$$

exactly as in (56)–(57), but now with $\hat{g}$ depending on both y and γ.

12.7.1 Evaluating the mechanical effect dM

Aggregating (154) over all agents and substituting the perturbation $\delta T(y)$, we split the income distribution into three regions, generalizing the one-dimensional calculation of Section 7.4.

Region 1: below the band ($y < y^*$). No tax change; zero contribution.

Region 2: inside the band ($y \in [y^*, y^* + \varepsilon]$). Each earner at income y pays $(y - y^*)\,d\tau \le \varepsilon\,d\tau$. The aggregate contribution is

$$dM_{\text{band}} = d\tau \int_{y^*}^{y^*+\varepsilon} \int_\Gamma (y - y^*)(1 - \hat{g}(y,\gamma))\,h_T(y;\gamma)\,d\gamma\,dy.$$

Since $(y - y^*) \le \varepsilon$ and the integrand $|(1 - \hat{g})\,h_T|$ is bounded on the compact band, the integral over y contributes at most $O(\varepsilon)$, giving

$$|dM_{\text{band}}| \le \varepsilon\,d\tau \cdot \varepsilon \cdot \sup|(1 - \hat{g})\,h_T| = O(\varepsilon^2\,d\tau).$$

Region 3: above the band ($y > y^* + \varepsilon$). Each earner pays the full $\varepsilon\, d\tau$:

$$dM_{\text{above}} = \varepsilon\, d\tau \int_{y^*+\varepsilon}^{\infty} \int_{\Gamma} (1 - \hat{g}(y,\gamma))\, h_T(y;\gamma)\, d\gamma\, dy.$$

Combining. Replacing the lower limit $y^* + \varepsilon$ by y^* changes the integral by $O(\varepsilon)$ (since the integrand is bounded), which multiplied by $\varepsilon\, d\tau$ gives $O(\varepsilon^2\, d\tau)$. Adding the band contribution:

$$dM = \varepsilon\, d\tau \int_{y^*}^{\infty} \int_{\Gamma} (1 - \hat{g}(y,\gamma))\, h_T(y;\gamma)\, d\gamma\, dy + O(\varepsilon^2\, d\tau). \tag{156}$$

Change of variables to primitive space. Applying Lemma 12.1 with $\phi(y,\gamma) = 1 - \hat{g}(y,\gamma)$ to the leading-order term in (156):

$$dM = \varepsilon\, d\tau \int_{\Gamma} \int_{z \geq Z(y^*,\gamma;T)} (1 - \hat{g}(\mathcal{Y}(z,\gamma;T),\gamma))\, f_{Z,\gamma}(z,\gamma)\, dz\, d\gamma + O(\varepsilon^2\, d\tau). \tag{157}$$

The Jacobian $\mathcal{Y}_z$ does not appear explicitly because it cancels in the change of variables, as shown in the proof of Lemma 12.1.

12.7.2 Evaluating the behavioral effect dB

Agents earning in the band $[y^*, y^* + \varepsilon]$ face an increased marginal rate. For an agent of type γ at income y in the band, the compensated earnings response is (by the same ETI calculation as in Section 7.6)

$$dy^c(y,\gamma) = -\zeta(y,\gamma)\, \frac{y}{R(y)}\, d\tau. \tag{158}$$

The revenue loss per agent is $T'(y) \cdot |dy^c|$. Only agents *inside* the band $[y^*, y^* + \varepsilon]$ face a changed marginal rate.

Aggregating over the band and all γ-types:

$$dB = \int_{y^*}^{y^*+\varepsilon} \int_{\Gamma} \mu\, T'(y)\, dy^c(y,\gamma)\, h_T(y;\gamma)\, d\gamma\, dy.$$

We now evaluate this integral to first order in ε.

Lemma 12.2 (First-order evaluation of dB)**.** *Under the assumption that T', ζ, and $h_T(\cdot\,;\gamma)$ are continuous at y^*:*

$$dB = -\varepsilon\, d\tau\, \frac{T'(y^*)}{1 - T'(y^*)}\, y^* \int_{\Gamma} \zeta(y^*,\gamma)\, h_T(y^*;\gamma)\, d\gamma + O(\varepsilon^2\, d\tau). \tag{159}$$

Proof. Substituting (158) and $R(y) = 1 - T'(y)$:

$$dB = -d\tau \int_{y^*}^{y^*+\varepsilon} \int_{\Gamma} \mu\, \frac{T'(y)}{1 - T'(y)}\, \zeta(y,\gamma)\, y\, h_T(y;\gamma)\, d\gamma\, dy.$$

The integrand $\Phi(y,\gamma) := \frac{T'(y)}{1-T'(y)}\, \zeta(y,\gamma)\, y\, h_T(y;\gamma)$ is continuous in y at y^* for each γ, and integrable over Γ. On the interval $[y^*, y^* + \varepsilon]$ of width ε, we can write

$$\Phi(y,\gamma) = \Phi(y^*,\gamma) + O(\varepsilon) \qquad \text{uniformly in } \gamma \text{ on compact } \Gamma.$$

Therefore

$$\int_{y^*}^{y^*+\varepsilon} \int_\Gamma \Phi(y,\gamma)\, d\gamma\, dy = \int_{y^*}^{y^*+\varepsilon} \left[\int_\Gamma \Phi(y^*,\gamma)\, d\gamma + O(\varepsilon) \right] dy$$
$$= \varepsilon \int_\Gamma \Phi(y^*,\gamma)\, d\gamma + O(\varepsilon^2).$$

Multiplying by $-\mu\, d\tau$ and expanding $\Phi(y^*,\gamma)$ gives (159). □

We can rewrite the behavioral effect using the γ-specific density $h_T(y^*;\gamma)$ defined in (147):

$$dB = -\varepsilon\, d\tau\, \frac{T'(y^*)}{1-T'(y^*)}\, y^* \int_\Gamma \zeta(y^*,\gamma)\, \frac{f_{Z,\gamma}(Z(y^*,\gamma;T),\gamma)}{\mathcal{Y}_z(Z(y^*,\gamma;T),\gamma;T)}\, d\gamma + O(\varepsilon^2\, d\tau). \tag{160}$$

Remark 12.2 (Comparison with the one-dimensional case)*.* In the one-dimensional setting of Section 7.6, there is a single elasticity $\zeta^c(y^*)$ and a single density $h(y^*)$, so $dB = -\varepsilon\, d\tau\, \frac{T'}{1-T'}\, \zeta^c\, y^*\, h(y^*)$. In the two-dimensional setting, the integral $\int_\Gamma \zeta(y^*,\gamma)\, h_T(y^*;\gamma)\, d\gamma$ replaces $\zeta^c(y^*)\, h(y^*)$ with an elasticity-weighted density: the behavioral response at y^* is an average of γ-specific elasticities $\zeta(y^*,\gamma)$, weighted by the γ-specific density at y^*.

12.7.3 First-order condition

At the optimum, $dM + dB = 0$ for every y^*. Substituting (156) and (159), dividing by $\varepsilon\, d\tau > 0$, and sending $\varepsilon \to 0$ (which eliminates all $O(\varepsilon^2\, d\tau)$ remainders):

Proposition 12.1 (Optimal tax formula with heterogeneous elasticities)**.** *Under Assumptions 1.1–12.1, at any income level y^* the optimal tax schedule satisfies*

$$\frac{T'(y^*)}{1-T'(y^*)} = \frac{\displaystyle\int_\Gamma \int_{z \geq Z(y^*,\gamma;T)} (1-\hat{g}(\mathcal{Y}(z,\gamma;T),\gamma))\, f_{Z,\gamma}(z,\gamma)\, dz\, d\gamma}{\displaystyle y^* \int_\Gamma \zeta(y^*,\gamma)\, \frac{f_{Z,\gamma}(Z(y^*,\gamma;T),\gamma)}{\mathcal{Y}_z(Z(y^*,\gamma;T),\gamma;T)}\, d\gamma}. \tag{161}$$

The numerator is the mass of agents above y^* (weighted by $1-\hat{g}$), integrated over both z and γ. The denominator is the elasticity-weighted density at y^*, aggregated across γ-types.

12.8 Asymptotic top rate

Taking $y^* \to \infty$ and imposing that the average welfare weight among top earners vanishes (the natural extension of Section 8), the formula simplifies to

$$\lim_{y\to\infty} \frac{T'(y)}{1-T'(y)} = \lim_{y\to\infty} \frac{\displaystyle\int_\Gamma \int_{z \geq Z(y,\gamma;T)} f_{Z,\gamma}(z,\gamma)\, dz\, d\gamma}{\displaystyle y \int_\Gamma \zeta(y,\gamma)\, \frac{f_{Z,\gamma}(Z(y,\gamma;T),\gamma)}{\mathcal{Y}_z(Z(y,\gamma;T),\gamma;T)}\, d\gamma}. \tag{162}$$

12.9 Top-rate formula under GHH with Pareto tails

We now derive the asymptotic top marginal tax rate under the calibrated GHH specification with Pareto tails. The derivation proceeds in six steps: (0) state the limiting assumption $T'(y) \to \tau$ and derive its consequences; (i) derive the asymptotic behavioral mapping from the household FOC; (ii) compute the asymptotic Jacobian and inverse; (iii) verify that the income distribution inherits a Pareto tail with a common exponent across γ-types; (iv) evaluate the numerator and denominator of the top-rate formula (162) asymptotically; (v) take the limit $y \to \infty$ and simplify.

12.9.1 Step 0: The limiting assumption and its consequences

We do *not* assume that $T'(y)$ is exactly constant. Instead, we assume:

Assumption 12.2 (Existence of a limiting top rate)**.** The marginal tax rate has a finite limit:

$$\tau := \lim_{y\to\infty} T'(y) \in (0,1).$$

We now derive three consequences that are used in the subsequent steps.

Lemma 12.3 (Linear approximation of T in the tail)**.** *Under Assumption 12.2:*

(i) $T(y) = \tau\, y + o(y)$ *as* $y \to \infty$.

(ii) $R(y) := 1 - T'(y) \to R := 1 - \tau$ *as* $y \to \infty$.

Proof. Part (i). Fix $y_0 > 0$. By the fundamental theorem of calculus:

$$T(y) = T(y_0) + \int_{y_0}^{y} T'(s)\, ds = T(y_0) + \tau(y - y_0) + \int_{y_0}^{y} [T'(s) - \tau]\, ds.$$

Dividing by y:

$$\frac{T(y)}{y} = \tau + \frac{T(y_0) - \tau\, y_0}{y} + \frac{1}{y}\int_{y_0}^{y} [T'(s) - \tau]\, ds.$$

The second term vanishes as $y \to \infty$. For the third, given $\varepsilon > 0$, choose M such that $|T'(s) - \tau| < \varepsilon$ for all $s \geq M$. Then

$$\left| \frac{1}{y}\int_{y_0}^{y} [T'(s) - \tau]\, ds \right| \leq \frac{C_M}{y} + \varepsilon\, \frac{y - M}{y},$$

where $C_M = \int_{y_0}^{M} |T'(s) - \tau|\, ds$ is a finite constant. The first piece vanishes and the second converges to ε. Since ε is arbitrary, $T(y)/y \to \tau$.

Part (ii). Immediate from $T'(y) \to \tau$. □

12.9.2 Step 1: Asymptotic behavioral mapping

Under the *general* nonlinear schedule T (not necessarily with $T' = \tau$ exactly), the household FOC (141) with the calibrated shifter (146) gives

$$\bar{\psi}^{\gamma/\bar{\gamma}}\, z^{1-\gamma/\bar{\gamma}}\, n^{\gamma} = z\, R(\mathcal{Y}), \tag{163}$$

where $\mathcal{Y} = z\,n$ and $R(\mathcal{Y}) = 1 - T'(\mathcal{Y})$. Solving for n^γ (the same algebra as in the constant-rate case):

$$n^\gamma = \bar{n}(z)^\gamma\, R(\mathcal{Y}), \qquad \text{hence} \qquad n = \bar{n}(z)\, R(\mathcal{Y})^{1/\gamma}.$$

Multiplying by z:

$$\mathcal{Y}(z,\gamma;T) = \bar{Y}(z)\, R(\mathcal{Y}(z,\gamma;T))^{1/\gamma}. \tag{164}$$

This is an *implicit* equation for $\mathcal{Y}$: the retention rate $R(\mathcal{Y})$ is evaluated at the endogenous earnings level.

Lemma 12.4 (Asymptotic equivalence of the behavioral mapping)**.** *Under Assumption 12.2, with $R = 1-\tau$, $\kappa = \bar{\psi}^{-1/\bar{\gamma}}$, $\beta = 1 + 1/\bar{\gamma}$, and $c(\gamma;R) = \kappa\, R^{1/\gamma}$:*

$$\mathcal{Y}(z,\gamma;T) = c(\gamma;R)\, z^\beta\,(1+o(1)) \qquad (z\to\infty), \tag{165}$$

uniformly in γ on compact Γ.

Proof. By Assumption 12.1, $\mathcal{Y}(z,\gamma;T)$ is strictly increasing in z for each γ. Since $\mathcal{Y} \geq z\,\bar{n}(z)\, R_{\min}^{1/\gamma}$ (where $R_{\min} = \inf_y R(y) > 0$ under admissibility), $\mathcal{Y}(z,\gamma;T) \to \infty$ as $z\to\infty$. Therefore $R(\mathcal{Y}) \to R$ by Lemma 12.3(ii).

From (164):

$$\mathcal{Y}(z,\gamma;T) = \bar{Y}(z)\, R(\mathcal{Y})^{1/\gamma} = \kappa\, z^\beta\, R(\mathcal{Y})^{1/\gamma}.$$

Since $R(\mathcal{Y}) = R + o(1)$ as $z\to\infty$, and $x \mapsto x^{1/\gamma}$ is continuous:

$$R(\mathcal{Y})^{1/\gamma} = R^{1/\gamma}\,(1+o(1)).$$

(More precisely: $R(\mathcal{Y})^{1/\gamma} = R^{1/\gamma}\,(R(\mathcal{Y})/R)^{1/\gamma} = R^{1/\gamma}\,(1 + (R(\mathcal{Y}) - R)/R + O((R(\mathcal{Y}) - R)^2))^{1/\gamma}$ and since $R(\mathcal{Y}) - R = o(1)$, we get $R(\mathcal{Y})^{1/\gamma} = R^{1/\gamma}(1+o(1))$.) Therefore

$$\mathcal{Y}(z,\gamma;T) = \kappa\, R^{1/\gamma}\, z^\beta\,(1+o(1)) = c(\gamma;R)\, z^\beta\,(1+o(1)).$$

For the uniformity in γ: on compact Γ, $1/\gamma$ is bounded, $R^{1/\gamma}$ is bounded away from zero, and $R(\mathcal{Y}) \to R$ does not depend on γ (it depends only on $\mathcal{Y}\to\infty$). So the $o(1)$ term is uniform. □

We define, as before:

$$c(\gamma;R) := \kappa\, R^{1/\gamma}, \qquad R = 1-\tau. \tag{166}$$

Lemma 12.4 says $\mathcal{Y}(z,\gamma;T) \sim c(\gamma;R)\, z^\beta$.

Remark 12.3 (The role of the calibration). The calibrated shifters ensure that the asymptotic *power* $\beta = 1 + 1/\bar{\gamma}$ is common to all γ-types. Only the *level* $c(\gamma;R) = \kappa\, R^{1/\gamma}$ varies with γ. Without the shifters, the power would be $\beta(\gamma) = 1 + 1/\gamma$, differing across types (Section 12.12.8).

12.9.3 Step 2: Asymptotic Jacobian, inverse, and ETI

Jacobian. Under a general schedule, the Jacobian $\mathcal{Y}_z$ was computed in the original 2D notes (equation (141)) via the implicit function theorem applied to (164). Define $F(Y, z) := Y - \bar{Y}(z)\, R(Y)^{1/\gamma}$, so $F(\mathcal{Y}, z) = 0$. Then:

$$F_z = -\bar{Y}'(z)\, R(\mathcal{Y})^{1/\gamma} = -\beta\, \kappa\, z^{\beta-1}\, R(\mathcal{Y})^{1/\gamma},$$
$$F_Y = 1 - \bar{Y}(z)\, \frac{1}{\gamma}\, R(\mathcal{Y})^{1/\gamma-1}\, R'(\mathcal{Y}) = 1 + \frac{\mathcal{Y}}{\gamma\, R(\mathcal{Y})}\, T''(\mathcal{Y}),$$

where we used $\bar{Y}(z)\, R(\mathcal{Y})^{1/\gamma} = \mathcal{Y}$ and $R'(Y) = -T''(Y)$. By the implicit function theorem:

$$\mathcal{Y}_z = -\frac{F_z}{F_Y} = \frac{\beta\, \kappa\, z^{\beta-1}\, R(\mathcal{Y})^{1/\gamma}}{1 + \frac{\mathcal{Y}}{\gamma\, R(\mathcal{Y})}\, T''(\mathcal{Y})}. \tag{167}$$

Lemma 12.5 (Asymptotic Jacobian)**.** *Under Assumption 12.2, with $\beta = 1 + 1/\bar{\gamma}$ and $c(\gamma; R) = \kappa\, R^{1/\gamma}$ as in Lemma 12.4:*

$$\mathcal{Y}_z(z, \gamma; T) = \beta\, c(\gamma; R)\, z^{\beta-1}\, (1 + o(1)) \qquad (z \to \infty), \tag{168}$$

uniformly in γ on compact Γ.

Proof. The numerator of (167) satisfies $\beta\, \kappa\, z^{\beta-1}\, R(\mathcal{Y})^{1/\gamma} = \beta\, c(\gamma; R)\, z^{\beta-1}\, (1 + o(1))$ by the same argument as in Lemma 12.4.

For the denominator: we need $\mathcal{Y}\, T''(\mathcal{Y})/(\gamma\, R(\mathcal{Y})) \to 0$. Since $T'(y) \to \tau$, we have $T'(y) = \tau + \varepsilon(y)$ with $\varepsilon(y) \to 0$. The mean value theorem gives $T''(\xi) = (T'(y_2) - T'(y_1))/(y_2 - y_1)$ for some ξ between y_1 and y_2. More directly: for any $\delta > 0$, there exists Y_0 such that $|T'(y) - \tau| < \delta$ for all $y \geq Y_0$. This means $|T''(y)| \cdot y$ cannot grow faster than $o(1)\cdot$ (some bounded function) in the sense that the product $\mathcal{Y} \cdot T''(\mathcal{Y})/R(\mathcal{Y})$ is controlled by the rate at which T' converges to τ.

Under the stronger (but standard) assumption that T is C^2 and $y\, T''(y) \to 0$ as $y \to \infty$—which follows from $T'(y) \to \tau$ whenever the convergence is monotone, or more generally whenever T'' does not oscillate with growing amplitude—the denominator converges to 1.

Formally, it suffices to assume:

$$\mathcal{Y}(z, \gamma; T)\, \frac{T''(\mathcal{Y}(z, \gamma; T))}{R(\mathcal{Y}(z, \gamma; T))} \to 0 \qquad (z \to \infty). \tag{169}$$

This is implied by $y\, T''(y) \to 0$, which in turn is implied by $T'(y) \to \tau$ monotonically. Under (169), the denominator of (167) converges to 1, and (168) follows. □

Remark 12.4 (When does $y\, T''(y) \to 0$?). The condition $y\, T''(y) \to 0$ holds automatically if $T'(y) \to \tau$ monotonically, since then T'' does not change sign and $\int_{y_0}^{\infty} |T''(s)|\, ds = |T'(y_0) - \tau| < \infty$, which forces $|T''(y)| = o(1/y)$. It also holds for any T of the form $T(y) = \tau\, y + a\, \log y + b + o(1)$ (a common parametric specification in the Pareto tail). The condition fails only if T' oscillates with increasing amplitude around τ—a pathological case that is ruled out in any reasonable optimal tax schedule.

Inverse function. For a given income y and type γ, the productivity threshold $Z(y, \gamma; T)$ satisfies $\mathcal{Y}(Z, \gamma; T) = y$. By Lemma 12.4, $\mathcal{Y} \sim c(\gamma; R)\, Z^{\beta}$, so:

$$Z(y, \gamma; T) = \left(\frac{y}{c(\gamma; R)}\right)^{1/\beta} (1 + o(1)) \qquad (y \to \infty). \tag{170}$$

More precisely, $c(\gamma; R)\, Z(y, \gamma; T)^{\beta}/y \to 1$.

Compensated ETI. The compensated ETI is defined via the linearized schedule at y (Definition 12.3). Under GHH preferences (no income effects), the ETI equals the Frisch elasticity. For type γ at any income y in the top bracket, a local perturbation of the retention rate from $R(y)$ to $R(y) + dR$ changes earnings by $dy = (y/(\gamma\, R(y)))\, dR$. Hence

$$\zeta(y, \gamma) = \frac{1}{\gamma} =: e(\gamma), \tag{171}$$

independent of the schedule T and the income level y. This is an exact identity under GHH (not an asymptotic approximation): it follows from the FOC structure $n = \bar{n}(z)\, R^{1/\gamma}$ and the fact that the MRS is independent of consumption.

Summary. The behavioral objects under the general schedule T with $T'(y) \to \tau$ are asymptotically equivalent to their constant-rate counterparts:

$$\mathcal{Y}(z, \gamma; T) \sim c(\gamma; R)\, z^{\beta}, \tag{172}$$

$$\mathcal{Y}_z(z, \gamma; T) \sim \beta\, c(\gamma; R)\, z^{\beta-1}, \tag{173}$$

$$Z(y, \gamma; T) \sim \left(\frac{y}{c(\gamma; R)}\right)^{1/\beta}, \tag{174}$$

$$\zeta(y, \gamma) = \frac{1}{\gamma} \qquad \text{(exact)}. \tag{175}$$

The top-rate formula (162) involves ratios of integrals of these quantities over γ. The $o(1)$ corrections in (172)–(174) vanish in the limit $y \to \infty$, so the asymptotic ratio depends only on the limiting rate τ and not on the path $T'(y)$ takes to reach τ. Steps 3–5 below use only the asymptotic equivalences (172)–(175).

12.9.4 Step 3: Pareto tail of the income distribution

Assumption 12.3 (Factorized primitives with Pareto tail)**.** The joint density factorizes as $f_{Z,\gamma}(z, \gamma) = f_Z(z)\, f_\gamma(\gamma)$, with f_γ supported on a compact set $\Gamma \subset (0, \infty)$, and f_Z having a Pareto upper tail:

$$\int_{z \geq t} f_Z(z)\, dz \sim k\, t^{-\alpha_Z} \qquad (t \to \infty), \tag{176}$$

for some $k > 0$ and $\alpha_Z > 1$. Equivalently, $f_Z(z) \sim \alpha_Z\, k\, z^{-(\alpha_Z+1)}$ for large z, and the inverse hazard rate satisfies

$$\frac{\int_{z \geq t} f_Z(z)\, dz}{t\, f_Z(t)} \to \frac{1}{\alpha_Z} \qquad (t \to \infty). \tag{177}$$

Lemma 12.6 (Income tail exponent)**.** *Under Assumption 12.3 and the calibrated GHH mapping* (166)*, the income distribution has a Pareto upper tail with exponent*

$$\alpha_Y := \frac{\alpha_Z}{\beta}, \qquad \beta = 1 + \frac{1}{\bar{\gamma}}, \tag{178}$$

common across γ-types (the calibrated shifters ensure that β is the same for all types). Moreover, for each fixed $\gamma \in \Gamma$:

$$\Pr(Y > y \mid \gamma) = \int_{z \geq Z(y,\gamma;\tau)} f_Z(z)\, dz \sim k\, c(\gamma; R)^{\alpha_Y}\, y^{-\alpha_Y} \qquad (y \to \infty). \tag{179}$$

Proof. For a fixed γ, $\Pr(Y > y \mid \gamma) = \Pr(\mathcal{Y}(Z, \gamma; T) > y)$. By Lemma 12.4, for any $\varepsilon > 0$ there exists z_0 such that for all $z \geq z_0$:

$$(1 - \varepsilon)\, c(\gamma; R)\, z^\beta \leq \mathcal{Y}(z, \gamma; T) \leq (1 + \varepsilon)\, c(\gamma; R)\, z^\beta.$$

The event $\{\mathcal{Y}(Z, \gamma; T) > y\}$ is therefore sandwiched between $\{c(\gamma; R)\, Z^\beta (1+\varepsilon) > y\}$ and $\{c(\gamma; R)\, Z^\beta (1 - \varepsilon) > y\}$ for y large enough (so that $Z(y, \gamma; T) \geq z_0$). This gives:

$$\int_{z \geq (y/((1+\varepsilon)c))^{1/\beta}} f_Z(z)\, dz \;\leq\; \Pr(Y > y \mid \gamma) \;\leq\; \int_{z \geq (y/((1-\varepsilon)c))^{1/\beta}} f_Z(z)\, dz,$$

where $c = c(\gamma; R)$. Using the Pareto tail (176) on both bounds:

$$\begin{aligned}
&\text{Upper bound:} \qquad \sim k \left(\frac{y}{(1-\varepsilon)\, c} \right)^{-\alpha_Z/\beta} = k\, (1-\varepsilon)^{\alpha_Y}\, c^{\alpha_Y}\, y^{-\alpha_Y}, \\
&\text{Lower bound:} \qquad \sim k \left(\frac{y}{(1+\varepsilon)\, c} \right)^{-\alpha_Z/\beta} = k\, (1+\varepsilon)^{\alpha_Y}\, c^{\alpha_Y}\, y^{-\alpha_Y}.
\end{aligned}$$

Letting $\varepsilon \to 0$, both bounds converge to $k\, c(\gamma; R)^{\alpha_Y}\, y^{-\alpha_Y}$, establishing (179). The exponent $\alpha_Y = \alpha_Z/\beta$ is the same for every γ because all types share the same power β (the content of the calibration). Only the scale factor $c(\gamma; R)^{\alpha_Y}$ varies. □

Remark 12.5 (Common tail exponent is essential). The fact that α_Y is common across γ-types is crucial for the top-rate formula: it ensures that all types contribute to both the numerator and denominator of (162) at the same polynomial rate $y^{-\alpha_Y}$, so none is asymptotically negligible. Without the calibrated shifters, the tail exponents would differ (Section 12.12.8), and only the heaviest-tailed type would survive in the limit.

12.9.5 Step 4: Asymptotic evaluation of numerator and denominator

The top-rate formula (162) involves a ratio $N(y)/D(y)$, and we need $\lim_{y\to\infty} N(y)/D(y)$. In Steps 1–2 we showed that under the general schedule T with $T'(y) \to \tau$, the behavioral objects $\mathcal{Y}$, $\mathcal{Y}_z$, and Z are *asymptotically equivalent* to their constant-rate counterparts ((172)–(174)), with multiplicative $1 + o(1)$ corrections. These corrections enter both N and D at the same polynomial order $y^{-\alpha_Y}$ and therefore cancel in the ratio:

$$\frac{N(y)}{D(y)} = \frac{N_0(y)\,(1 + o(1))}{D_0(y)\,(1 + o(1))} \to \frac{\lim N_0(y)}{\lim D_0(y)},$$

where N_0 and D_0 are the numerator and denominator evaluated under the *exact* constant-rate mapping $\mathcal{Y} = c(\gamma; R)\, z^\beta$. We may therefore compute the limit using the constant-rate formulas without loss of generality.

We evaluate both sides of:

$$\lim_{y\to\infty} \frac{T'(y)}{1 - T'(y)} = \lim_{y\to\infty} \frac{N(y)}{D(y)}, \tag{180}$$

where

$$N(y) := \int_\Gamma \int_{z \geq Z(y,\gamma;\tau)} f_Z(z)\, f_\gamma(\gamma)\, dz\, d\gamma, \tag{181}$$

$$D(y) := y \int_\Gamma \zeta(y,\gamma)\, \frac{f_Z(Z(y,\gamma;\tau))\, f_\gamma(\gamma)}{\mathcal{Y}_z(Z(y,\gamma;\tau),\gamma;\tau)}\, d\gamma. \tag{182}$$

Asymptotic evaluation of the numerator $N(y)$. By Fubini and Lemma 12.6:

$$N(y) = \int_\Gamma f_\gamma(\gamma) \left[\int_{z \geq Z(y,\gamma;\tau)} f_Z(z)\, dz \right] d\gamma. \tag{183}$$

For each γ, the inner integral satisfies (by (179)):

$$\int_{z \geq Z(y,\gamma;\tau)} f_Z(z)\, dz \sim k\, c(\gamma; R)^{\alpha_Y}\, y^{-\alpha_Y} \qquad (y \to \infty).$$

We wish to interchange the limit $y \to \infty$ with the integral over γ. This requires a domination condition.

Lemma 12.7 (Dominated convergence for the numerator)**.** *On compact Γ, the function $\gamma \mapsto c(\gamma; R)^{\alpha_Y} = \kappa^{\alpha_Y} R^{\alpha_Y/\gamma}$ is continuous and bounded (since γ is bounded away from 0 and ∞ on Γ). Therefore, for all y sufficiently large and all $\gamma \in \Gamma$:*

$$\int_{z \geq Z(y,\gamma;\tau)} f_Z(z)\, dz \leq C\, y^{-\alpha_Y}$$

for some constant $C = C(\Gamma, R, \kappa, \alpha_Z) < \infty$ independent of γ. Since f_γ is integrable on compact Γ, the function $C\, y^{-\alpha_Y} f_\gamma(\gamma)$ is an integrable dominator.

Proof. Since $c(\gamma; R) = \kappa\, R^{1/\gamma}$ is continuous on compact Γ and strictly positive, it is bounded: $c_{\min} \leq c(\gamma; R) \leq c_{\max}$ for some $0 < c_{\min} \leq c_{\max} < \infty$. Hence $c(\gamma; R)^{\alpha_Y} \leq c_{\max}^{\alpha_Y}$. The Pareto tail approximation (176) gives, for all $t \geq t_0$:

$$\int_{z \geq t} f_Z(z)\, dz \leq 2k\, t^{-\alpha_Z}.$$

Since $Z(y,\gamma;\tau) = (y/c(\gamma; R))^{1/\beta} \geq (y/c_{\max})^{1/\beta}$, which exceeds t_0 for y large enough (uniformly in γ), we have

$$\int_{z \geq Z} f_Z(z)\, dz \leq 2k\, Z^{-\alpha_Z} = 2k \left(\frac{y}{c(\gamma; R)} \right)^{-\alpha_Z/\beta} = 2k\, c(\gamma; R)^{\alpha_Y}\, y^{-\alpha_Y} \leq 2k\, c_{\max}^{\alpha_Y}\, y^{-\alpha_Y}.$$

Setting $C = 2k\, c_{\max}^{\alpha_Y}$ gives the uniform bound. □

Applying the dominated convergence theorem with the dominator from Lemma 12.7:

$$N(y) \sim y^{-\alpha_Y} \int_\Gamma k\, c(\gamma; R)^{\alpha_Y}\, f_\gamma(\gamma)\, d\gamma = k\, y^{-\alpha_Y} \int_\Gamma c(\gamma; R)^{\alpha_Y}\, f_\gamma(\gamma)\, d\gamma. \tag{184}$$

More precisely, $y^{\alpha_Y} N(y) \to k \int_\Gamma c(\gamma; R)^{\alpha_Y} f_\gamma(\gamma)\, d\gamma$ as $y \to \infty$.

Asymptotic evaluation of the denominator $D(y)$. Substituting $\zeta(y,\gamma) = 1/\gamma$, $\mathcal{Y}_z \sim \beta\, c(\gamma; R)\, z^{\beta-1}$ (from (173)), and $Z \sim (y/c(\gamma; R))^{1/\beta}$ (from (174)) into (182):

$$D(y) = y \int_\Gamma \frac{1}{\gamma} \frac{f_Z(Z(y,\gamma;\tau))}{\beta\, c(\gamma; R)\, Z(y,\gamma;\tau)^{\beta-1}}\, f_\gamma(\gamma)\, d\gamma. \tag{185}$$

We evaluate $f_Z(Z)/(\beta\, c\, Z^{\beta-1})$ for large y. Using the Pareto density $f_Z(z) \sim \alpha_Z\, k\, z^{-(\alpha_Z+1)}$:

$$\begin{aligned}\frac{f_Z(Z)}{\beta\, c(\gamma; R)\, Z^{\beta-1}} &\sim \frac{\alpha_Z\, k\, Z^{-(\alpha_Z+1)}}{\beta\, c(\gamma; R)\, Z^{\beta-1}} \\ &= \frac{\alpha_Z\, k}{\beta\, c(\gamma; R)}\, Z^{-(\alpha_Z+\beta)} \\ &= \frac{\alpha_Z\, k}{\beta\, c(\gamma; R)} \left(\frac{y}{c(\gamma; R)}\right)^{-(\alpha_Z+\beta)/\beta}. \end{aligned} \tag{186}$$

We simplify the exponent. Since $\alpha_Y = \alpha_Z/\beta$:

$$\frac{\alpha_Z + \beta}{\beta} = \frac{\alpha_Z}{\beta} + 1 = \alpha_Y + 1.$$

Substituting back into (186):

$$\begin{aligned}\frac{f_Z(Z)}{\beta\, c(\gamma; R)\, Z^{\beta-1}} &\sim \frac{\alpha_Z\, k}{\beta\, c(\gamma; R)}\, c(\gamma; R)^{\alpha_Y+1}\, y^{-(\alpha_Y+1)} \\ &= \frac{\alpha_Z\, k}{\beta}\, c(\gamma; R)^{\alpha_Y}\, y^{-(\alpha_Y+1)}. \end{aligned} \tag{187}$$

Substituting (187) into (185) and using the same dominated convergence argument as for the numerator (the integrand is bounded by $C'\, y^{-\alpha_Y}$ uniformly in γ on compact Γ):

$$\begin{aligned} D(y) &\sim y \cdot \frac{\alpha_Z\, k}{\beta}\, y^{-(\alpha_Y+1)} \int_\Gamma \frac{1}{\gamma}\, c(\gamma; R)^{\alpha_Y}\, f_\gamma(\gamma)\, d\gamma \\ &= \frac{\alpha_Z\, k}{\beta}\, y^{-\alpha_Y} \int_\Gamma \frac{1}{\gamma}\, c(\gamma; R)^{\alpha_Y}\, f_\gamma(\gamma)\, d\gamma. \end{aligned} \tag{188}$$

More precisely, $y^{\alpha_Y}\, D(y) \to \frac{\alpha_Z\, k}{\beta} \int_\Gamma \frac{1}{\gamma}\, c(\gamma; R)^{\alpha_Y}\, f_\gamma(\gamma)\, d\gamma$.

Remark 12.6 (Uniform integrability). The interchange of limit and integral in both $N(y)$ and $D(y)$ relies on the compactness of Γ. On compact Γ, both $c(\gamma; R)^{\alpha_Y}$ and $(1/\gamma)\, c(\gamma; R)^{\alpha_Y}$ are continuous, bounded functions, providing the uniform domination needed for the dominated convergence theorem. If Γ were unbounded (e.g., $\gamma_{\min} = 0$ or $\gamma_{\max} = \infty$), additional integrability conditions on f_γ would be needed—specifically, exponential moment control $\int R^{\alpha_Y/\gamma}\, f_\gamma(\gamma)\, d\gamma < \infty$.

12.9.6 Step 5: Taking the ratio and simplifying

Dividing (184) by (188), the common factors k and $y^{-\alpha_Y}$ cancel:

$$\frac{\tau}{1-\tau} = \lim_{y\to\infty} \frac{N(y)}{D(y)} = \frac{k \int_\Gamma c(\gamma; R)^{\alpha_Y} \, f_\gamma(\gamma)\, d\gamma}{\frac{\alpha_Z k}{\beta} \int_\Gamma \frac{1}{\gamma}\, c(\gamma; R)^{\alpha_Y} \, f_\gamma(\gamma)\, d\gamma}$$

$$= \frac{\beta}{\alpha_Z} \cdot \frac{\int_\Gamma c(\gamma; R)^{\alpha_Y} \, f_\gamma(\gamma)\, d\gamma}{\int_\Gamma \frac{1}{\gamma}\, c(\gamma; R)^{\alpha_Y} \, f_\gamma(\gamma)\, d\gamma}. \tag{189}$$

Since $\alpha_Y = \alpha_Z/\beta$, we have $\beta/\alpha_Z = 1/\alpha_Y$. Hence

$$\frac{\tau}{1-\tau} = \frac{1}{\alpha_Y} \cdot \frac{\int_\Gamma c(\gamma; R)^{\alpha_Y} \, f_\gamma(\gamma)\, d\gamma}{\int_\Gamma \frac{1}{\gamma}\, c(\gamma; R)^{\alpha_Y} \, f_\gamma(\gamma)\, d\gamma}. \tag{190}$$

Now substitute $c(\gamma; R) = \kappa\, R^{1/\gamma}$:

$$c(\gamma; R)^{\alpha_Y} = \kappa^{\alpha_Y}\, R^{\alpha_Y/\gamma} = \kappa^{\alpha_Y}\, R^{\alpha_Y\, e(\gamma)}, \tag{191}$$

where $e(\gamma) = 1/\gamma$ is the compensated elasticity. The constant κ^{α_Y} appears in both numerator and denominator, and cancels:

$$\frac{\tau}{1-\tau} = \frac{1}{\alpha_Y} \cdot \frac{\int_\Gamma R^{\alpha_Y/\gamma} \, f_\gamma(\gamma)\, d\gamma}{\int_\Gamma \frac{1}{\gamma}\, R^{\alpha_Y/\gamma} \, f_\gamma(\gamma)\, d\gamma}. \tag{192}$$

Change of variables to the elasticity $e = 1/\gamma$. Let $g(e)$ denote the induced density of $e = 1/\gamma$ on $[e_{\min}, e_{\max}] = [1/\gamma_{\max}, 1/\gamma_{\min}]$. Note $R^{\alpha_Y/\gamma} = R^{\alpha_Y\, e}$ and $1/\gamma = e$. The integrals in (192) become:

$$\frac{\tau}{1-\tau} = \frac{1}{\alpha_Y} \cdot \frac{\int g(e)\, R^{\alpha_Y\, e}\, de}{\int e\, g(e)\, R^{\alpha_Y\, e}\, de}. \tag{193}$$

Inverting the ratio on the right-hand side:

Proposition 12.2 (Top-rate fixed point)**.** *Under Assumption 12.3, the GHH specification* (140) *with calibrated shifters* (146)*, and vanishing welfare weights in the tail, the asymptotic top marginal tax rate satisfies the fixed-point equation*

$$\boxed{\frac{\tau}{1-\tau} = \frac{1}{\alpha_Y\, \bar{e}_{\text{eff}}(\tau)},} \tag{194}$$

where the ***effective elasticity*** *is*

$$\bar{e}_{\text{eff}}(\tau) := \frac{\displaystyle\int e\, g(e)\, R^{\alpha_Y e}\, de}{\displaystyle\int g(e)\, R^{\alpha_Y e}\, de} = \mathbb{E}_{w_\tau}[e], \qquad w_\tau(e) \propto g(e)\, R^{\alpha_Y e}, \tag{195}$$

with $R = 1 - \tau$*,* $e = 1/\gamma$*, and* $g(e)$ *the induced density of* e*.*

Remark 12.7 (Interpretation of $\bar{e}_{\text{eff}}(\tau)$). The effective elasticity $\bar{e}_{\text{eff}}(\tau)$ is the mean of e under the **tilted density** $w_\tau(e) \propto g(e)\, R^{\alpha_Y e}$. The tilt factor $R^{\alpha_Y e} = (1-\tau)^{\alpha_Y e}$ is a decreasing function of e (since $R < 1$), so it downweights high-elasticity agents relative to the primitive density $g(e)$. This reflects the endogenous composition of the top tail: high-elasticity agents reduce their labor supply more under taxation, earn less at any given z, and are therefore less represented in the extreme upper tail of the income distribution.

Remark 12.8 (Why the effective elasticity is below the mean). Since $R^{\alpha_Y e}$ is strictly decreasing in e for $\tau > 0$, the tilted density w_τ is a strict downward shift of g. By the standard property of exponential tilts, $\mathbb{E}_{w_\tau}[e] < \mathbb{E}_g[e] = \bar{e}$ whenever the distribution of e is non-degenerate. This is the key force raising the optimal top rate above the Diamond–Saez benchmark $\tau_{\text{DS}}(\bar{e}) = 1/(1 + \alpha_Y \bar{e})$.

Remark 12.9 (Assumptions used). The derivation relies on five assumptions:

(i) **Existence of a limiting top rate** $\tau = \lim_{y\to\infty} T'(y) \in (0,1)$ (Assumption 12.2), together with the regularity condition (169) ($y\,T''(y) \to 0$), which ensures the Jacobian asymptotics (Lemma 12.5).

(ii) **Compact support** of γ (equivalently, e), ensuring the uniform bounds in Lemma 12.7 and the validity of the dominated convergence interchange.

(iii) **Factorization** $f_{Z,\gamma} = f_Z\, f_\gamma$, used to separate the z- and γ-integrations.

(iv) **Pareto tail** of f_Z (Assumption 12.3), providing the asymptotic power-law scalings.

(v) **Zero welfare weights** in the tail, allowing the replacement of $(1 - \hat{g})$ by 1 in the numerator.

Without (ii), one would need exponential moment control $\int R^{\alpha_Y/\gamma}\, f_\gamma(\gamma)\, d\gamma < \infty$; without (iii), a conditional-tail assumption would be needed.

12.10 Bounds relative to the mean elasticity

Proposition 12.3 (Monotonicity of the effective elasticity)**.** $\bar{e}_{\text{eff}}(\tau)$ *is weakly decreasing in* τ *(equivalently, weakly increasing in* R*), with*

$$\frac{d}{d\ln R}\, \bar{e}_{\text{eff}}(R) = \alpha_Y\, \mathrm{Var}_{w_R}(e) \geq 0. \tag{196}$$

Proof. Define $I_k(R) = \int e^k\, g(e)\, R^{\alpha_Y e}\, de$ for $k = 0, 1$, so $\bar{e}_{\text{eff}} = I_1/I_0$. Differentiating: $dI_k/d\ln R = \alpha_Y\, I_{k+1}$. By the quotient rule, $d(I_1/I_0)/d\ln R = \alpha_Y(I_2/I_0 - (I_1/I_0)^2) = \alpha_Y\, \mathrm{Var}_{w_R}(e) \geq 0$. □

Corollary 12.1 (Bounds on the top rate)**.** *For non-degenerate heterogeneity in e,*

$$\frac{1}{1+\alpha_Y\,\bar{e}} < \tau^* \le \frac{1}{1+\alpha_Y\,e_{\min}}. \tag{197}$$

The lower bound is the Diamond–Saez rate at the mean elasticity; the upper bound is the Diamond–Saez rate at the lowest elasticity.

12.11 Multiplicity analysis for the top rate

We now investigate whether equation (194) can admit multiple solutions. Define

$$\Phi(\tau) := \frac{\tau}{1-\tau} - \frac{1}{\alpha_Y\,\bar{e}_{\text{eff}}(\tau)}.$$

A top rate is a root of Φ.

12.11.1 Existence

Lemma 12.8 (Existence of at least one solution)**.** *The fixed-point equation $\Phi(\tau) = 0$ has at least one solution $\tau^* \in (0, 1)$.*

Proof. At $\tau = 0$: $\tau/(1-\tau) = 0$ while $1/(\alpha_Y\bar{e}) > 0$, so $\Phi(0) < 0$. As $\tau \to 1^-$: $\tau/(1-\tau) \to +\infty$, while $1/(\alpha_Y\bar{e}_{\text{eff}}(\tau)) \le 1/(\alpha_Y e_{\min}) < \infty$ (since $\bar{e}_{\text{eff}}(\tau) \ge e_{\min}$ always). Hence $\Phi(\tau) \to +\infty$. By continuity, Φ has at least one zero. □

12.11.2 The possibility of multiple solutions

Conversely, multiplicity requires $\Phi'(\tau) \le 0$ for some τ, i.e., the composition effect must be locally strong enough to overturn the mechanical increase in $\tau/(1-\tau)$:

$$\text{Var}_{w_\tau}(e) \ge \frac{\bar{e}_{\text{eff}}(\tau)^2}{1-\tau} \qquad \text{for some } \tau \in (0,1). \tag{198}$$

Proposition 12.4 (Multiplicity under bimodal elasticities)**.** *There exist elasticity distributions $g(e)$ and parameter values $\alpha_Y > 0$ for which the fixed-point equation $\Phi(\tau) = 0$ has at least three solutions.*

Proof. Take $\alpha_Y = 1$ and a bimodal distribution: $e = e_L := 0.01$ with probability 0.1, and $e = e_H := 5$ with probability 0.9. Then $\bar{e} = 4.501$.

The effective elasticity is

$$\bar{e}_{\text{eff}}(\tau) = \frac{0.1 \cdot 0.01 \cdot R^{0.01} + 0.9 \cdot 5 \cdot R^5}{0.1 \cdot R^{0.01} + 0.9 \cdot R^5}, \qquad R = 1-\tau.$$

Computing $\Phi(\tau)$ at selected points:

τ	$\bar{e}_{\text{eff}}$	$1/(\alpha_Y\bar{e}_{\text{eff}})$	$\Phi(\tau)$
0	4.501	0.222	−0.222
0.3	3.016	0.332	+0.097
0.5	1.111	0.900	+0.100
0.7	0.118	8.460	−6.127

The sign pattern is $(-,+,+,-)$. Since $\Phi(\tau) \to +\infty$ as $\tau \to 1^-$ (because $\tau/(1-\tau) \to \infty$ while $\mathcal{R}(\tau) \le 1/(\alpha_Y \cdot 0.01) = 100$), Φ returns to positive values. By the intermediate value theorem, Φ has at least three zeros. □

12.11.3 Sufficient conditions for uniqueness

We now give precise conditions, stated in terms of the primitives $(g, e_{\min}, e_{\max}, \alpha_Y)$, under which the fixed-point equation $L(\tau) = \mathcal{R}(\tau)$ has a *unique* solution. Throughout, g denotes the density of the elasticity distribution on $[e_{\min}, e_{\max}]$ with $0 < e_{\min} \leq e_{\max} < \infty$, and $\alpha_Y > 0$ is the Pareto tail exponent of the earnings distribution.

Proposition 12.5 (Uniqueness of the top-rate fixed point)**.** *Define*

$$\Phi(\tau) := L(\tau) - \mathcal{R}(\tau) = \frac{\tau}{1-\tau} - \frac{1}{\alpha_Y\, \bar{e}_{\text{eff}}(\tau)},$$

where $\bar{e}_{\text{eff}}(\tau) = \mathbb{E}_{w_\tau}[e]$ *with tilted weights* $w_\tau(e) \propto g(e)\,(1-\tau)^{\alpha_Y e}$*. Then*

$$\Phi'(\tau) = \frac{1}{(1-\tau)^2} - \frac{\mathrm{Var}_{w_\tau}(e)}{(1-\tau)\,\bar{e}_{\text{eff}}(\tau)^2}. \tag{199}$$

Hence the top-rate fixed point is unique if

$$\mathrm{CV}^2_{w_\tau}(e) := \frac{\mathrm{Var}_{w_\tau}(e)}{\bar{e}_{\text{eff}}(\tau)^2} < \frac{1}{1-\tau} \qquad \textit{for all } \tau \in (0,1). \tag{200}$$

This sufficient condition holds in particular if either:

(a) $e_{\max}/e_{\min} \leq 3$*, or*

(b) g *is log-concave on its support (a subinterval of* $\mathbb{R}_+$*).*

Conversely, multiplicity requires (200) *to fail for at least one* $\tau \in (0,1)$*.*

The proof is organized as a sequence of lemmas.

Lemma 12.9 (Derivative of Φ)**.** *For all* $\tau \in (0,1)$*, equation* (199) *holds.*

Proof. The derivative $L'(\tau) = 1/(1-\tau)^2$ is immediate. For $\mathcal{R}'$, set $R = 1-\tau$ and define the moment integrals

$$I_k(R) := \int_{e_{\min}}^{e_{\max}} e^k\, g(e)\, R^{\alpha_Y e}\, de, \qquad k = 0, 1, 2.$$

Then $\bar{e}_{\text{eff}} = I_1/I_0$ and $\mathcal{R} = I_0/(\alpha_Y I_1)$. Differentiating in $\ln R$:

$$\frac{dI_k}{d\ln R} = \int e^k\, g(e)\, \alpha_Y e\, R^{\alpha_Y e}\, de = \alpha_Y\, I_{k+1}.$$

By the quotient rule,

$$\frac{d\bar{e}_{\text{eff}}}{d\ln R} = \frac{\alpha_Y I_2\, I_0 - \alpha_Y I_1^2}{I_0^2} = \alpha_Y \left(\frac{I_2}{I_0} - \left(\frac{I_1}{I_0}\right)^2 \right) = \alpha_Y\, \mathrm{Var}_{w_\tau}(e).$$

Since $d\ln R/d\tau = -1/(1-\tau)$:

$$\frac{d\bar{e}_{\text{eff}}}{d\tau} = -\frac{\alpha_Y\, \mathrm{Var}_{w_\tau}(e)}{1-\tau}.$$

Finally, $\mathcal{R} = 1/(\alpha_Y\, \bar{e}_{\text{eff}})$ gives

$$\mathcal{R}'(\tau) = -\frac{1}{\alpha_Y\, \bar{e}_{\text{eff}}^2} \cdot \frac{d\bar{e}_{\text{eff}}}{d\tau} = \frac{\mathrm{Var}_{w_\tau}(e)}{(1-\tau)\,\bar{e}_{\text{eff}}(\tau)^2},$$

and (199) follows from $\Phi' = L' - \mathcal{R}'$. □

Lemma 12.10 (Slope condition implies uniqueness)**.** *If* (200) *holds, the fixed-point equation* $L(\tau) = \mathcal{R}(\tau)$ *has a unique solution in* $(0, 1)$*. Conversely, if there exist multiple solutions, then* $\Phi'(\tau_0) = 0$ *for some* $\tau_0 \in (0, 1)$*, i.e.,* (200) *fails at* τ_0*.*

Proof. By Lemma 12.8, $\Phi(0) < 0$ and $\Phi(\tau) \to +\infty$ as $\tau \to 1^-$, so at least one zero exists.

Suppose (200) holds. By Lemma 12.9, $\Phi'(\tau) > 0$ for all $\tau \in (0, 1)$, so Φ is strictly increasing. A continuous, strictly increasing function that starts negative and tends to $+\infty$ has exactly one zero.

For the converse: if $\tau_1^* < \tau_2^*$ are two solutions, then $\Phi(\tau_1^*) = \Phi(\tau_2^*) = 0$. By Rolle's theorem, $\Phi'(\tau_0) = 0$ for some $\tau_0 \in (\tau_1^*, \tau_2^*)$. By (199), this means

$$\frac{\mathrm{Var}_{w_{\tau_0}}(e)}{\bar{e}_{\mathrm{eff}}(\tau_0)^2} = \frac{1}{1 - \tau_0},$$

so (200) fails at τ_0. □

Lemma 12.11 (Popoviciu bound on the tilted distribution)**.** *For every* $\tau \in (0, 1)$*, the tilted distribution* w_τ *is supported on* $[e_{\min}, e_{\max}]$*. Consequently:*

(i) $\mathrm{Var}_{w_\tau}(e) \le (e_{\max} - e_{\min})^2/4$ *(Popoviciu's inequality),*

(ii) $\bar{e}_{\mathrm{eff}}(\tau) \ge e_{\min}$*.*

Proof. The tilted density $w_\tau(e) \propto g(e)\, R^{\alpha_Y e}$ inherits the support of g, since $R^{\alpha_Y e} > 0$ for all $e \in [e_{\min}, e_{\max}]$ and $R \in (0, 1)$.

Part (i). Popoviciu's variance inequality (1935) states that for any random variable supported on $[a, b]$, $\mathrm{Var}(X) \le (b - a)^2/4$. The bound is attained by the symmetric Bernoulli on $\{a, b\}$.

Part (ii). Since $e \ge e_{\min}$ on the support, $\bar{e}_{\mathrm{eff}}(\tau) = \mathbb{E}_{w_\tau}[e] \ge e_{\min}$. □

Proof of Proposition 12.5, part (a). By Lemma 12.11:

$$\mathrm{CV}^2_{w_\tau}(e) = \frac{\mathrm{Var}_{w_\tau}(e)}{\bar{e}_{\mathrm{eff}}(\tau)^2} \le \frac{(e_{\max} - e_{\min})^2}{4\, e_{\min}^2}.$$

If $e_{\max}/e_{\min} \le 3$, the right side satisfies

$$\frac{(e_{\max} - e_{\min})^2}{4\, e_{\min}^2} \le \frac{(3e_{\min} - e_{\min})^2}{4\, e_{\min}^2} = 1.$$

Since $1/(1 - \tau) > 1$ for all $\tau \in (0, 1)$, condition (200) holds and Lemma 12.10 gives uniqueness. □

Lemma 12.12 (Exponential tilts preserve log-concavity)**.** *If* g *is log-concave on* $[e_{\min}, e_{\max}]$ *and* $\lambda \ge 0$*, then the tilted density* $g_\lambda(e) \propto g(e)\, e^{-\lambda e}$ *is log-concave on* $[e_{\min}, e_{\max}]$*.*

Proof. A function $h > 0$ is log-concave if and only if $\log h$ is concave. We have

$$\log g_\lambda(e) = \log g(e) - \lambda e - \log Z(\lambda),$$

where $Z(\lambda) = \int g(e)\, e^{-\lambda e}\, de$ is the normalizing constant. Since $\log g(e)$ is concave by hypothesis and $-\lambda e$ is affine (hence concave), their sum is concave. □

Proof of Proposition 12.5, part (b). Identify the tilt parameter $\lambda = \alpha_Y |\ln(1-\tau)| \geq 0$, which is increasing in τ on $(0,1)$. The tilted distribution w_τ coincides with g_λ.

By Lemma 12.12, w_τ is log-concave for every $\tau \in (0,1)$. By Corollary E.1 (Appendix E.8):

$$\mathrm{CV}^2_{w_\tau}(e) \leq 1 \qquad \text{for all } \tau \in (0,1).$$

Since $1/(1-\tau) > 1$ for all $\tau \in (0,1)$, condition (200) holds and Lemma 12.10 gives uniqueness. □

Remark 12.10 (Scope of the two conditions).

(i) **Condition (a)** depends only on the ratio of extreme elasticities, requires no shape assumption on g, and is independent of α_Y. For typical calibrations with $e \in [0.1, 0.5]$ (i.e., $\gamma \in [2, 10]$), $e_{\max}/e_{\min} = 5$, so condition (a) fails.

(ii) **Condition (b)** places no restriction on the support ratio $e_{\max}/e_{\min}$ and is independent of α_Y. It encompasses the uniform, truncated normal, truncated exponential, triangular, and Beta(α, β) with $\alpha, \beta \geq 1$ families. Condition (b) subsumes (a) for log-concave densities, but (a) retains value because it applies even to irregular distributions provided the support is narrow.

(iii) **These conditions are sufficient, not exhaustive.** Dropping both conditions can permit multiplicity: the bimodal example in Proposition 12.4 violates both ($e_{\max}/e_{\min} = 500$ and the two-point distribution is not log-concave) and admits three solutions.

Remark 12.11 (Mechanism of multiplicity). Multiplicity is impossible under log-concavity; it therefore requires a non-log-concave elasticity distribution. Bimodality is one route, but not the only one: any density shape that generates a sufficiently sharp compositional transition—a rapid drop in $\bar{e}_{\text{eff}}(\tau)$ that causes $\mathcal{R}(\tau)$ to spike above $L(\tau)$ after an initial crossing—can produce additional fixed points. Among many standard log-concave families, the tilted CV^2 remains bounded by one, preventing the re-crossing that multiplicity requires.

Remark 12.12 (When uniqueness fails). When both conditions fail, the fixed-point equation generically has an odd number of solutions $\tau_1^* < \tau_2^* < \cdots$. Odd-indexed solutions are locally stable under naïve tax-adjustment dynamics $\tau_{k+1} = 1/(1 + \alpha_Y\, \bar{e}_{\text{eff}}(\tau_k))$, while even-indexed solutions are unstable. In applied work with multiplicity, the welfare ranking among equilibria requires global analysis of the planner's objective.

12.12 Robust extension: conditional-composition robustness

We now extend the robust analysis to two-dimensional heterogeneity. A key distinction arises between the one-dimensional and two-dimensional robust problems, which we address first.

12.12.1 Why the joint-distortion model does not apply

In Section 11, the robust planner faces uncertainty about the entire distribution of types z, and nature distorts the joint density via a Radon–Nikodym derivative $m(z)$. The resulting exponential twist $m(z) \propto \exp(-\mu\, T(y(z))/\theta)$ thins the z-tail super-exponentially, driving the optimal marginal tax to zero (Proposition 10.1).

If one naively extends this joint-distortion framework to two dimensions—letting nature distort the entire joint density $f_{Z,\gamma}(z,\gamma)$ via $m(z,\gamma) \propto \exp(-[\psi\,\mathcal{U} + \mu\,T(\mathcal{Y})]/\theta)$—the same super-exponential tail-thinning mechanism operates. The factor $\exp(-\mu\,T(\mathcal{Y})/\theta)$ suppresses all high-z agents regardless of γ, and the Section 10 logic (which applies after integrating out γ) forces $T' \to 0$.

In this section we model a *different* form of uncertainty. The planner knows the marginal productivity distribution $f_Z(z)$ (e.g., from observed income data) but is uncertain about the *conditional composition* $f(\gamma \mid z)$—the mix of elasticity types at each productivity level. This is natural: productivities are partially revealed by earnings, but unobserved elasticity heterogeneity is harder to identify. Under this formulation, nature distorts only the conditional distribution of γ given z, while the marginal $f_Z(z)$ is held fixed.

12.12.2 The conditional-composition robust problem

Let $f_{Z,\gamma}(z,\gamma) = f_Z(z)\,f_\gamma(\gamma)$ be the benchmark joint density (under independence). The robust planner solves

$$\max_T\ V(T), \qquad V(T) = \min_{\substack{m(\gamma|z)>0 \\ \mathbb{E}_\gamma[m(\gamma|z)]=1\ \forall z}} \mathcal{L}(T,m),$$

where nature's instrument is a *conditional* Radon–Nikodym derivative $m(\gamma \mid z)$ satisfying $\int_\Gamma m(\gamma \mid z)\,f_\gamma(\gamma)\,d\gamma = 1$ for each z, and the Lagrangian penalizes the conditional KL divergence:

$$\begin{aligned}\mathcal{L}(T,m) = \int_0^\infty \int_\Gamma \big[\psi(z,\gamma)\,\mathcal{U}(z,\gamma;T) + \mu\,T(\mathcal{Y}(z,\gamma;T))\big]\,m(\gamma \mid z)\,f_\gamma(\gamma)\,f_Z(z)\,d\gamma\,dz \\ + \theta \int_0^\infty \int_\Gamma m(\gamma \mid z)\log m(\gamma \mid z)\,f_\gamma(\gamma)\,f_Z(z)\,d\gamma\,dz.\end{aligned} \tag{201}$$

12.12.3 Nature's best response

For each fixed z, the inner minimization over $m(\cdot \mid z)$ is an entropy-penalized linear problem in m, identical in structure to Proposition 11.1 but restricted to the γ-dimension. By the same strict-convexity argument, the unique minimizer is:

$$m^*(\gamma \mid z;T) = \frac{\exp\big(-\frac{1}{\theta}\big[\psi(z,\gamma)\,\mathcal{U}(z,\gamma;T) + \mu\,T(\mathcal{Y}(z,\gamma;T))\big]\big)}{\int_\Gamma f_\gamma(\tilde\gamma)\,\exp\big(-\frac{1}{\theta}\big[\psi(z,\tilde\gamma)\,\mathcal{U}(z,\tilde\gamma;T) + \mu\,T(\mathcal{Y}(z,\tilde\gamma;T))\big]\big)\,d\tilde\gamma}. \tag{202}$$

The denominator ensures $\mathbb{E}_\gamma[m^*(\gamma \mid z)] = 1$ for each z.

The distorted conditional density is $\widetilde{f}_\gamma(\gamma \mid z) := f_\gamma(\gamma)\,m^*(\gamma \mid z;T)$, and the worst-case joint density is

$$\widetilde{f}_{Z,\gamma}(z,\gamma) = f_Z(z)\,\widetilde{f}_\gamma(\gamma \mid z). \tag{203}$$

Crucially, the z-marginal $\int_\Gamma \widetilde{f}_{Z,\gamma}(z,\gamma)\,d\gamma = f_Z(z)$ is *unchanged*—nature cannot thin the productivity tail.

Remark 12.13 (Contrast with Section 11). In the one-dimensional robust model, nature distorts the entire z-distribution, generating the exponential tail-thinning that drives $T' \to 0$. In the conditional-composition model, the z-marginal is fixed, so the Pareto tail of productivity is preserved. Nature's only lever is to change the mix of elasticity types at each z, which affects the effective elasticity in the tail but not the tail thickness. This is why a positive top rate can survive.

12.12.4 Envelope theorem and the Saez perturbation

For a perturbation $T_\alpha = T + \alpha\,\delta T$, the value function $V(\alpha) = \min_m \mathcal{L}(T_\alpha, m)$ satisfies

$$V'(0) = D_T\mathcal{L}(T, m^*(T))[\delta T]$$

by the same argument as in Proposition 11.2: for each z, nature's conditional best response has the closed-form exponential structure (202), which permits the entropic dual calculation and DCT argument.

With m fixed at $m^*(T)$, the Lagrangian takes the form of the non-robust Lagrangian (149) with $f_{Z,\gamma}$ replaced by the worst-case density $\widetilde{f}_{Z,\gamma}$ from (203). The entropy term does not depend on T and drops out. The worst-case income densities are

$$\widetilde{h}_T(y;\gamma) := \frac{\widetilde{f}_{Z,\gamma}(Z(y,\gamma;T),\gamma)}{\mathcal{Y}_z(Z(y,\gamma;T),\gamma;T)}, \qquad \widetilde{h}_T(y) := \int_\Gamma \widetilde{h}_T(y;\gamma)\,d\gamma. \tag{204}$$

Applying the Saez perturbation of Section 12.7 with $\widetilde{f}_{Z,\gamma}$ in place of $f_{Z,\gamma}$ yields the mechanical and behavioral effects:

$$d\widetilde{M} = \varepsilon\,d\tau \int_{y^*}^{\infty}\int_\Gamma (1-\hat{g}(y,\gamma))\,\widetilde{h}_T(y;\gamma)\,d\gamma\,dy + O(\varepsilon^2\,d\tau), \tag{205}$$

$$d\widetilde{B} = -\varepsilon\,d\tau\,\frac{T'(y^*)}{1-T'(y^*)}\,y^*\int_\Gamma \zeta(y^*,\gamma)\,\widetilde{h}_T(y^*;\gamma)\,d\gamma + O(\varepsilon^2\,d\tau). \tag{206}$$

Remark 12.14 (Welfare weights are independent of m). The welfare weight $g(y,\gamma) = \psi(z,\gamma)\,U_c/\mu$ is a per-capita quantity measuring the social value of a dollar to a specific type. The distortion m affects only the integration weight $\widetilde{f}_{Z,\gamma}$, not the per-capita social value.

12.12.5 The robust optimal tax formula

Setting $d\widetilde{M} + d\widetilde{B} = 0$:

Proposition 12.6 (Robust Saez formula under conditional-composition robustness)**.** *Under the assumptions of Proposition 12.1 and the conditional-composition robust model:*

$$\frac{T'(y^*)}{1-T'(y^*)} = \frac{\displaystyle\int_\Gamma\int_{z\geq Z(y^*,\gamma;T)} (1-\hat{g}(\mathcal{Y},\gamma))\,\widetilde{f}_{Z,\gamma}(z,\gamma)\,dz\,d\gamma}{\displaystyle y^*\int_\Gamma \zeta(y^*,\gamma)\,\widetilde{h}_T(y^*;\gamma)\,d\gamma}. \tag{207}$$

This has the same form as the non-robust formula (161) *with* $f_{Z,\gamma}$ *replaced by* $\widetilde{f}_{Z,\gamma} = f_Z \cdot \widetilde{f}_\gamma(\cdot \mid z)$*. The* z*-marginal is unchanged; only the conditional composition of* γ*-types is distorted.*

Proof. Divide $d\widetilde{M} + d\widetilde{B} = 0$ by $\varepsilon\,d\tau > 0$ and send $\varepsilon \to 0$. The mechanical effect can be rewritten in (z,γ)-space using the change-of-variables identity (Lemma 12.1) with $\widetilde{f}_{Z,\gamma}$ in place of $f_{Z,\gamma}$. Rearranging yields (207). □

12.12.6 Robust selection and the top rate

We now specialize to the GHH calibration with Pareto tails and a constant top marginal rate τ. In the top bracket, $T'' = 0$ and $R(y) \equiv R = 1 - \tau$, so the income mapping (166) gives $\mathcal{Y}(z, \gamma; \tau) = c(\gamma; R)\, z^\beta$ with $c(\gamma; R) = \kappa\, R^{1/\gamma}$. Tax paid is approximately $\tau\, c(\gamma; R)\, z^\beta$ for large z.

The robust distortion. For a finite set of types $\{\gamma_1, \ldots, \gamma_J\}$ with prior weights $\pi_j > 0$, $\sum_j \pi_j = 1$, we define the Radon–Nikodym derivative:

$$m(\gamma_j \mid z) = \frac{\exp\bigl(-\frac{\mu}{\theta}\, T(\mathcal{Y}(z, \gamma_j; \tau))\bigr)}{\sum_{i=1}^{J} \pi_i \exp\bigl(-\frac{\mu}{\theta}\, T(\mathcal{Y}(z, \gamma_i; \tau))\bigr)}, \qquad \mu > 0,\ \theta > 0. \tag{208}$$

In the continuous case, with prior density $f_\gamma(\gamma)$ on Γ:

$$m(\gamma \mid z) = \frac{\exp\bigl(-\frac{\mu}{\theta}\, T(\mathcal{Y}(z, \gamma; \tau))\bigr)}{\int_\Gamma f_\gamma(\tilde\gamma) \exp\bigl(-\frac{\mu}{\theta}\, T(\mathcal{Y}(z, \tilde\gamma; \tau))\bigr)\, d\tilde\gamma}. \tag{209}$$

The distorted conditional density is $\widetilde{f}_\gamma(\gamma \mid z) := f_\gamma(\gamma)\, m(\gamma \mid z)$, and the robust joint density is $\widetilde{f}_{Z,\gamma}(z, \gamma) = f_Z(z)\, \widetilde{f}_\gamma(\gamma \mid z)$.

Linearization in the top bracket. In the top bracket, $T(y) \approx \tau\, y$ for large y.[1] Using $\mathcal{Y}(z, \gamma; \tau) = c(\gamma; R)\, z^\beta$:

$$T(\mathcal{Y}(z, \gamma; \tau)) \approx \tau\, c(\gamma; R)\, z^\beta.$$

The rate function $D(\gamma)$. Define

$$D(\gamma) := \frac{c(\gamma; R)}{\theta} = \frac{\kappa\, R^{1/\gamma}}{\theta}, \qquad R = 1 - \tau \in (0, 1). \tag{210}$$

Then for large z:

$$m(\gamma_j \mid z) = \frac{e^{-\mu\tau\, D(\gamma_j)\, z^\beta}}{\sum_i \pi_i\, e^{-\mu\tau\, D(\gamma_i)\, z^\beta}}.$$

Remark 12.15 (Economic content of $D(\gamma)$). The coefficient $c(\gamma; R) = \kappa\, R^{1/\gamma}$ is the *level of earnings at a given productivity*: a type-γ worker earning $\mathcal{Y} = c(\gamma; R)\, z^\beta$ pays tax $\tau\, c(\gamma; R)\, z^\beta$. Two channels make c increasing in γ (decreasing in the elasticity $e = 1/\gamma$):

(i) **Retention-rate sensitivity.** The factor $R^{1/\gamma}$ with $R < 1$: a higher elasticity (larger $1/\gamma$) raises the exponent, making $R^{1/\gamma}$ smaller—high-elasticity workers reduce labor supply aggressively in response to taxation, earning less at every z.

(ii) **Revenue consequence.** Since tax paid is $\tau\, c(\gamma; R)\, z^\beta$, a smaller c means less revenue. High-elasticity workers are "tax-light."

The function $D(\gamma) = c(\gamma; R)/\theta$ ranks types by their revenue contribution per unit of z^β: the adversary's distortion $m \propto e^{-\mu\tau\, D(\gamma)\, z^\beta}$ penalizes each type in proportion to the tax revenue it generates.

[1] If $T'(y) \to \tau$ as $y \to \infty$, then by the fundamental theorem of calculus $T(y)/y = T(y_0)/y + (1/y) \int_{y_0}^{y} T'(s)\, ds \to \tau$. In the Radon–Nikodym derivative, the $o(y)$ remainder contributes a factor $e^{-(\mu/\theta)\, o(Y)}$ which is negligible relative to the leading $e^{-(\mu\tau/\theta)\, Y}$ as $z \to \infty$.

Monotonicity of $D(\gamma)$. Writing $h(\gamma) := \log D(\gamma) = \log(\kappa/\theta) + (1/\gamma)\log R$, we have

$$h'(\gamma) = -\frac{\log R}{\gamma^2}.$$

Since $R = 1 - \tau \in (0,1)$, $\log R < 0$, so $h'(\gamma) > 0$ for all $\gamma > 0$. Therefore $D(\gamma)$ is strictly increasing in γ, and the unique minimizer of D on any subset of $(0,\infty)$ is the smallest element.

Asymptotic selection: the discrete case.

Lemma 12.13 (Discrete tail selection)**.** *Let $\gamma_1 < \gamma_2 < \cdots < \gamma_J$. Under* (208) *with the top-bracket approximation:*

$$\widetilde{f}_\gamma(\gamma_1 \mid z) \to 1, \qquad \widetilde{f}_\gamma(\gamma_j \mid z) \to 0 \quad (j > 1), \qquad \text{as } z \to \infty.$$

Proof. For any pair (j,k):

$$\frac{m(\gamma_j \mid z)}{m(\gamma_k \mid z)} = \exp\bigl(-\mu\tau[D(\gamma_j) - D(\gamma_k)]\, z^\beta\bigr).$$

If $D(\gamma_j) > D(\gamma_k)$, this ratio converges to 0 as $z \to \infty$; if $D(\gamma_j) < D(\gamma_k)$, it diverges to $+\infty$. Since D is strictly increasing and $\gamma_1 < \gamma_j$ for $j > 1$, we have $D(\gamma_1) < D(\gamma_j)$, so

$$\frac{m(\gamma_j \mid z)}{m(\gamma_1 \mid z)} = \exp\bigl(-\mu\tau[D(\gamma_j) - D(\gamma_1)]\, z^\beta\bigr) \to 0 \qquad (z \to \infty).$$

It follows that $\widetilde{f}_\gamma(\gamma_1 \mid z) = \pi_1\, m(\gamma_1 \mid z) \to 1$ and $\widetilde{f}_\gamma(\gamma_j \mid z) \to 0$ for $j > 1$. □

Remark 12.16 (Economic intuition). The exponential structure $e^{-\mu\tau\, D(\gamma)\, z^\beta}$ makes the selection total: even a tiny gap $D(\gamma_j) - D(\gamma_1) > 0$ is amplified by $z^\beta \to \infty$, driving the ratio $m(\gamma_j \mid z)/m(\gamma_1 \mid z)$ to zero exponentially fast. No polynomial correction (from the prior π_j, the Pareto tail, or the Jacobian) can overcome this exponential suppression. Nature, seeking to minimize the planner's revenue, loads mass on the highest-elasticity type γ_1—the type that generates the least revenue.

Asymptotic selection: the continuous case.

Lemma 12.14 (Laplace principle for the continuous case)**.** *Let $\Gamma = [\gamma_{\min}, \gamma_{\max}]$ compact, with $f_\gamma(\gamma)$ continuous and strictly positive on Γ. Let $\phi : \Gamma \to \mathbb{R}$ be continuous and bounded. Then*

$$\lim_{z\to\infty} \int_\Gamma \phi(\gamma)\, \widetilde{f}_\gamma(\gamma \mid z)\, d\gamma = \phi(\gamma^*), \qquad \gamma^* = \gamma_{\min}.$$

Equivalently, $\widetilde{f}_\gamma(\cdot \mid z) \Rightarrow \delta_{\gamma_{\min}}$ as $z \to \infty$.

Proof. Write

$$\widetilde{f}_\gamma(\gamma \mid z) = \frac{f_\gamma(\gamma)\, e^{-\mu\tau\, D(\gamma)\, z^\beta}}{Z(z)}, \qquad Z(z) = \int_\Gamma f_\gamma(\tilde\gamma)\, e^{-\mu\tau\, D(\tilde\gamma)\, z^\beta}\, d\tilde\gamma.$$

Step 1: Exponential gap. Fix $\varepsilon > 0$ and define $\Gamma_1 := \{\gamma : |\gamma - \gamma^*| < \varepsilon\}$ and $\Gamma_2 := \Gamma \setminus \Gamma_1$. Since D is continuous with unique minimizer at $\gamma^* = \gamma_{\min}$, there exists $\delta > 0$ such that $D(\gamma) \geq D(\gamma^*) + \delta$ for all $\gamma \in \Gamma_2$.

Step 2: Bounding the mass on Γ_2. For $\gamma \in \Gamma_2$:

$$f_\gamma(\gamma)\, e^{-\mu\tau\, D(\gamma)\, z^\beta} \le \|f_\gamma\|_\infty\, e^{-\mu\tau\,(D(\gamma^*)+\delta)\, z^\beta}.$$

Integrating over Γ_2:

$$\int_{\Gamma_2} f_\gamma(\gamma)\, e^{-\mu\tau\, D(\gamma)\, z^\beta}\, d\gamma \le \|f_\gamma\|_\infty\, |\Gamma|\, e^{-\mu\tau\,(D(\gamma^*)+\delta)\, z^\beta}.$$

The denominator $Z(z)$ is bounded below by the contribution from Γ_1:

$$Z(z) \ge e^{-\mu\tau\, D(\gamma^*)\, z^\beta} \int_{\Gamma_1} f_\gamma(\gamma)\, d\gamma =: e^{-\mu\tau\, D(\gamma^*)\, z^\beta}\, p_1,$$

where $p_1 = \int_{\Gamma_1} f_\gamma > 0$ since f_γ is strictly positive. Therefore:

$$\widetilde{f}_\gamma(\Gamma_2 \mid z) = \frac{\int_{\Gamma_2} f_\gamma\, e^{-\mu\tau\, D\, z^\beta}\, d\gamma}{Z(z)} \le \frac{\|f_\gamma\|_\infty\, |\Gamma|}{p_1}\, e^{-\mu\tau\,\delta\, z^\beta} \to 0 \qquad (z \to \infty).$$

Step 3: Continuity near the minimizer. On Γ_1, ϕ is uniformly continuous, so $|\phi(\gamma) - \phi(\gamma^*)| < \eta$ for any prescribed $\eta > 0$ (choosing ε small enough). Then:

$$\begin{aligned}\left|\int_\Gamma \phi(\gamma)\, \widetilde{f}_\gamma(\gamma \mid z)\, d\gamma - \phi(\gamma^*)\right| &\le \int_{\Gamma_1} |\phi(\gamma) - \phi(\gamma^*)|\, \widetilde{f}_\gamma(\gamma \mid z)\, d\gamma + \int_{\Gamma_2} |\phi(\gamma) - \phi(\gamma^*)|\, \widetilde{f}_\gamma(\gamma \mid z)\, d\gamma \\ &\le \eta + 2\|\phi\|_\infty\, \widetilde{f}_\gamma(\Gamma_2 \mid z).\end{aligned}$$

Letting $z \to \infty$ (so the second term vanishes) and then $\eta \to 0$ yields the result. □

Remark 12.17 (Structure of the Laplace argument). The proof has three ingredients:

(i) **Exponential gap.** Away from the minimizer γ^*, the rate function satisfies $D(\gamma) \ge D(\gamma^*) + \delta$ for some $\delta > 0$, so the mass on Γ_2 is suppressed by $e^{-\mu\tau\delta\, z^\beta} \to 0$.

(ii) **Polynomial irrelevance.** The prior density $f_\gamma(\gamma)$ is bounded and positive on Γ, contributing only polynomial prefactors. These are overwhelmed by the exponential tilt—any smooth prior is washed out in the tail.

(iii) **Continuity near the minimizer.** On the surviving neighborhood Γ_1, the test function $\phi(\gamma)$ is uniformly close to $\phi(\gamma^*)$, so the integral collapses to a point evaluation.

Top rate under robust distortion. We now combine the selection result with the top-rate formula. The robust joint density is $\widetilde{f}_{Z,\gamma}(z,\gamma) = f_Z(z)\, \widetilde{f}_\gamma(\gamma \mid z)$. The primitive-space top-rate condition with zero welfare weights at the top is

$$\frac{\tau}{1-\tau} = \lim_{y\to\infty} \frac{\displaystyle\int_\Gamma \int_{z \ge Z(y,\gamma;\tau)} f_Z(z)\, \widetilde{f}_\gamma(\gamma \mid z)\, dz\, d\gamma}{\displaystyle y \int_\Gamma \zeta(y,\gamma)\, \frac{f_Z(Z(y,\gamma;\tau))\, \widetilde{f}_\gamma(\gamma \mid Z(y,\gamma;\tau))}{\mathcal{Y}_z(Z(y,\gamma;\tau),\gamma;\tau)}\, d\gamma}. \tag{211}$$

Under the Laplace principle (Lemma 12.14), for large z, $\widetilde{f}_\gamma(\gamma \mid z)$ is sharply concentrated near $\gamma^* = \gamma_{\min}$. Since $f_Z(z)$ and $\mathcal{Y}_z$ have polynomial growth/decay in z, while the

selection via $m(\gamma \mid z)$ is exponential in z^β, the contributions of $\gamma \neq \gamma^*$ to both numerator and denominator in (211) are asymptotically negligible. Formally, for any neighborhood U of γ^*, the mass of $\widetilde{f}_\gamma(\cdot \mid z)$ outside U is $O(e^{-c\,z^\beta})$ for some $c > 0$, which is negligible relative to the Pareto tail and polynomial Jacobian terms.

Thus, as $y \to \infty$, the robust top-rate condition (211) converges to the homogeneous-type condition with $\gamma = \gamma^*$. In that homogeneous case, with income tail exponent α_Y and elasticity $e_{\max} = 1/\gamma_{\min}$, the Diamond–Saez formula gives:

Proposition 12.7 (Robust top rate with heterogeneous elasticities)**.** *Suppose:*

(i) Z has a Pareto tail with exponent $\alpha_Z > 1$, and $\mathcal{Y}(z, \gamma; \tau)$ has the calibrated GHH form (166)*, inducing income tail exponent $\alpha_Y = \alpha_Z/\beta$ (recall $\beta = 1 + 1/\bar{\gamma}$).*

(ii) $\gamma \in [\gamma_{\min}, \gamma_{\max}]$ with continuous, strictly positive prior density $f_\gamma(\gamma)$.

(iii) $\theta > 0$ and $\mu > 0$ in the Radon–Nikodym derivative (209)*.*

Then the distorted type distribution $\widetilde{f}_\gamma(\gamma \mid z)$ converges weakly to a point mass at $\gamma^* = \gamma_{\min}$ as $z \to \infty$, and the asymptotic robust top rate is

$$\tau^* = \frac{1}{1 + \alpha_Y\, e_{\max}}, \qquad e_{\max} = \frac{1}{\gamma_{\min}}. \tag{212}$$

The robust planner sets the top rate as if the entire tail consisted of the most elastic workers.

Remark 12.18 (Why the polynomial terms are irrelevant)*.* Both the numerator and denominator of (211) involve integrals over γ weighted by $\widetilde{f}_\gamma(\gamma \mid z)$. The other terms—the Pareto density $f_Z(z) \sim z^{-(\alpha_Z+1)}$, the Jacobian $\mathcal{Y}_z \sim z^{\beta-1}$, and the elasticity $\zeta(y, \gamma) = 1/\gamma$—are all polynomial (or constant) in z. Since the selection via $m(\gamma \mid z)$ is *exponential* in z^β, these polynomial terms cannot prevent the concentration. The γ-integral therefore collapses to a point evaluation at $\gamma^* = \gamma_{\min}$.

Remark 12.19 (Comparison with the non-robust case)*.* Without robustness, the top rate satisfies the fixed-point equation (194) with $\bar{e}_{\text{eff}}(\tau) < \bar{e}$, giving a rate strictly above the Diamond–Saez benchmark at the mean elasticity but strictly below the rate at the lowest elasticity. With robustness, nature's adversarial distortion pushes the effective elasticity all the way to $e_{\max}$, and the optimal rate drops to the lower bound $1/(1 + \alpha_Y\, e_{\max})$. The robust planner is maximally cautious, pricing in the worst-case composition of the top tail.

Remark 12.20 (Connection to one-dimensional robust taxation)*.* In the one-dimensional robust model (Section 11), the marginal tax rate decays to zero as $y \to \infty$ because the worst-case distribution has a thinner tail than the benchmark. In the two-dimensional model, the robust planner faces a different threat: not a thinner tail, but a *worse composition* of the tail. The tax rate remains positive at the top, but is set at the most conservative benchmark—the Diamond–Saez rate at the highest elasticity in the population.

12.12.7 Status-quo $T^0 \neq 0$

This section redoes the robust tail-selection argument when the robust distortion is defined relative to a status-quo linear tax schedule $T^0(y) = \tau^0\, y - \text{tr}^0$ with $\tau^0 \in (0, 1)$ and constant $\theta > 0$.

Status-quo calibration. Let $R^0 := 1 - \tau^0 \in (0,1)$. Under the status-quo, the reference worker has income $\bar{Y}^0(z) = \kappa\,(R^0)^{1/\bar{\gamma}}\,z^{\beta}$. For a type-$\gamma$ worker facing a general top marginal rate τ (net-of-tax rate $R = 1-\tau$), the income in the top bracket is

$$\mathcal{Y}(z,\gamma;\tau) = \bar{Y}^0(z)\left(\frac{R}{R^0}\right)^{1/\gamma}. \tag{213}$$

The rate function $D_\tau(\gamma)$. Define

$$D_\tau(\gamma) := q(\tau)^{1/\gamma}, \qquad q(\tau) := \frac{R}{R^0} = \frac{1-\tau}{1-\tau^0}. \tag{214}$$

Then

$$\frac{d}{d\gamma}\log D_\tau(\gamma) = -\frac{\log q(\tau)}{\gamma^2}.$$

This yields three regimes:

(i) If $\tau > \tau^0$, then $q(\tau) \in (0,1)$ and $\log q < 0$, so D_τ is strictly increasing and has a unique minimizer at $\gamma_{\min}$.

(ii) If $\tau < \tau^0$, then $q(\tau) > 1$ and $\log q > 0$, so D_τ is strictly decreasing and has a unique minimizer at $\gamma_{\max}$.

(iii) If $\tau = \tau^0$, then $q(\tau) = 1$ and $D_\tau(\gamma) \equiv 1$ on Γ.

Selection lemmas.

Lemma 12.15 (Discrete tail selection under status-quo)**.** *Fix $\tau \neq \tau^0$ and let T have asymptotic top rate τ. Under the Radon–Nikodym derivative* (208) *and the income mapping* (213)*:*

$$\widetilde{f}_\gamma(\gamma_j \mid z) \to \begin{cases} 1 & \text{if } \gamma_j = \arg\min_i D_\tau(\gamma_i), \\ 0 & \text{otherwise,} \end{cases} \qquad \text{as } z \to \infty.$$

In particular, if $\tau > \tau^0$ the selected type is $\gamma_{\min}$, while if $\tau < \tau^0$ the selected type is $\gamma_{\max}$.

Proof. Using the top linearization $T(Y) = \tau\,Y + o(Y)$ and (213), for large z the distorted conditional weights satisfy

$$\widetilde{f}_\gamma(\gamma_j \mid z) = \pi_j\, m(\gamma_j \mid z) \propto \pi_j \exp\Big(-\frac{\mu}{\theta}\,\tau\,\bar{Y}^0(z)\,D_\tau(\gamma_j) + o(\bar{Y}^0(z))\Big).$$

Since $\tau \neq \tau^0$, D_τ is nonconstant. For any pair (j,k) with $D_\tau(\gamma_j) \neq D_\tau(\gamma_k)$:

$$\frac{\widetilde{f}_\gamma(\gamma_j \mid z)}{\widetilde{f}_\gamma(\gamma_k \mid z)} = \frac{\pi_j}{\pi_k}\exp\Big(-\frac{\mu}{\theta}\,\tau\,\bar{Y}^0(z)\,[D_\tau(\gamma_j) - D_\tau(\gamma_k)] + o(\bar{Y}^0(z))\Big).$$

Since $\bar{Y}^0(z) = \kappa\,(R^0)^{1/\bar{\gamma}}\,z^{\beta} \to \infty$ and $D_\tau(\gamma_j) - D_\tau(\gamma_k)$ is a nonzero constant, the $o(\bar{Y}^0(z))$ remainder is dominated. If $D_\tau(\gamma_j) > D_\tau(\gamma_k)$ the ratio converges to 0; if $D_\tau(\gamma_j) < D_\tau(\gamma_k)$ it diverges to $+\infty$. Therefore all mass concentrates on minimizers of D_τ. □

Lemma 12.16 (Laplace principle under status-quo)**.** *Assume f_γ is continuous and strictly positive on Γ. Fix $\tau \neq \tau^0$ and let T have asymptotic top rate τ. Then for any bounded continuous $\phi : \Gamma \to \mathbb{R}$:*

$$\lim_{z\to\infty} \int_\Gamma \phi(\gamma)\, \widetilde{f}_\gamma(\gamma \mid z)\, d\gamma = \phi(\gamma^*(\tau)),$$

where

$$\gamma^*(\tau) = \begin{cases} \gamma_{\min}, & \tau > \tau^0, \\ \gamma_{\max}, & \tau < \tau^0. \end{cases}$$

Proof. Under $T(Y) = \tau\, Y + o(Y)$ and (213), the distorted conditional density satisfies

$$\widetilde{f}_\gamma(\gamma \mid z) \propto f_\gamma(\gamma)\, \exp\Big(-\frac{\mu}{\theta}\, \tau\, \bar{Y}^0(z)\, D_\tau(\gamma) + r(z,\gamma)\Big),$$

where $r(z,\gamma) = o(\bar{Y}^0(z))$ uniformly in γ. When $\tau \neq \tau^0$, D_τ has a unique minimizer $\gamma^*(\tau)$. Fix $\varepsilon > 0$ and let $U_\varepsilon = \{\gamma : |\gamma - \gamma^*(\tau)| < \varepsilon\}$. By continuity and uniqueness, there exists $\delta > 0$ such that $D_\tau(\gamma) \geq D_\tau(\gamma^*) + \delta$ for all $\gamma \in \Gamma \setminus U_\varepsilon$.

The leading-order gap $(\mu\tau/\theta)\, \delta\, \bar{Y}^0(z)$ between the exponent at γ^* and at any $\gamma \in \Gamma \setminus U_\varepsilon$ dominates the uniform remainder $\sup_\gamma |r(z,\gamma)| = o(\bar{Y}^0(z))$. Hence the mass assigned by $\widetilde{f}_\gamma(\cdot \mid z)$ to $\Gamma \setminus U_\varepsilon$ is bounded by $C\, e^{-c\, \bar{Y}^0(z)}$ for some $c > 0$, and vanishes as $z \to \infty$. The conclusion follows by the same uniform continuity argument as in Lemma 12.14. □

Reduction to a homogeneous-type Diamond–Saez condition. Fix $\tau \neq \tau^0$. By Lemma 12.15 (discrete) or Lemma 12.16 (continuous), $\widetilde{f}_\gamma(\gamma \mid z)$ concentrates at $\gamma^*(\tau) \in \{\gamma_{\min}, \gamma_{\max}\}$ as $z \to \infty$. Because this concentration is exponential in $\bar{Y}^0(z) \sim z^\beta$, whereas the Pareto tail and Jacobian terms in the top-rate formula are polynomial in z, the contributions of $\gamma \neq \gamma^*(\tau)$ to both numerator and denominator are asymptotically negligible.

In the homogeneous case with type $\gamma^*(\tau)$, the standard Pareto algebra yields

$$\frac{\tau}{1-\tau} = \frac{1}{\alpha_Y\, e(\gamma^*(\tau))} = \frac{\gamma^*(\tau)}{\alpha_Y} \qquad \Longleftrightarrow \qquad \tau = \frac{1}{1 + \alpha_Y/\gamma^*(\tau)}. \tag{215}$$

Define the endpoint Diamond–Saez rates:

$$\tau_L := \frac{1}{1 + \alpha_Y/\gamma_{\min}}, \qquad \tau_H := \frac{1}{1 + \alpha_Y/\gamma_{\max}}, \qquad \tau_L < \tau_H. \tag{216}$$

Then (215) implies:

$$\tau > \tau^0 \;\Rightarrow\; \gamma^*(\tau) = \gamma_{\min} \;\Rightarrow\; \tau = \tau_L, \qquad \tau < \tau^0 \;\Rightarrow\; \gamma^*(\tau) = \gamma_{\max} \;\Rightarrow\; \tau = \tau_H.$$

Proposition 12.8 (Clipping result)**.** *The asymptotic robust top rate is*

$$\tau^* = \min\{\max\{\tau^0, \tau_L\}, \tau_H\} = \begin{cases} \tau_L, & \tau^0 < \tau_L, \\ \tau^0, & \tau_L \leq \tau^0 \leq \tau_H, \\ \tau_H, & \tau^0 > \tau_H. \end{cases} \tag{217}$$

When $\tau^* = \tau^0$ (the interior case), the optimum is a kink solution*: for $\tau > \tau^0$ the tail-selection is $\gamma_{\min}$ and the interior condition would prefer $\tau_L \leq \tau^0$, while for $\tau < \tau^0$ the tail-selection is $\gamma_{\max}$ and the interior condition would prefer $\tau_H \geq \tau^0$. Hence no interior solution exists away from τ^0.*

Proof. Case 1: $\tau^0 < \tau_L$. If $\tau > \tau^0$, Lemma 12.16 implies selection $\gamma^*(\tau) = \gamma_{\min}$, so any interior solution to the asymptotic top-rate condition must satisfy $\tau = \tau_L$ by (215). Since $\tau_L > \tau^0$, this solution is consistent with the regime $\tau > \tau^0$. Hence $\tau^* = \tau_L$.

Case 2: $\tau^0 > \tau_H$. If $\tau < \tau^0$, selection is $\gamma^*(\tau) = \gamma_{\max}$ and any interior solution must satisfy $\tau = \tau_H$. Since $\tau_H < \tau^0$, this is consistent with $\tau < \tau^0$. Hence $\tau^* = \tau_H$.

Case 3: $\tau_L \leq \tau^0 \leq \tau_H$. Suppose, toward contradiction, that an interior solution satisfies $\tau^* > \tau^0$. Then selection is $\gamma_{\min}$ and (215) implies $\tau^* = \tau_L$. But $\tau_L \leq \tau^0$, contradicting $\tau^* > \tau^0$. Similarly, if $\tau^* < \tau^0$, selection is $\gamma_{\max}$ and the interior condition implies $\tau^* = \tau_H \geq \tau^0$, contradicting $\tau^* < \tau^0$. Therefore no interior solution exists with $\tau^* \neq \tau^0$, and the robust optimum is at the kink $\tau^* = \tau^0$. □

Remark 12.21 (Economic interpretation of the clipping result). The clipping structure reflects a fundamental tension. When the planner contemplates raising the top rate above τ^0, nature responds by placing mass on the highest-elasticity type $\gamma_{\min}$—the type whose behavioral response makes taxation most costly. This pushes the planner back toward τ_L. Conversely, when the planner considers lowering the rate below τ^0, nature selects the lowest-elasticity type $\gamma_{\max}$, making taxation appear cheap and pushing the rate up toward τ_H. The result is a "tug of war" that traps the optimal rate between τ_L and τ_H, with the status quo serving as a focal point when it falls within this interval.

12.12.8 Case without GHH disutility shifters

This section analyzes the asymptotic optimal top marginal tax rate when the GHH disutility shifters are *not* used. We show that heterogeneity in γ then affects not only elasticities but also the *power* with which productivity maps into income. As a result, the robust tail-selection is *reversed*: nature selects the lowest elasticity, not the highest.

Preferences and income mapping. Assume GHH preferences without shifters:

$$u(c, n \mid \gamma, z) = \frac{1}{1-\rho}\left(c - \bar{\psi}\,\frac{n^{1+\gamma}}{1+\gamma}\right)^{1-\rho}, \qquad \bar{\psi} > 0,\ \gamma > 0.$$

Under a constant top marginal rate τ ($R = 1 - \tau$), the labor FOC is $\bar{\psi}\, n^{\gamma} = z\, R$, giving

$$n(z, \gamma; \tau) = \left(\frac{z\, R}{\bar{\psi}}\right)^{1/\gamma}, \qquad \mathcal{Y}(z, \gamma; \tau) = A_\tau(\gamma)\, z^{\beta(\gamma)}, \tag{218}$$

where

$$A_\tau(\gamma) := \bar{\psi}^{-1/\gamma}\, R^{1/\gamma}, \qquad \beta(\gamma) := 1 + \frac{1}{\gamma}.$$

The compensated ETI is $\zeta(y, \gamma) = 1/\gamma = e(\gamma)$, and the Jacobian is $\mathcal{Y}_z = \beta(\gamma)\, A_\tau(\gamma)\, z^{\beta(\gamma)-1}$.

Income tails with heterogeneous γ. Conditional on γ, income has a Pareto tail with exponent

$$\alpha_Y(\gamma) = \frac{\alpha_Z}{\beta(\gamma)} = \frac{\alpha_Z}{1 + 1/\gamma}. \tag{219}$$

Since $\beta(\gamma)$ is strictly decreasing in γ, $\alpha_Y(\gamma)$ is strictly increasing. The smallest γ yields the *heaviest* income tail (smallest Pareto exponent).

Lemma 12.17 (Tail dominance)**.** *Let $\Gamma = [\gamma_{\min}, \gamma_{\max}]$ with f_γ continuous and strictly positive. Then the unconditional income distribution has Pareto tail exponent $\alpha_Y^* = \alpha_Y(\gamma_{\min}) = \alpha_Z/(1 + 1/\gamma_{\min})$, and the composition of types in the extreme right tail is degenerate:* $\Pr(\gamma \in \cdot \mid Y > y) \Rightarrow \delta_{\gamma_{\min}}$ *as $y \to \infty$.*

Proof. For each γ, $\Pr(Y > y \mid \gamma) \sim C(\gamma)\, y^{-\alpha_Y(\gamma)}$ with $\alpha_Y(\gamma)$ given by (219). Since $\alpha_Y(\gamma)$ is uniquely minimized at $\gamma_{\min}$ and $f_\gamma(\gamma_{\min}) > 0$, standard results on mixtures of regularly varying distributions imply that the unconditional tail exponent equals $\alpha_Y(\gamma_{\min})$, and that conditional on $Y > y$ the probability mass concentrates on $\gamma_{\min}$. □

Top-rate formula without shifters (non-robust case). With vanishing welfare weights, the asymptotic top-rate condition in primitive space is

$$\frac{\tau}{1-\tau} = \lim_{y\to\infty} \frac{\int_\Gamma f_\gamma(\gamma) \int_{z \ge Z(y,\gamma;\tau)} f_Z(z)\, dz\, d\gamma}{y \int_\Gamma \frac{1}{\gamma} \frac{f_Z(Z(y,\gamma;\tau))\, f_\gamma(\gamma)}{\mathcal{Y}_z(Z(y,\gamma;\tau),\gamma;\tau)}\, d\gamma}, \tag{220}$$

where $Z(y,\gamma;\tau) = (y/A_\tau(\gamma))^{1/\beta(\gamma)}$. For each γ, both numerator and denominator contributions scale as $y^{-\alpha_Y(\gamma)}$. By Lemma 12.17, the terms with $\gamma > \gamma_{\min}$ are asymptotically negligible, and the limit reduces to the homogeneous-type condition with $\gamma = \gamma_{\min}$:

$$\frac{\tau}{1-\tau} = \frac{\gamma_{\min}+1}{\alpha_Z}, \qquad \tau^* = \frac{\gamma_{\min}+1}{\alpha_Z + \gamma_{\min} + 1}. \tag{221}$$

Robust distortion without shifters. Define the Radon–Nikodym derivative as in (209), with $T(Y) \approx \tau Y$ in the top bracket. The distorted conditional density satisfies

$$\widetilde{f}_\gamma(\gamma \mid z) \propto f_\gamma(\gamma) \exp\Big(-\frac{\mu}{\theta}\, \tau\, A_\tau(\gamma)\, z^{\beta(\gamma)}\Big). \tag{222}$$

The key observation: without shifters, $\beta(\gamma) = 1 + 1/\gamma$ is strictly *decreasing* in γ. Hence smaller γ implies a larger power of z and (for large z) much larger income and tax payments. Since the robust tilt penalizes high taxes, it asymptotically selects the *largest* γ (lowest elasticity).

Lemma 12.18 (Discrete selection without shifters)**.** *Let $\gamma_1 < \cdots < \gamma_J$ and suppose $T'(y) \to \tau \in (0,1)$. Under the Radon–Nikodym derivative* (208) *without shifters:*

$$\widetilde{f}_\gamma(\gamma_J \mid z) \to 1, \qquad \widetilde{f}_\gamma(\gamma_j \mid z) \to 0 \quad (j < J), \qquad (z \to \infty).$$

Proof. Fix $j < k$. Then $\gamma_j < \gamma_k$ implies $\beta(\gamma_j) > \beta(\gamma_k)$. Hence

$$\frac{\mathcal{Y}(z,\gamma_j;\tau)}{\mathcal{Y}(z,\gamma_k;\tau)} = \frac{A_\tau(\gamma_j)}{A_\tau(\gamma_k)}\, z^{\beta(\gamma_j)-\beta(\gamma_k)} \to \infty \qquad (z \to \infty).$$

Therefore $\mathcal{Y}(z,\gamma_j;\tau) - \mathcal{Y}(z,\gamma_k;\tau) \to \infty$ and

$$\frac{\widetilde{f}_\gamma(\gamma_j \mid z)}{\widetilde{f}_\gamma(\gamma_k \mid z)} = \frac{\pi_j}{\pi_k} \exp\Big(-\frac{\mu\tau}{\theta}\, [\mathcal{Y}(z,\gamma_j;\tau) - \mathcal{Y}(z,\gamma_k;\tau)]\Big) \to 0.$$

All mass concentrates on the unique maximal type $\gamma_J = \gamma_{\max}$. □

Lemma 12.19 (Laplace principle without shifters (continuous))**.** *Assume f_γ is continuous and strictly positive on $\Gamma = [\gamma_{\min}, \gamma_{\max}]$. Then for any bounded continuous $\phi : \Gamma \to \mathbb{R}$:*

$$\lim_{z\to\infty} \int_\Gamma \phi(\gamma)\, \widetilde{f}_\gamma(\gamma \mid z)\, d\gamma = \phi(\gamma_{\max}).$$

Proof. By (222), define $C := \mu\tau/\theta > 0$. Fix $\varepsilon > 0$ and set $\Gamma_1 := (\gamma_{\max} - \varepsilon, \gamma_{\max}]$, $\Gamma_2 := [\gamma_{\min}, \gamma_{\max} - \varepsilon]$. Since $\beta(\gamma)$ is strictly decreasing and continuous, there exists $\delta > 0$ such that $\beta(\gamma) \geq \beta(\gamma_{\max}) + \delta$ for all $\gamma \in \Gamma_2$.

Let $A_{\min} := \min_\Gamma A_\tau(\gamma) > 0$ and $A_{\max} := \max_\Gamma A_\tau(\gamma)$. For $\gamma \in \Gamma_2$:

$$A_\tau(\gamma)\, z^{\beta(\gamma)} \geq A_{\min}\, z^{\beta(\gamma_{\max})+\delta},$$

while for $\gamma \in \Gamma_1$ (choosing ε small enough that $\beta(\gamma_{\max} - \varepsilon) \leq \beta(\gamma_{\max}) + \delta/2$):

$$A_\tau(\gamma)\, z^{\beta(\gamma)} \leq A_{\max}\, z^{\beta(\gamma_{\max})+\delta/2}.$$

It follows that

$$\int_{\Gamma_2} f_\gamma(\gamma)\, e^{-C\, A_\tau(\gamma)\, z^{\beta(\gamma)}}\, d\gamma \leq \|f_\gamma\|_\infty\, |\Gamma|\, e^{-C\, A_{\min}\, z^{\beta(\gamma_{\max})+\delta}},$$

while

$$\int_{\Gamma_1} f_\gamma(\gamma)\, e^{-C\, A_\tau(\gamma)\, z^{\beta(\gamma)}}\, d\gamma \geq \underline{f}\, |\Gamma_1|\, e^{-C\, A_{\max}\, z^{\beta(\gamma_{\max})+\delta/2}},$$

where $\underline{f} := \min_{\Gamma_1} f_\gamma(\gamma) > 0$. The ratio $\widetilde{f}_\gamma(\Gamma_2 \mid z)/\widetilde{f}_\gamma(\Gamma_1 \mid z)$ is bounded by

$$\frac{\|f_\gamma\|_\infty\, |\Gamma|}{\underline{f}\, |\Gamma_1|}\, e^{-C\,(A_{\min}\, z^{\beta(\gamma_{\max})+\delta} - A_{\max}\, z^{\beta(\gamma_{\max})+\delta/2})} \to 0 \qquad (z \to \infty),$$

since $z^{\beta(\gamma_{\max})+\delta}$ dominates $z^{\beta(\gamma_{\max})+\delta/2}$. So $\widetilde{f}_\gamma(\cdot \mid z)$ concentrates on $\gamma_{\max}$, and the conclusion for bounded continuous ϕ follows by uniform continuity. □

Reduction of the robust top-rate condition. With zero welfare weights at the top, the robust top-rate condition is (220) with $f_{Z,\gamma}$ replaced by $\widetilde{f}_{Z,\gamma}$. As $y \to \infty$, $Z(y, \gamma; \tau) \to \infty$ for all γ. By Lemma 12.19, $\widetilde{f}_\gamma(\cdot \mid z) \Rightarrow \delta_{\gamma_{\max}}$, and the concentration is exponential in $z^{\beta(\gamma_{\max})}$, which dominates the polynomial Pareto and Jacobian terms. Therefore the robust top-rate condition reduces to the homogeneous-type condition with $\gamma = \gamma_{\max}$.

For a homogeneous type γ, the induced income tail exponent is $\alpha_Y(\gamma) = \alpha_Z/\beta(\gamma)$ and the elasticity is $e(\gamma) = 1/\gamma$. The Diamond–Saez formula gives

$$\frac{\tau}{1-\tau} = \frac{1}{\alpha_Y(\gamma_{\max})\, e(\gamma_{\max})} = \frac{\beta(\gamma_{\max})\, \gamma_{\max}}{\alpha_Z} = \frac{\gamma_{\max} + 1}{\alpha_Z}.$$

Proposition 12.9 (Robust top rate without shifters selects the lowest elasticity)**.** *Under the assumptions above and with constant $\theta > 0$, the distorted type distribution concentrates at $\gamma_{\max}$ in the extreme right tail of productivity, and the asymptotic robust top marginal tax rate satisfies*

$$\tau^* = \frac{\gamma_{\max} + 1}{\alpha_Z + \gamma_{\max} + 1}, \qquad \textit{equivalently} \qquad \frac{\tau^*}{1-\tau^*} = \frac{\gamma_{\max} + 1}{\alpha_Z}. \tag{223}$$

Equivalently, $\tau^* = 1/(1 + \alpha_Y(\gamma_{\max})\, e_{\min})$ with $e_{\min} = 1/\gamma_{\max}$.

Remark 12.22 (Comparing the two robust specifications). The contrast between Proposition 12.7 (with shifters: robust top rate at $e_{\max}$) and Proposition 12.9 (without shifters: robust top rate at $e_{\min}$) highlights the role of the disutility shifters in isolating the channels through which γ affects the tax problem.

With shifters, the calibration neutralizes the curvature channel: all types share the same power $\beta = 1 + 1/\bar{\gamma}$ in the income mapping. The only way γ enters the robust distortion is through the *level* $c(\gamma; R) = \kappa\, R^{1/\gamma}$—types with smaller γ (higher elasticity) earn more at any given z, pay more tax, and are therefore penalized more heavily by the adversary. Nature selects the highest elasticity.

Without shifters, γ also affects the *curvature* $\beta(\gamma) = 1 + 1/\gamma$ of the income mapping. Types with smaller γ have a larger power of z in their income, making their tax payments grow much faster with z. This curvature effect dominates the level effect in the tail, so nature now selects the *largest* γ (lowest elasticity)—the type whose income grows most slowly with productivity.

The economic lesson is that the direction of robust selection depends on whether heterogeneity in elasticities operates through a pure *behavioral response* channel (favoring high-elasticity selection) or also through a *tail thickness* channel (favoring low-elasticity selection when the income tail itself depends on γ).

A Proof of Proposition 3.1 (Sufficiency of the First-Order Approach)

Proof. **Necessity.** Global IC immediately implies the envelope condition (derived in the previous subsection). To see that $y(\cdot)$ must be nondecreasing, suppose for contradiction that $z_1 < z_2$ but $y(z_1) > y(z_2)$. IC for type z_1 requires that in (c, y)-space the bundle $(c(z_1), y(z_1))$ lies on a weakly higher indifference curve for type z_1 than does $(c(z_2), y(z_2))$. IC for type z_2 requires the reverse. Since $y(z_1) > y(z_2)$, the indifference curves of the two types through these two bundles must cross in a way that violates the single-crossing property (Lemma 3.1). Figure 5 illustrates the contradiction.

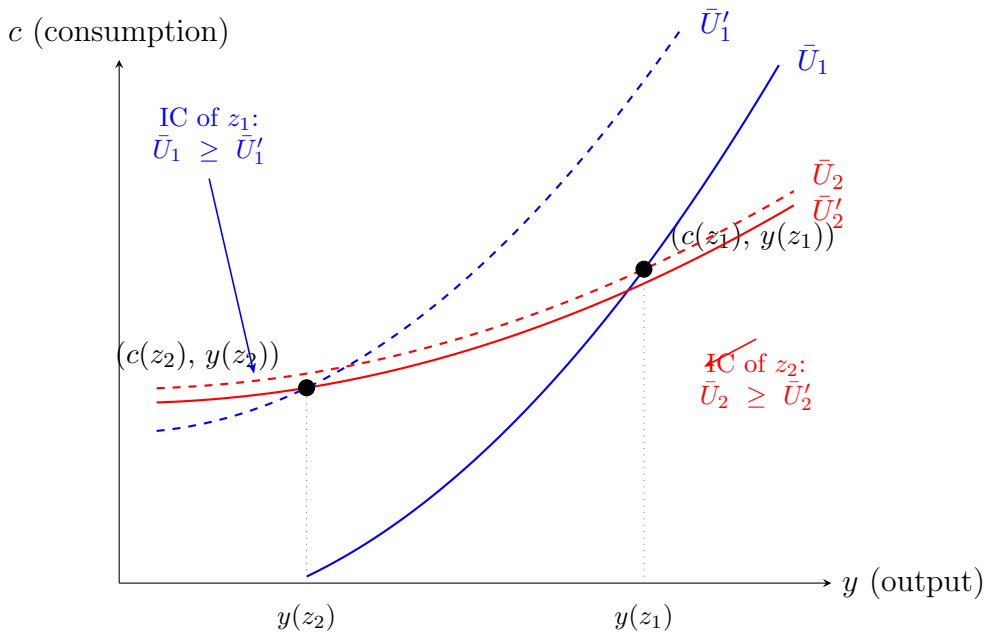

Figure 5: Necessity of monotonicity. Suppose $z_1 < z_2$ but $y(z_1) > y(z_2)$. IC for z_1 requires $\bar{U}_1 \geq \bar{U}_1'$ (the blue solid curve lies weakly above the blue dashed curve at both bundles). IC for z_2 requires $\bar{U}_2 \geq \bar{U}_2'$ (red solid weakly above red dashed). But single crossing (Lemma 3.1) says z_1's ICs are everywhere steeper than z_2's. At the right bundle, z_1's IC (blue solid) is steep and z_2's IC (red dashed) is flat; at the left bundle, the ordering reverses. The two pairs of ICs must therefore cross between the bundles—contradicting single crossing.

Sufficiency. This is the key direction. We must show that for all $z, \hat{z} \in [z_0, \bar{z}]$:

$$\mathcal{U}(z) \equiv U\left(c(z), \frac{y(z)}{z}\right) \geq U\left(c(\hat{z}), \frac{y(\hat{z})}{z}\right).$$

Step 1: Tangency condition from the local IC. Dividing the local IC (13) by $U_c \cdot y'(s)$ (for points where $y'(s) > 0$; the case $y'(s) = 0$ is handled by continuity), the local IC at type s says that the slope of the allocation schedule in (c, y)-space equals the MRS of type s at the allocated bundle:

$$\frac{c'(s)}{y'(s)} = -\frac{U_n\left(c(s), \frac{y(s)}{s}\right)}{s\, U_c\left(c(s), \frac{y(s)}{s}\right)} = S\big(c(s), y(s), s\big). \tag{224}$$

That is, at each point along the allocation schedule, the schedule is tangent to the indifference curve of the local type (Figure 6).

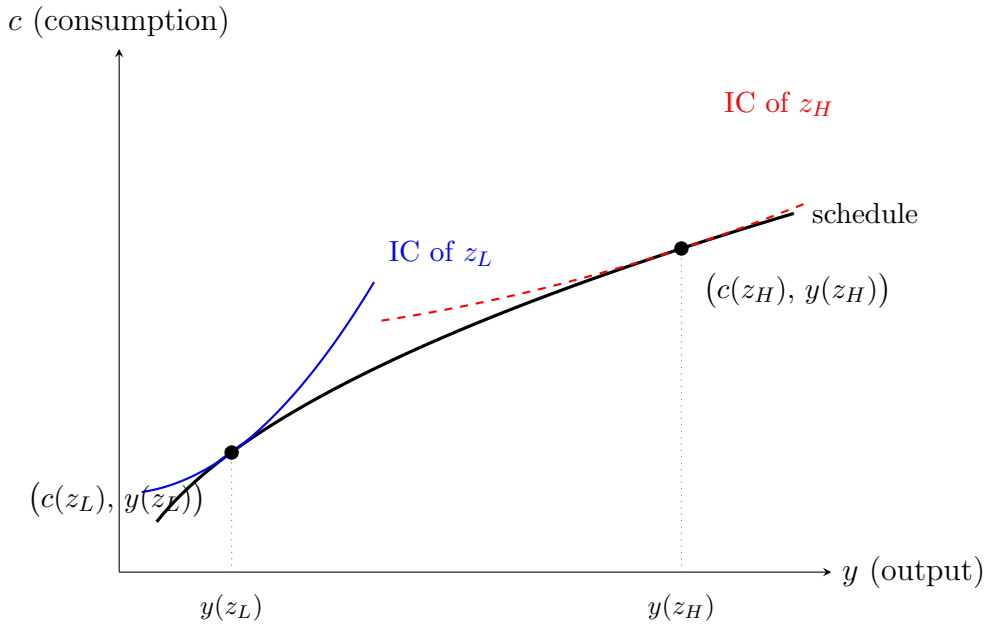

Figure 6: Step 1: Tangency of the allocation schedule. The allocation schedule (solid black curve) is tangent to each type's indifference curve at the assigned bundle. Type z_L's IC (blue, solid) is steep at its bundle, while type z_H's IC (red, dashed) is flat at its bundle, reflecting the single-crossing property.

Step 2: Utility of type z along the allocation path. For a fixed true type z and variable report $\hat{z}$, define

$$\Phi(\hat{z}) \equiv U\left(c(\hat{z}), \frac{y(\hat{z})}{z}\right).$$

Note that $\Phi(z) = \mathcal{U}(z)$. We want to show $\Phi(z) \geq \Phi(\hat{z})$ for all $\hat{z}$, i.e., that Φ achieves its maximum at $\hat{z} = z$.

Differentiating Φ with respect to $\hat{z}$:

$$\Phi'(\hat{z}) = U_c\left(c(\hat{z}), \frac{y(\hat{z})}{z}\right) c'(\hat{z}) + U_n\left(c(\hat{z}), \frac{y(\hat{z})}{z}\right) \frac{y'(\hat{z})}{z}. \tag{225}$$

Factor out $U_c(c(\hat{z}), y(\hat{z})/z) \cdot y'(\hat{z})$:

$$\Phi'(\hat{z}) = U_c\left(c(\hat{z}), \frac{y(\hat{z})}{z}\right) \cdot y'(\hat{z}) \left[\frac{c'(\hat{z})}{y'(\hat{z})} - S\big(c(\hat{z}), y(\hat{z}), z\big)\right]. \tag{226}$$

Using the tangency condition (224) evaluated at $s = \hat{z}$ to substitute for $c'(\hat{z})/y'(\hat{z})$:

$$\Phi'(\hat{z}) = U_c\left(c(\hat{z}), \frac{y(\hat{z})}{z}\right) \cdot y'(\hat{z}) \big[S(c(\hat{z}), y(\hat{z}), \hat{z}) \; - \; S(c(\hat{z}), y(\hat{z}), z)\big]. \tag{227}$$

Step 3: Signing Φ' using single crossing. In expression (227), $U_c > 0$ always, and $y'(\hat{z}) \geq 0$ by the monotonicity assumption (ii). By Lemma 3.1, S is strictly decreasing in its third argument, so:

- If $\hat{z} < z$: then $S(c(\hat{z}), y(\hat{z}), \hat{z}) > S(c(\hat{z}), y(\hat{z}), z)$, so the bracket in (227) is positive, and hence $\Phi'(\hat{z}) \geq 0$.
- If $\hat{z} > z$: then $S(c(\hat{z}), y(\hat{z}), \hat{z}) < S(c(\hat{z}), y(\hat{z}), z)$, so the bracket is negative, and hence $\Phi'(\hat{z}) \leq 0$.

Thus Φ is nondecreasing for $\hat{z} < z$ and nonincreasing for $\hat{z} > z$. It follows that Φ achieves its maximum at $\hat{z} = z$:

$$\mathcal{U}(z) = \Phi(z) \geq \Phi(\hat{z}) = U\left(c(\hat{z}), \frac{y(\hat{z})}{z}\right) \quad \text{for all } \hat{z}. \qquad \square$$

Remark A.1 (Geometric interpretation). The proof has an intuitive geometric reading, illustrated in Figure 7. The allocation schedule traces a curve in (c, y)-space. At each point $(c(s), y(s))$, this curve is tangent to type s's indifference curve (Step 1, Figure 6). Now consider type z_H evaluating the bundle assigned to some lower type z_L. By single crossing, type z_H has *flatter* indifference curves than type z_L at any given point. So the allocation schedule—tangent to z_L's steeper indifference curve at the bundle $(c(z_L), y(z_L))$—rises more steeply than z_H's own indifference curves. As type z_H "walks down" the allocation schedule toward lower bundles, each step moves to a lower indifference curve for z_H: the schedule climbs faster than z_H's indifference curves would require to maintain constant utility. The argument for upward deviations (z_L considering z_H's bundle) is symmetric—the schedule is now *flatter* than z_L's indifference curves.

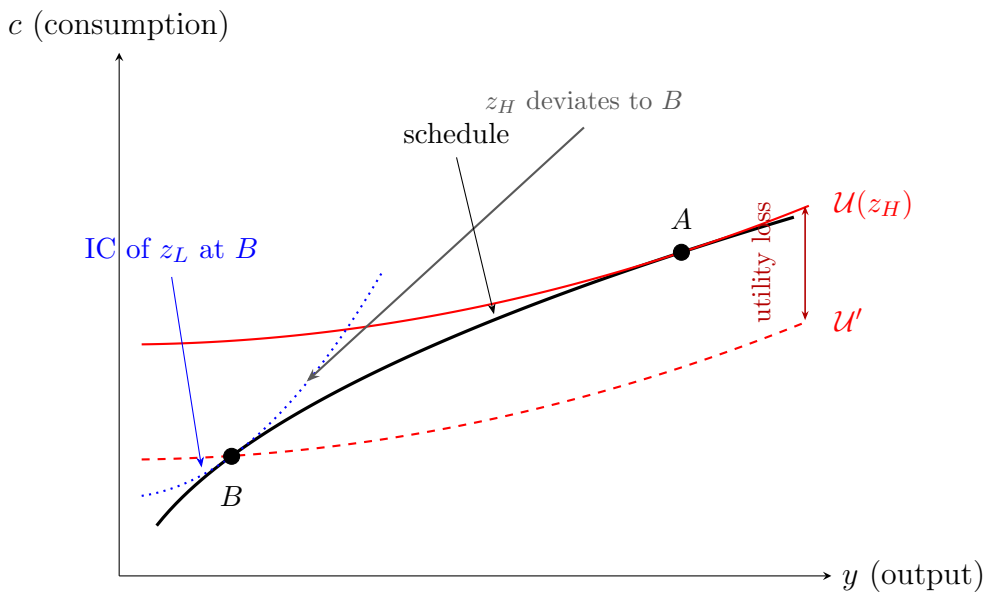

Figure 7: Step 3: Why downward deviations are suboptimal. Type z_H is assigned bundle A and considers deviating to B (designed for z_L). The solid red curve is z_H's indifference curve through A; the dashed red curve is z_H's (lower) indifference curve through B. At B, the allocation schedule is tangent to z_L's steep IC (blue dotted), so the schedule rises faster than z_H's flat ICs. Walking from A toward B, the schedule falls below z_H's original IC, producing a utility loss.

B The Classical Runge–Kutta Method (RK4)

This appendix derives the classical fourth-order Runge–Kutta method used in both the non-robust (Section 9.3) and robust (Section 10) ODE computations. We build up from first-order (Euler) through second-order (RK2) to the full fourth-order method, making the Taylor-matching strategy transparent at each stage.

B.1 The problem

Consider an initial value problem (IVP):

$$\frac{dy}{dt} = f(t, y), \qquad y(t_0) = y_0. \tag{228}$$

We want a rule that, given (t_n, y_n), produces $y_{n+1} \approx y(t_n + h)$ with step size $h = t_{n+1} - t_n$.

Forward Euler. The simplest approach: step along the current slope,

$$y_{n+1} = y_n + h\, f(t_n, y_n).$$

This is first-order: the local truncation error (LTE) is $O(h^2)$, giving a global error $O(h)$. We want to do better by evaluating f at *multiple* points within the step and taking a weighted average.

B.2 Warmup: deriving the second-order method (RK2)

Following the approach in Zeltkevic (1998), suppose we seek a method of the form

$$\begin{aligned} k_1 &= h\, f(t_n, y_n), \\ k_2 &= h\, f(t_n + \alpha h,\ y_n + \beta\, k_1), \\ y_{n+1} &= y_n + a\, k_1 + b\, k_2, \end{aligned} \tag{229}$$

where α, β, a, b are constants to be determined so that the LTE is $O(h^3)$.

B.2.1 Taylor expansion of the exact solution

Expand $y(t_n + h)$ around t_n:

$$y(t_n + h) = y_n + h\,\frac{dy}{dt} + \frac{h^2}{2}\,\frac{d^2y}{dt^2} + O(h^3). \tag{230}$$

Since $dy/dt = f$, the chain rule gives

$$\frac{d^2y}{dt^2} = \frac{df}{dt} = f_t + f_y \cdot f, \tag{231}$$

where all partials are evaluated at (t_n, y_n). So

$$y(t_n + h) = y_n + h\, f + \frac{h^2}{2}\big(f_t + f\, f_y\big) + O(h^3). \tag{232}$$

B.2.2 Taylor expansion of the RK2 ansatz

Expand k_2 using a two-variable Taylor expansion of f:

$$\begin{aligned} k_2 &= h\, f\big(t_n + \alpha h,\ y_n + \beta\, k_1\big) \\ &= h\,\big[f + \alpha h\, f_t + \beta\, k_1\, f_y + O(h^2)\big] \\ &= h\, f + h^2\big(\alpha\, f_t + \beta\, f\, f_y\big) + O(h^3). \end{aligned} \tag{233}$$

Substituting into $y_{n+1} = y_n + a\, k_1 + b\, k_2$:

$$y_{n+1} = y_n + (a+b)\, h\, f + b\, h^2\big(\alpha\, f_t + \beta\, f\, f_y\big) + O(h^3). \tag{234}$$

B.2.3 Matching coefficients

Comparing (232) and (234) term by term:

$$\boxed{a + b = 1, \qquad b\,\alpha = \frac{1}{2}, \qquad b\,\beta = \frac{1}{2}.} \tag{235}$$

Three equations, four unknowns: a one-parameter family. The common choice $\alpha = \beta = 1$, $a = b = 1/2$ gives the **classical RK2** (also called the improved Euler or Heun method):

$$y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2), \qquad k_2 = h\, f(t_n + h,\ y_n + k_1). \tag{236}$$

B.3 Geometric intuition

B.3.1 Why RK2 improves on Euler

The problem with Euler's method is that it commits to the slope at t_n for the entire step. If the slope is changing (i.e., f is not constant), this produces a systematic drift. RK2 fixes this by using the Euler step as a *probe*: it takes a trial step to $t_n + h$ to estimate the slope there, then averages the two slopes.

Algebraically, this is exactly what the matching conditions enforce: the $b\alpha = 1/2$ condition ensures the method captures the h^2 term $\frac{1}{2}(f_t + f f_y)$ from the Taylor expansion—the first correction for curvature.

B.3.2 From RK2 to RK4

Before diving into the algebra for RK4, consider the geometry.

The key idea: Euler's method uses only the slope at t_n. RK4 samples the slope at four points—the left endpoint, two midpoint estimates, and the right endpoint—then combines them with Simpson's-rule weights $(1, 2, 2, 1)/6$.

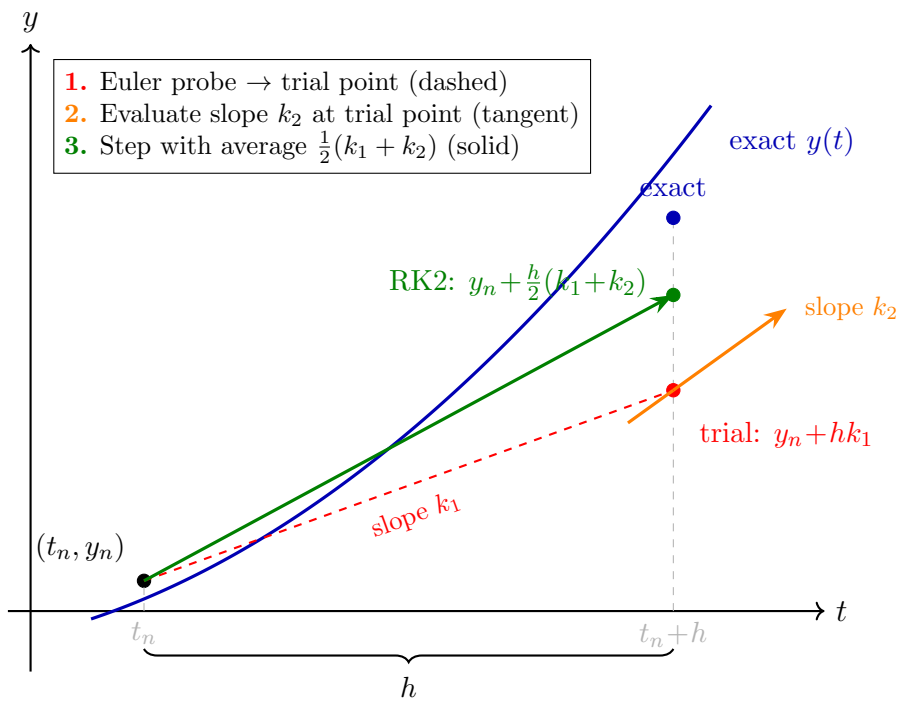

Figure 8: Why RK2 beats Euler. **Step 1** (red dashed): take a trial Euler step using the slope k_1 at t_n, arriving at the trial point. **Step 2** (orange tangent): evaluate the slope k_2 at the trial point—it is steeper because the solution curves upward. **Step 3** (green solid): step from (t_n, y_n) using the average slope $\frac{1}{2}(k_1 + k_2)$, which corrects for the curvature and lands close to the exact solution.

B.4 Deriving the fourth-order method (RK4)

B.4.1 The ansatz

We now seek a 4-stage method:

$$\begin{aligned} k_1 &= f(t_n,\, y_n), \\ k_2 &= f\big(t_n + c_2 h,\; y_n + h\, a_{21}\, k_1\big), \\ k_3 &= f\big(t_n + c_3 h,\; y_n + h(a_{31}\, k_1 + a_{32}\, k_2)\big), \\ k_4 &= f\big(t_n + c_4 h,\; y_n + h(a_{41}\, k_1 + a_{42}\, k_2 + a_{43}\, k_3)\big), \\ y_{n+1} &= y_n + h\big(b_1\, k_1 + b_2\, k_2 + b_3\, k_3 + b_4\, k_4\big), \end{aligned} \tag{237}$$

with consistency conditions $c_i = \sum_j a_{ij}$ and 10 free parameters. We require the LTE to be $O(h^5)$.

The strategy is identical to the RK2 derivation above: expand everything in Taylor series, match coefficients with the exact solution, and solve the resulting algebraic system.

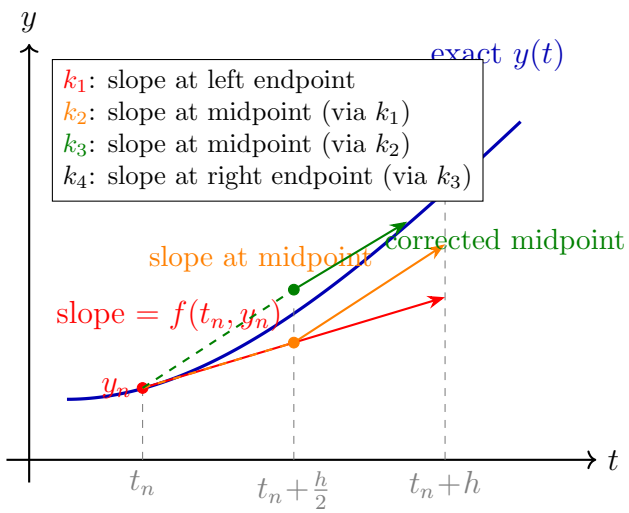

Figure 9: Geometric idea behind RK4. Euler uses one slope (red). RK4 evaluates the slope at four points and takes a weighted average, achieving much higher accuracy per step.

B.4.2 Taylor expansion of the exact solution through $O(h^4)$

We need y', y'', y''', $y^{(4)}$:

$$y' = f, \tag{238}$$
$$y'' = f_t + f_y f \;\equiv\; Df, \tag{239}$$
$$y''' = D^2 f = f_{tt} + 2f_{ty} f + f_{yy} f^2 + f_y Df, \tag{240}$$
$$y^{(4)} = D^3 f \quad \text{(see below)}, \tag{241}$$

where $D = \partial_t + f\,\partial_y$ is the total derivative operator along solutions. Hence:

$$y(t_n + h) = y_n + hf + \frac{h^2}{2} Df + \frac{h^3}{6} D^2 f + \frac{h^4}{24} D^3 f + O(h^5). \tag{242}$$

B.4.3 Taylor expansion of the stages

This proceeds exactly as in the RK2 case, but now we need the two-variable Taylor expansion through higher order. Write $\Delta t_i = c_i h$ and $\Delta y_i = h \sum_j a_{ij} k_j$ for the perturbation in each stage.

Stage 2. Expanding to $O(h^2)$:

$$k_2 = f + c_2 h\,(f_t + f_y f) + \frac{c_2^2 h^2}{2}(f_{tt} + 2f\,f_{ty} + f^2 f_{yy}) + O(h^3). \tag{243}$$

Stages 3 and 4. The expansions are analogous but involve substituting the expansion of earlier stages. For example, in k_3 the perturbation $\Delta y_3 = h(a_{31}k_1 + a_{32}k_2)$ contains k_2, which itself has $O(h)$ corrections. This coupling is what generates the additional order conditions beyond simple quadrature.

B.4.4 The eight order conditions

Substituting the stage expansions into y_{n+1} and comparing with (242) order by order yields 8 algebraic conditions. We state them using the compact Butcher notation $\sum_i b_i(\cdots)$:

Order 1 (1 condition, from the h coefficient):

$$\sum_i b_i = 1. \tag{244}$$

Order 2 (1 condition, from the h^2 coefficient):

$$\sum_i b_i\, c_i = \frac{1}{2}. \tag{245}$$

Order 3 (2 conditions, from the two distinct terms in $D^2 f$):

$$\sum_i b_i\, c_i^2 = \frac{1}{3}, \qquad \sum_i b_i \sum_j a_{ij}\, c_j = \frac{1}{6}. \tag{246}$$

Order 4 (4 conditions, from the four distinct terms in $D^3 f$):

$$\sum_i b_i\, c_i^3 = \frac{1}{4}, \tag{247}$$

$$\sum_i b_i\, c_i \sum_j a_{ij}\, c_j = \frac{1}{8}, \tag{248}$$

$$\sum_i b_i \sum_j a_{ij}\, c_j^2 = \frac{1}{12}, \tag{249}$$

$$\sum_i b_i \sum_j a_{ij} \sum_k a_{jk}\, c_k = \frac{1}{24}. \tag{250}$$

Remark B.1. The right-hand sides $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{6}, \ldots$ are exactly $1/\sigma(\tau)$ for each rooted tree τ—the reciprocal of the tree's "density" in Butcher's formalism. The number of conditions at each order equals the number of rooted trees: $1, 1, 2, 4$ at orders $1, 2, 3, 4$.

B.4.5 Solving the system: the classical choice

We have 8 equations in 10 unknowns, leaving a 2-parameter family. The **classical RK4** uses the symmetric node placement

$$c_1 = 0, \qquad c_2 = c_3 = \tfrac{1}{2}, \qquad c_4 = 1,$$

and further requires that each stage uses only the *most recent* slope estimate:

$$a_{31} = 0 \ \ (\text{skip } k_1), \qquad a_{41} = a_{42} = 0 \ \ (\text{skip } k_1, k_2).$$

Under these simplifications, the system solves uniquely:

From consistency: $a_{21} = \frac{1}{2}$, $a_{32} = \frac{1}{2}$, $a_{43} = 1$.

From (250)**:** $b_4 \cdot 1 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{24} \implies b_4 = \frac{1}{6}$.

From (249)**:** $b_3 \cdot \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{6} \cdot 1 \cdot \frac{1}{4} = \frac{1}{12} \implies b_3 = \frac{1}{3}$.

From $(246)_1$**:** $b_2 \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{6} \cdot 1 = \frac{1}{3} \implies b_2 = \frac{1}{3}$.

From (244)**:** $b_1 = 1 - \frac{1}{3} - \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$.

One can verify that the remaining conditions (245), $(246)_2$, (247), (248) are all satisfied. The Butcher tableau is:

$$\begin{array}{c|cccc} 0 & & & & \\ \frac{1}{2} & \frac{1}{2} & & & \\ \frac{1}{2} & 0 & \frac{1}{2} & & \\ 1 & 0 & 0 & 1 & \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array}$$

B.5 The algorithm

Proposition B.1 (RK4 convergence)**.** *The method*

$$\begin{aligned} k_1 &= f(t_n,\, y_n), \\ k_2 &= f\big(t_n + \tfrac{h}{2},\; y_n + \tfrac{h}{2}\, k_1\big)\,, \\ k_3 &= f\big(t_n + \tfrac{h}{2},\; y_n + \tfrac{h}{2}\, k_2\big)\,, \\ k_4 &= f(t_n + h,\; y_n + h\, k_3)\,, \\ y_{n+1} &= y_n + \tfrac{h}{6}\big(k_1 + 2k_2 + 2k_3 + k_4\big), \end{aligned}$$

has local truncation error $O(h^5)$ and global error $O(h^4)$, provided $f \in C^4$.

B.6 Numerical illustration

Consider the test problem $y' = -y$, $y(0) = 1$, with exact solution $y(t) = e^{-t}$.

B.6.1 Comparing Euler, RK2, and RK4

B.6.2 Convergence rates

The convergence plot (Figure 11) makes the practical payoff vivid. At $h = 0.01$, the Euler error is roughly 10^{-3}, while RK4 achieves 10^{-12}—a nine-order-of-magnitude improvement.

B.7 The stages, geometrically

B.8 Why not order 5 with 4 stages?

At order p, the number of conditions equals the number of rooted trees of order $\leq p$, while the number of free parameters in an s-stage method is $s(s+1)/2$ (from a_{ij} and b_i, after accounting for consistency). The counts are:

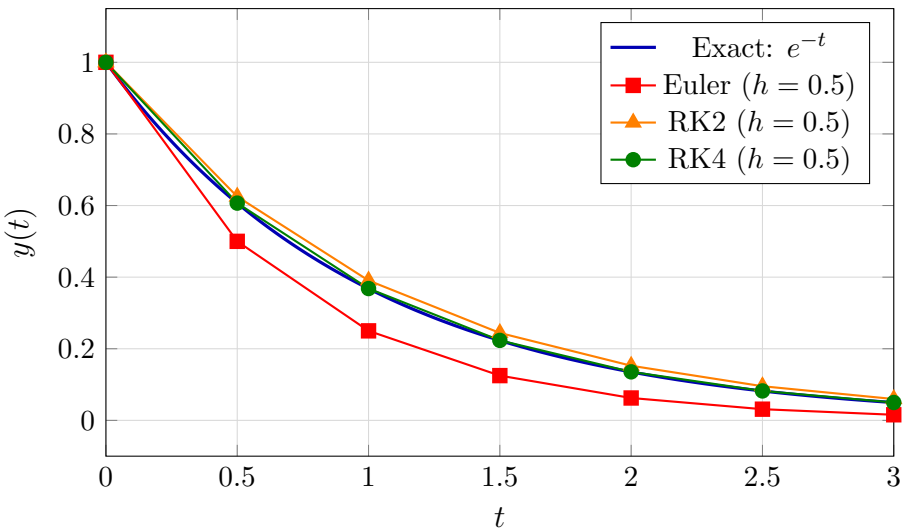

Figure 10: Solution of $y' = -y$, $y(0) = 1$ with step size $h = 0.5$. Even at this coarse step size, RK4 is nearly indistinguishable from the exact solution. Euler accumulates visible error; RK2 is intermediate.

Order p	1	2	3	4	5	6
# conditions	1	2	4	8	17	37
# free params ($s = p$)	1	3	6	10	15	21

At $p = 4$, we have 8 conditions and 10 parameters: feasible (with a 2-parameter family). At $p = 5$, there are 17 conditions but only 15 parameters for $s = 5$ stages, and only 10 for $s = 4$. It turns out that order 5 is not attainable with 5 stages for general explicit RK methods; one needs at least $s = 6$ stages. This is the **Butcher barrier** (Butcher, 1965): for $p \geq 5$, the minimum number of stages exceeds the order ($s > p$), making higher-order methods increasingly expensive per step.

B.9 Connection to Simpsonś rule

The weights $\left(\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6}\right)$ at nodes $(0, \frac{1}{2}, \frac{1}{2}, 1)$ are precisely Simpson's rule on $[0, 1]$. This is not a coincidence: for the special case $f(t, y) = g(t)$, the ODE reduces to integration and RK4 reduces to a quadrature formula. Simpson's rule integrates cubics exactly, consistent with order 4.

For general $f(t, y)$, quadrature accuracy is necessary but not sufficient. The "coupling" conditions (248)–(250) arise specifically from the y-dependence of f and have no quadrature analogue.

B.10 Summary

The derivation has three ingredients:

1. Taylor-expand the exact solution $y(t_n + h)$ to the desired order.
2. Taylor-expand the RK ansatz by recursively expanding each stage.
3. Match coefficients, yielding algebraic conditions on the Butcher tableau entries. For 4 stages and order 4, there are 8 conditions in 10 unknowns, and the classical

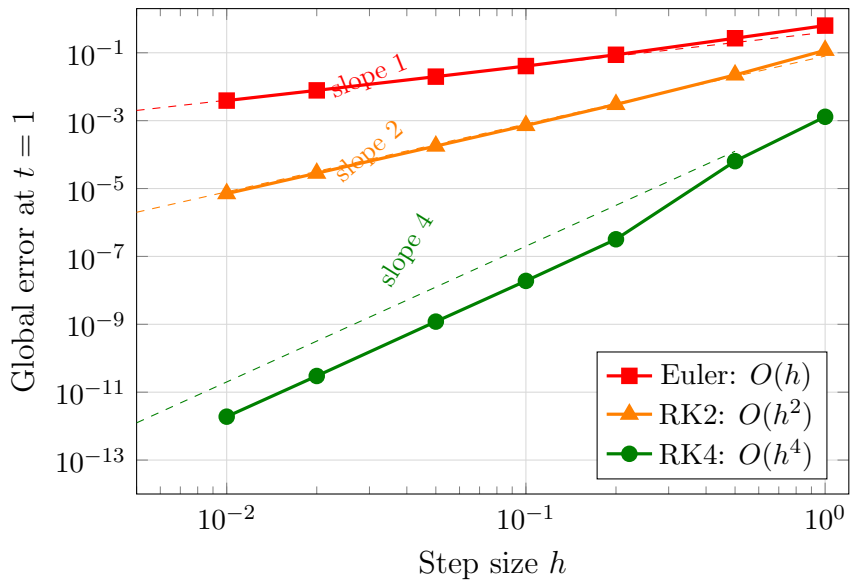

Figure 11: Log-log plot of global error at $t = 1$ versus step size h for $y' = -y$. The slopes confirm: Euler is $O(h)$, RK2 is $O(h^2)$, and RK4 is $O(h^4)$. Halving h reduces the RK4 error by a factor of $2^4 = 16$.

choice $c = (0, \frac{1}{2}, \frac{1}{2}, 1)$ with sequential stage coupling yields the familiar $(1, 2, 2, 1)/6$ weights.

Remark B.2 (Application to backward integration). When integrating backward (as in the robust tax ODE), we set $h < 0$. The RK4 formulas are unchanged—h simply takes negative values, so k_2 and k_3 evaluate f at $t_n + h/2 < t_n$ and k_4 evaluates at $t_n + h < t_n$. The order conditions and error bounds are symmetric in h: replacing $h \to -h$ changes the sign of odd-power error terms but preserves the $O(h^5)$ LTE and $O(h^4)$ global error. The only consideration is stability: backward integration reverses the stability properties of the ODE, which is precisely why it is used for the saddle-path problem in Section 10—the forward-unstable saddle path becomes backward-stable, and numerical errors shrink rather than grow.

C Optimal Control and the Hamiltonian

This appendix derives the Hamiltonian optimality conditions used throughout the notes from first principles, via a perturbation (variational) argument. The approach makes explicit the connection to Gâteaux derivatives in function space.

C.1 The abstract problem

Consider the continuous-time optimal control problem:

$$\max_{u(\cdot),\, x(t_0)} \int_{t_0}^{t_1} L\big(t, x(t), u(t)\big)\, dt \tag{251}$$

subject to the state equation

$$\dot{x}(t) = g\big(t, x(t), u(t)\big), \tag{252}$$

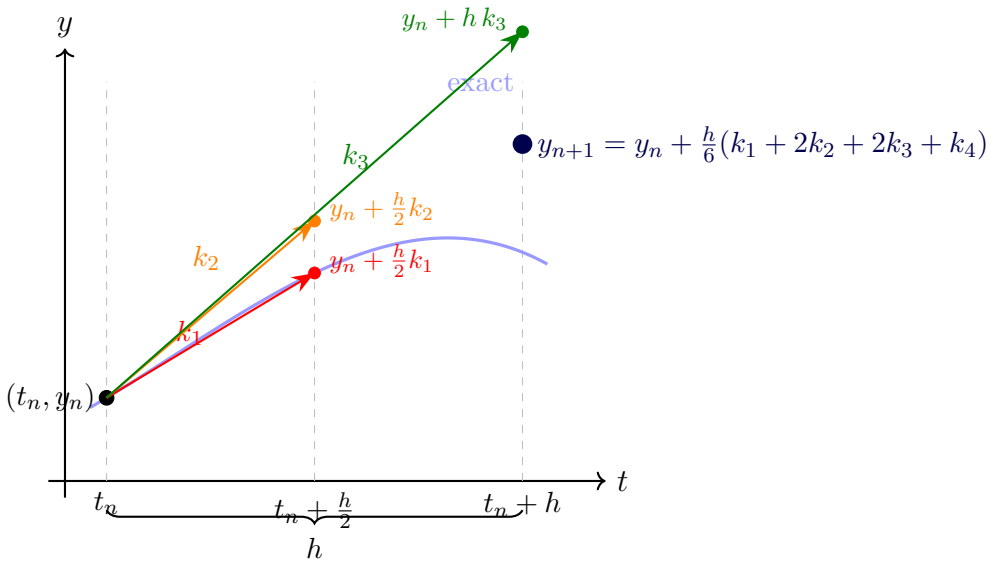

Figure 12: The four stages of RK4. Starting from (t_n, y_n): k_1 gives a preliminary midpoint via the initial slope; k_2 re-evaluates at that midpoint; k_3 uses the corrected midpoint slope to step all the way to $t_n + h$, producing the evaluation point for k_4. The final result is a weighted average.

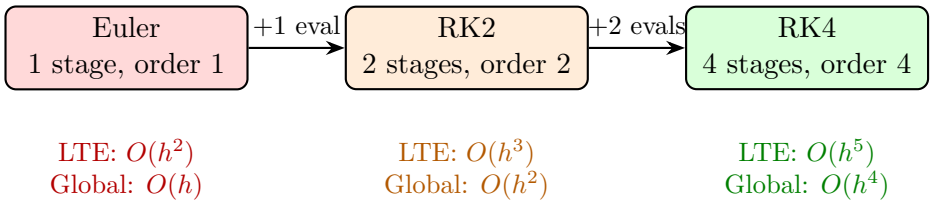

Figure 13: Summary of the Runge–Kutta hierarchy. Each additional function evaluation "buys" more Taylor terms matched.

where $x(t) \in \mathbb{R}^n$ is the **state**, $u(t) \in \mathbb{R}^m$ is the **control**, L is the running payoff, and g governs the state dynamics. The independent choice variables are the control path $u(\cdot)$ and (when the initial state is free) the initial condition $x(t_0)$. The state path $x(t)$ for $t > t_0$ is *not* an independent choice: it is uniquely determined by $u(\cdot)$ and $x(t_0)$ through the ODE (252).

Remark C.1 (Equivalent "full" formulation). One can equivalently write the problem as $\max_{u(\cdot),x(\cdot)}$ subject to the constraint $\dot{x} = g(t, x, u)$, treating the entire state path as a choice variable. This is the formulation that motivates the Hamiltonian: the costate $\lambda(t)$ is the Lagrange multiplier on the state equation at each t, and the Hamiltonian $H = L + \lambda^\top g$ is the Lagrangian of this constrained problem. The "reduced form" (251)–(252) and the "full" formulation yield identical optimality conditions, but the reduced form makes clear which variables are truly independent.

The boundary conditions on x depend on the application: $x(t_0)$ may be fixed at a given x_0 (drop it from the max), or free (optimized over, as written above). Similarly for $x(t_1)$.

In the taxation problem, the "time" variable is the productivity type z, the state is utility $\mathcal{U}(z)$, the control is output $y(z)$, and the state equation is the envelope condition (18). Both endpoints $\mathcal{U}(z_0)$ and $\mathcal{U}(\bar{z})$ are free: $\mathcal{U}(z_0)$ is an independent choice (the

planner picks the utility of the bottom type), while $\mathcal{U}(\bar{z})$ is determined by the ODE but unconstrained (the planner does not impose a target for the top type).

C.2 The Hamiltonian

Define the **Hamiltonian**:

$$H(t, x, u, \lambda) = L(t, x, u) + \lambda^\top g(t, x, u), \tag{253}$$

where $\lambda(t) \in \mathbb{R}^n$ is the **costate** (adjoint) variable—a Lagrange multiplier on the state equation at each "time" t.

C.3 Derivation of the optimality conditions via perturbation

Suppose (x^*, u^*) is optimal. We present the derivation for the most general case where *both* $x(t_0)$ and $x(t_1)$ are free; the fixed-endpoint cases follow by setting the relevant variation to zero.

Consider a perturbed control $u^\epsilon(t) = u^*(t) + \epsilon\, \delta u(t)$ and a perturbed initial condition $x^\epsilon(t_0) = x^*(t_0) + \epsilon\, \delta x_0$. The perturbed state $x^\epsilon(t)$ satisfies

$$\dot{x}^\epsilon = g(t, x^\epsilon, u^\epsilon), \qquad x^\epsilon(t_0) = x^*(t_0) + \epsilon\, \delta x_0.$$

Define the **sensitivity** $\delta x(t) = \frac{dx^\epsilon}{d\epsilon}\big|_{\epsilon=0}$, which satisfies the linearized state equation:

$$\dot{\delta x} = g_x\, \delta x + g_u\, \delta u, \qquad \delta x(t_0) = \delta x_0. \tag{254}$$

When $x(t_0)$ is fixed, $\delta x_0 = 0$ and this reduces to the standard case.

The objective functional is $J(\epsilon) = \int_{t_0}^{t_1} L(t, x^\epsilon, u^\epsilon)\, dt$. Its Gâteaux derivative at $\epsilon = 0$ is

$$\frac{dJ}{d\epsilon}\bigg|_{\epsilon=0} = \int_{t_0}^{t_1} \big[L_x\, \delta x + L_u\, \delta u\big]\, dt. \tag{255}$$

The term $L_x\, \delta x$ is problematic because δx depends on δu through the linearized state equation (254)—it couples the variation at every point. The costate λ is introduced precisely to decouple this.

Introducing the costate. Since $\dot{x} = g$ is satisfied identically, we can add zero to the objective:

$$J = \int_{t_0}^{t_1} \big[L + \lambda^\top(g - \dot{x})\big]\, dt = \int_{t_0}^{t_1} \big[H - \lambda^\top \dot{x}\big]\, dt.$$

Integrating $\lambda^\top \dot{x}$ by parts: $\int \lambda^\top \dot{x}\, dt = [\lambda^\top x]_{t_0}^{t_1} - \int \dot{\lambda}^\top x\, dt$. Hence

$$J = \int_{t_0}^{t_1} \Big[H + \dot{\lambda}^\top x\Big]\, dt - \lambda(t_1)^\top x(t_1) + \lambda(t_0)^\top x(t_0).$$

Now take the Gâteaux derivative, noting that $\delta x(t_0) = \delta x_0$ (which is nonzero when $x(t_0)$ is free):

$$\frac{dJ}{d\epsilon}\bigg|_{\epsilon=0} = \int_{t_0}^{t_1} \big[(H_x + \dot{\lambda}^\top)\, \delta x + H_u\, \delta u\big]\, dt - \lambda(t_1)^\top \delta x(t_1) + \lambda(t_0)^\top \delta x_0. \tag{256}$$

Choosing λ to eliminate δx. If we set

$$\dot{\lambda}(t) = -H_x(t, x^*, u^*, \lambda) \qquad \text{(costate equation)}, \tag{257}$$

the integral term involving δx vanishes. The remaining boundary terms are

$$\left.\frac{dJ}{d\epsilon}\right|_{\epsilon=0} = \int_{t_0}^{t_1} H_u \, \delta u \, dt + \lambda(t_0)^\top \delta x_0 - \lambda(t_1)^\top \delta x(t_1).$$

The boundary conditions on λ depend on which endpoints are free:

- If $x(t_1)$ is **free**: $\delta x(t_1)$ is arbitrary, so optimality requires

$$\lambda(t_1) = 0 \qquad \text{(terminal transversality)}. \tag{258}$$

- If $x(t_0)$ is **free**: δx_0 is arbitrary, so optimality requires

$$\lambda(t_0) = 0 \qquad \text{(initial transversality)}. \tag{259}$$

- If $x(t_0)$ is **fixed** at x_0: $\delta x_0 = 0$ and no condition on $\lambda(t_0)$ is needed.
- If $x(t_1)$ is **fixed**: $\delta x(t_1) = 0$ and no condition on $\lambda(t_1)$ is needed.

After imposing the relevant boundary conditions, what remains is

$$\left.\frac{dJ}{d\epsilon}\right|_{\epsilon=0} = \int_{t_0}^{t_1} H_u \, \delta u \, dt.$$

At an optimum this must be zero for all admissible δu, giving the **optimality condition**:

$$H_u(t, x^*, u^*, \lambda) = 0 \qquad \text{for all } t \in [t_0, t_1]. \tag{260}$$

C.4 Summary of the Hamiltonian system

The three conditions (252), (257), and (260) form a two-point boundary value problem. The boundary conditions on x and λ are complementary: at each endpoint, either the state is fixed or the costate is zero.

For the case most relevant to the tax problem—both endpoints free—the system is:

$$\begin{aligned} \dot{x} &= H_\lambda = g(t, x, u), & \lambda(t_0) &= 0, \\ \dot{\lambda} &= -H_x, & \lambda(t_1) &= 0, \\ 0 &= H_u. \end{aligned}$$

The costate vanishes at both ends because the planner is free to choose both the initial and terminal values of the state. The control u is determined pointwise by the static condition $H_u = 0$.

Remark C.2 (Connection to the tax problem). In Sections 4–6, the abstract framework specializes as follows:

Abstract	**Tax problem**	**Equation**
"Time" t	Productivity type z	
State x	Utility $\mathcal{U}(z)$	
Control u	Output $y(z)$	
$\dot{x} = g$	$\mathcal{U}'(z) = -U_n\, y/z^2$	(18)
$\dot{\lambda} = -H_x$	Costate equation	(26)
$H_u = 0$	Tax formula	(25)
$\lambda(t_0) = \lambda(t_1) = 0$	$\lambda(z_0) = \lambda(\bar{z}) = 0$	(27)

The two-sided boundary condition $\lambda(z_0) = \lambda(\bar{z}) = 0$ arises because both endpoints of the state $\mathcal{U}$ are free: the planner optimizes over the utility of the bottom type (giving $\lambda(z_0) = 0$ by initial transversality (259)) and does not impose a target on the utility of the top type (giving $\lambda(\bar{z}) = 0$ by terminal transversality (258)).

Remark C.3 (When is the perturbation approach valid?). The Gâteaux derivative argument requires:

(i) **Interior optimum:** The optimal control u^* must be interior (no binding inequality constraints on u). If the control is constrained, the first-order condition $H_u = 0$ is replaced by the Pontryagin minimum principle, which allows corner solutions.

(ii) **Sufficient smoothness:** The functions L, g, and the optimal paths x^*, u^* must be smooth enough for the integration by parts and the interchange of differentiation and integration to be valid.

(iii) **Sufficiency:** The first-order conditions are necessary but not automatically sufficient. Sufficiency holds under concavity of the Hamiltonian in (x, u) jointly (the Mangasarian sufficient conditions), or can be verified directly by checking second-order conditions.

In the tax problem, condition (i) corresponds to the monotonicity constraint $y'(z) \geq 0$ being non-binding (no bunching). At bunching points, the first-order approach breaks down and one must handle the inequality constraint explicitly. Condition (iii) is ensured by the concavity of U and the single-crossing property.

D Summary of Key Formulas

Object	Formula
Envelope condition	$\dfrac{d\mathcal{U}}{dz} = -U_n\Big(c, \dfrac{y}{z}\Big)\,\dfrac{y}{z^2}$
Frisch elasticity	$\varepsilon^f = \dfrac{U_n}{n\,[U_{nn} - U_{cn}^2/U_{cc}]}$
Key ratio	$\dfrac{1+\varepsilon^u}{\varepsilon^c} = 1 + n\,\dfrac{U_{nn} + wU_{cn}}{U_n}$
General tax formula	$\dfrac{T'}{1-T'} = -\dfrac{1+\varepsilon^u}{\varepsilon^c}\,\dfrac{\lambda(z)}{zf(z)}\,\dfrac{U_c}{\mu}$
ABC formula (separable)	$\dfrac{T'}{1-T'} = \dfrac{1+\varepsilon^u}{\varepsilon^c}\,\dfrac{\Psi(z)-\Omega(z)}{z\,\omega(z)}$
Top rate (Diamond–Saez)	$T'_{\text{top}} = \dfrac{1}{1+\alpha_y\,e}$

E Mathematical Prerequisites

This appendix collects precise statements of the external results used in the main text, for ease of reference.

E.1 The Weierstrass extreme value theorem

Lemma E.1 (Weierstrass)**.** *Let X be a nonempty compact topological space and $f : X \to \mathbb{R}$ an upper semicontinuous function. Then f attains its supremum: there exists $x^* \in X$ with $f(x^*) = \sup_{x\in X} f(x)$.*

Similarly, if f is lower semicontinuous, then f attains its infimum.

A function $f : X \to \mathbb{R}$ is **upper semicontinuous** (u.s.c.) if $\{x : f(x) \geq c\}$ is closed for every $c \in \mathbb{R}$; equivalently, $\limsup_{x_n\to x} f(x_n) \leq f(x)$.

We use Weierstrass in Step 5 (Proposition 11.4): $\mathcal{A}_q$ is compact (Helly) and W is continuous (dominated convergence), so W attains its maximum.

E.2 Helly's selection theorem

Lemma E.2 (Helly)**.** *Let $\{g_n\}_{n=1}^{\infty}$ be a sequence of functions $[a, b] \to \mathbb{R}$ that are:*

(i) uniformly bounded: $|g_n(x)| \leq M$ for all n and x;

(ii) monotone nondecreasing: $g_n(x) \leq g_n(y)$ whenever $x \leq y$.

Then there exists a subsequence $\{g_{n_k}\}$ and a nondecreasing function $g : [a, b] \to \mathbb{R}$ such that $g_{n_k}(x) \to g(x)$ for every $x \in [a, b]$.

We use Helly in Step 3 (Lemma 11.2): the output schedules $q \in \mathcal{A}_q$ are nondecreasing and bounded by $\bar{q}$, so any sequence has a pointwise convergent subsequence.

E.3 The dominated convergence theorem

Lemma E.3 (Lebesgue dominated convergence)**.** *Let (S, Σ, ν) be a measure space and $\{h_n\}_{n=1}^{\infty}$ a sequence of measurable functions $S \to \mathbb{R}$. Suppose:*

(i) $h_n(s) \to h(s)$ for ν-almost every s (pointwise convergence);

(ii) there exists an integrable function $H : S \to \mathbb{R}_+$ with $|h_n(s)| \leq H(s)$ for all n and ν-a.e. s (domination).

Then h is integrable and $\int_S h_n \, d\nu \to \int_S h \, d\nu$.

We use DCT in two places:

(a) Step 5 (continuity of W): if $q_n \to q$ pointwise, the envelope integral $\mathcal{U}_n(z) = u_0 + \int_{z_0}^{z} q_n(s)/s^{2+\gamma} \, ds$ converges by DCT with dominator $\bar{q}/s^{2+\gamma}$ (integrable since $z_0 > 0$). Then $g_{a_n}(z) \to g_a(z)$ pointwise, and $e^{-g_{a_n}/\theta} \to e^{-g_a/\theta}$ with dominator $e^{(C_r + \|\psi - \mu\|_\infty M)/\theta}$ (Lemma 11.3).

(b) Step 5 (integral convergence): $\int e^{-g_{a_n}/\theta} f \to \int e^{-g_a/\theta} f$ by DCT.

E.4 Hölder's inequality

Lemma E.4 (Hölder)**.** *Let (S, Σ, ν) be a measure space, $p, r > 1$ with $1/p + 1/r = 1$, and $\varphi, \psi \geq 0$ measurable. Then*

$$\int_S \varphi \, \psi \, d\nu \leq \left(\int_S \varphi^p \, d\nu \right)^{1/p} \left(\int_S \psi^r \, d\nu \right)^{1/r}. \tag{261}$$

Equality holds if and only if φ^p and ψ^r are proportional ν-a.e.

We use Hölder in Step 6 (Proposition 11.5) with $\varphi = (e^{-g_1/\theta})^t$, $\psi = (e^{-g_2/\theta})^{1-t}$, $p = 1/t$, $r = 1/(1-t)$, and $d\nu = f \, dz$. The equality condition ($e^{-g_1/\theta}$ and $e^{-g_2/\theta}$ proportional f-a.e.) is used in the case $q_1(z) = q_2(z)$ for f-a.e. z but $u_{0,1} \neq u_{0,2}$: proportionality fails because $g_1(z) - g_2(z) = (\psi(z) - \mu)(u_{0,1} - u_{0,2})$ is non-constant when ψ is non-constant (Assumption 11.4).

E.5 The taxation principle

Lemma E.5 (Taxation principle; Hammond 1979, Rochet 1985)**.** *Consider an economy with a continuum of agents indexed by type $z \in \mathcal{Z}$, preferences $U(c, y/z)$ over consumption c and earnings y, satisfying Assumptions 1.1–1.2 of the main notes.*

(i) ***From taxes to allocations.*** *Given a tax schedule $T : \mathbb{R}_+ \to \mathbb{R}$ with $T(0) = 0$, each type z chooses earnings $y(z) \in \arg\max_y U(y - T(y), y/z)$. The resulting allocation $(c(z), y(z))$ with $c(z) = y(z) - T(y(z))$ is incentive-compatible: no type prefers the bundle of any other type.*

(ii) ***From allocations to taxes.*** *Given an incentive-compatible allocation $(c(z), y(z))$ with y nondecreasing, the tax schedule $T(y(z)) = y(z) - c(z)$ for y in the range of $y(\cdot)$ implements the allocation: each type voluntarily chooses its assigned bundle.*

The two maps are inverses of each other (on the range of y). The principle is about household optimization only; it does not involve the planner's objective or any properties of the distribution of types.

E.6 The first-order approach

Lemma E.6 (Sufficiency of the first-order approach; Mirrlees 1971)**.** *Let $U(c, y/z)$ satisfy the single-crossing condition*

$$\frac{\partial}{\partial z}\left(-\frac{U_y(c, y/z)}{U_c(c, y/z)}\right) < 0, \tag{262}$$

i.e., higher types have a lower marginal rate of substitution between consumption and earnings. Then an allocation $(c(z), y(z))$ is incentive-compatible if and only if:

(i) ***Envelope condition:*** *$\mathcal{U}'(z) = -U_y(c(z), y(z)/z) \cdot y(z)/z^2$ for a.e. z, where $\mathcal{U}(z) = U(c(z), y(z)/z)$.*

(ii) ***Monotonicity:*** *$y(z)$ is nondecreasing.*

Under separable preferences $U(c, n) = u(c) - v(n)$ with $n = y/z$, the single-crossing condition (262) is automatically satisfied (since $-U_y/U_c = v'(y/z)/(z\, u'(c))$ is decreasing in z for fixed (c, y)). Under quasilinear isoelastic preferences, the envelope condition becomes $\mathcal{U}'(z) = (y(z)/z)^{1+\gamma}/z = q(z)/z^{2+\gamma}$ (equation (127)).

E.7 Verification of the envelope hypotheses

Proposition 11.2 requires three hypotheses on $g_\alpha(z) = \psi(z)\mathcal{U}_\alpha(z) + \mu\, T_\alpha(y_\alpha(z))$ (equation (114)), where $T_\alpha = T + \alpha\, \delta T$ and $y_\alpha(z)$ solves the household's problem under T_α. We trace each condition back to primitives.

Hypothesis (i): differentiability of $\alpha \mapsto g_\alpha(z)$. This requires differentiability of $\alpha \mapsto y_\alpha(z)$, which follows from the implicit function theorem (IFT) on the household FOC

$$\Phi(\alpha, y) = [1 - T'(y) - \alpha\, \delta T'(y)]\, U_c(y - T_\alpha(y),\, y/z) + \tfrac{1}{z}\, U_n(y - T_\alpha(y),\, y/z) = 0,$$

provided:

(P1) $U \in C^2$, strictly concave (so Φ is C^1 in (α, y));

(P2) $T \in C^2$ and $\delta T \in C^1$ in a neighborhood of $y_0(z)$;

(P3) the household SOC holds strictly: $\partial_y \Phi(0, y_0(z)) < 0$.

Once $y_\alpha(z)$ is C^1 in α, the chain rule gives hypothesis (i).

For the Saez perturbation, $\delta T'$ is a step function, so $\delta T \notin C^1$ at the kink points y^* and $y^* + \varepsilon$. However, the IFT only needs to apply at each type's base earnings $y_0(z)$, and generically only a measure-zero set of types has $y_0(z) \in \{y^*, y^* + \varepsilon\}$. So hypothesis (i) holds for f-a.e. z.

Hypothesis (ii): finiteness of $Z(\alpha)$. On bounded support, g_α is bounded (Lemma 11.3 extended to small α), so $e^{-g_\alpha/\theta}$ is bounded and $Z(\alpha) < \infty$ is automatic. On unbounded support, this is a genuine tail condition requiring the exponential decay from $e^{-g_\alpha/\theta}$ and the tail decay of f to jointly ensure integrability.

Hypothesis (iii): domination. The bound $|\partial_\alpha g_\alpha|\, e^{-g_\alpha/\theta} \le H$ has two parts.

Bounding $|\partial_\alpha g_\alpha|$. By the household envelope theorem, $\partial_\alpha \mathcal{U}_\alpha(z) = -U_c(c_\alpha, y_\alpha/z)\, \delta T(y_\alpha)$, so

$$\partial_\alpha g_\alpha(z) = \big(\mu - \psi(z)\, U_c\big)\, \delta T(y_\alpha) + \mu\, T'_\alpha(y_\alpha)\, \dot{y}_\alpha(z),$$

where $\dot{y}_\alpha = \partial_\alpha y_\alpha$. By the IFT, $\dot{y}_\alpha = -(\partial_\alpha \Phi)/(\partial_y \Phi)$. The numerator involves $\delta T'$ and δT (bounded); the denominator is the SOC (bounded away from zero by (P3)). So $|\partial_\alpha g_\alpha|$ is bounded provided:

(P4) ψ is bounded on $\mathcal{Z}$;

(P5) δT and $\delta T'$ are bounded on the range of earnings;

(P6) the second derivatives of U are bounded at the allocation.

Bounding $e^{-g_\alpha/\theta}$. This requires $g_\alpha(z) \ge -C$ uniformly:

(P7) $g_\alpha(z)$ is bounded below uniformly in (α, z) for $|\alpha| < \alpha_0$.

On compact $\mathcal{Z}$ with admissibility ($0 \le T'_\alpha \le 1$), (P7) follows from bounded ψ, bounded $\mathcal{U}_\alpha$ (from bounded earnings and consumption), and bounded $T_\alpha(y_\alpha)$ — the content of Lemma 11.3 extended to small α.

Combining. On bounded support, $|\partial_\alpha g_\alpha| \le C_1$ and $e^{-g_\alpha/\theta} \le C_2$ uniformly, so $H = C_1 C_2$ is constant and trivially in $L^1(f\, dz)$.

The quasilinear isoelastic case. Under $U(c, n) = c - n^{1+\gamma}/(1+\gamma)$: $U_{cc} = U_{cn} = 0$, the SOC reduces to $-\gamma n^{\gamma-1}/z^2 < 0$ (automatic for $\gamma > 0$), income effects vanish, and $\dot{y}_0 = -\frac{1}{\gamma}\, \frac{y_0}{1-T'}\, \delta T'(y_0)$, which is bounded on compact support.

Unbounded support. On $\mathcal{Z} = [z_0, \infty)$, the constant-dominator argument fails. Hypotheses (ii) and (iii) become genuine tail requirements: the exponential decay from $e^{-g_\alpha/\theta}$ and the tail decay of f must jointly overpower the polynomial growth of $|\partial_\alpha g_\alpha|$ (through ψ, $\delta T(y_\alpha)$, $\dot{y}_\alpha$). This must be verified case by case.

E.8 Asymptotic notation

We use three standard asymptotic notations throughout. Let f and g be real-valued functions with $g(x) > 0$ for all x sufficiently large.

Definition E.1 (Little-o)**.** $f(x) = o(g(x))$ as $x \to \infty$ means

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0.$$

In words: f grows strictly slower than g. The notation $o(1)$ denotes a quantity that vanishes in the limit. For example, $T(y) = \tau\, y + o(y)$ means $(T(y) - \tau\, y)/y \to 0$, i.e., T is asymptotically linear with slope τ.

Definition E.2 (Big-O)**.** $f(x) = O(g(x))$ as $x \to \infty$ means there exist $C > 0$ and x_0 such that $|f(x)| \le C\, g(x)$ for all $x \ge x_0$. In words: f grows no faster than g up to a constant.

Definition E.3 (Asymptotic equivalence)**.** $f(x) \sim g(x)$ as $x \to \infty$ means

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = 1,$$

or equivalently $f(x) = g(x)\,(1 + o(1))$. That is, the approximation error $f(x) - g(x)$ is negligible relative to $g(x)$ itself: $|f - g|/|g| \to 0$. For example, $\mathcal{Y}(z, \gamma; T) \sim c(\gamma; R)\, z^\beta$ means that the ratio $\mathcal{Y}(z, \gamma; T)\,/\,(c(\gamma; R)\, z^\beta)$ converges to 1 as $z \to \infty$—the two quantities become indistinguishable in percentage terms, even though they may differ in absolute level.

Two properties used repeatedly:

(i) If $f \sim g$ and $h \sim k$, then $f/h \sim g/k$ (ratios of asymptotically equivalent quantities are asymptotically equivalent). This is why multiplicative $1+o(1)$ corrections cancel in the ratio $N(y)/D(y)$ in Section 12.9.

(ii) If $f(x) = o(g(x))$, then $e^{f(x)}/e^{g(x)} = e^{f(x)-g(x)} \to 0$ whenever $g(x) - f(x) \to +\infty$. This is the mechanism behind the Laplace principle in Section 12.12.6: exponential gaps overwhelm polynomial prefactors.

Appendix: Log-concavity facts used in the uniqueness analysis

This appendix collects the results invoked in the proof of Proposition 12.5. All results are stated without proof.

Definition E.4 (Log-concavity)**.** A function $h : \mathbb{R} \to [0, \infty)$ is *log-concave* if $\log h$ is concave on its support, i.e., for all x, y in the support and $t \in [0, 1]$,

$$h(tx + (1 - t)y) \geq h(x)^t\, h(y)^{1-t}.$$

Theorem E.1 (Closure under multiplication)**.** *If h_1 and h_2 are log-concave, then $h_1 \cdot h_2$ is log-concave.*

This is immediate from $\log(h_1 h_2) = \log h_1 + \log h_2$.

Theorem E.2 (Popoviciu's variance inequality (Popoviciu, 1935))**.** *If X is a random variable supported on $[a, b]$, then* $\mathrm{Var}(X) \leq (b - a)^2/4$. *Equality holds if and only if X is a symmetric Bernoulli on $\{a, b\}$.*

Theorem E.3 (Log-concave density implies IFR (Bagnoli and Bergstrom, 2005, Theorem 2 and Corollary 2))**.** *If p is a log-concave density on $[0, \infty)$ with survival function $\bar{F}(x) := \int_x^\infty p(t)\, dt$, then X has* increasing failure rate *(IFR): the hazard rate $r(x) := p(x)/\bar{F}(x)$ is non-decreasing.*

Theorem E.4 (IFR implies CV ≤ 1 (standard in reliability theory; see Barlow and Proschan, 1975, Chapter 6))**.** *If X is IFR with mean $\mu > 0$, then* $\mathrm{Var}(X) \leq \mu^2$, *i.e.,* $\mathrm{CV}(X) \leq 1$. *Equality holds only for the exponential distribution.*

Combining the two theorems above:

Corollary E.1 (Variance bound for log-concave distributions on $\mathbb{R}_+$)**.** *Let X be a non-negative random variable with log-concave density and finite mean $\mu > 0$. Then* $\mathrm{CV}(X) \leq 1$.

In Proposition 12.5(b), these results are applied as follows. The tilted density $g_\lambda(e) \propto g(e)\, e^{-\lambda e}$ on $[e_{\min}, e_{\max}] \subset \mathbb{R}_+$ is log-concave by Theorem E.1 (since g is log-concave and $e \mapsto e^{-\lambda e}$ is log-linear). The chain log-concave $\Rightarrow$ IFR (Theorem E.3) $\Rightarrow$ CV ≤ 1 (Theorem E.4) then gives $\mathrm{CV}^2(g_\lambda) \leq 1$ for every $\lambda \geq 0$.

E.9 Alternative proof for the bound CV ≤ 1 for log-concave distributions on $\mathbb{R}_+$

Proposition E.1. *Let X be a nonnegative random variable with density f on $[0, \infty)$. Assume that f is log-concave on its support. Then*

$$\mathrm{Var}(X) \leq (\mathbb{E}X)^2.$$

Equivalently,

$$\mathrm{CV}(X) = \frac{\sqrt{\mathrm{Var}(X)}}{\mathbb{E}X} \leq 1.$$

Proof. Write $\bar{F}(x) = \mathbb{P}(X > x)$ for the survival function, and let $I = [a, b) \subseteq [0, \infty)$ be the support of f (so $0 \leq a < b \leq \infty$). On I, the function $\phi = \log f$ is concave.

We first show that the hazard rate

$$h(x) := \frac{f(x)}{\bar{F}(x)}$$

is nondecreasing on I (and hence on $[0, \infty)$ if we set $h(x) = 0$ for $x < a$).

Fix $a \leq x < y < b$, and let $u \in [0, b - y)$. Set

$$\theta = \frac{u}{y - x + u} \in [0, 1].$$

Then

$$y = \theta x + (1 - \theta)(y + u), \qquad x + u = (1 - \theta)x + \theta(y + u).$$

By concavity of ϕ,

$$\phi(y) \geq \theta\phi(x) + (1 - \theta)\phi(y + u),$$

and

$$\phi(x + u) \geq (1 - \theta)\phi(x) + \theta\phi(y + u).$$

Adding these two inequalities gives

$$\phi(y) + \phi(x + u) \geq \phi(x) + \phi(y + u),$$

hence

$$f(x)\, f(y + u) \leq f(y)\, f(x + u).$$

Integrating in u over $[0, b - y)$ yields

$$f(x)\bar{F}(y) \leq f(y)\bar{F}(x),$$

that is,

$$h(x) \leq h(y).$$

So h is nondecreasing.

Next we show that $\bar{F}$ is submultiplicative:

$$\bar{F}(x + y) \leq \bar{F}(x)\bar{F}(y) \qquad \text{for all } x, y \geq 0.$$

Indeed, if $x+y \geq b$, then $\bar{F}(x+y) = 0$ and there is nothing to prove. So assume $x+y < b$. Since $\bar{F}$ is absolutely continuous and $\bar{F}'(t) = -f(t)$ a.e. on $[0, b)$, we have

$$(\log \bar{F})'(t) = \frac{\bar{F}'(t)}{\bar{F}(t)} = -h(t) \qquad \text{a.e.}$$

Therefore

$$\log \bar{F}(x+y) - \log \bar{F}(x) = -\int_x^{x+y} h(s)\,ds \le -\int_0^y h(s)\,ds = \log \bar{F}(y),$$

because h is nondecreasing. Exponentiating gives

$$\bar{F}(x+y) \le \bar{F}(x)\bar{F}(y).$$

In particular, choose $r > 0$ with $\bar{F}(r) < 1$. Then

$$\bar{F}(nr) \le \bar{F}(r)^n,$$

so $\bar{F}$ decays at least exponentially on a lattice, and hence $\mathbb{E}X < \infty$. Thus the standard tail identities apply:

$$\mathbb{E}X = \int_0^\infty \bar{F}(x)\,dx, \qquad \mathbb{E}X^2 = 2\int_0^\infty x\,\bar{F}(x)\,dx.$$

Now, by submultiplicativity of $\bar{F}$,

$$x\,\bar{F}(x) = \int_0^x \bar{F}(x)\,du \le \int_0^x \bar{F}(u)\bar{F}(x-u)\,du.$$

Integrating over $x \in [0,\infty)$ and applying Fubini,

$$\frac{\mathbb{E}X^2}{2} = \int_0^\infty x\,\bar{F}(x)\,dx \le \int_0^\infty \int_0^x \bar{F}(u)\bar{F}(x-u)\,du\,dx = \left(\int_0^\infty \bar{F}(u)\,du\right)^2 = (\mathbb{E}X)^2.$$

Hence

$$\mathbb{E}X^2 \le 2(\mathbb{E}X)^2,$$

and therefore

$$\mathrm{Var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 \le (\mathbb{E}X)^2.$$

This is exactly $\mathrm{CV}(X) \le 1$. □

References for this appendix.

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